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# Koyon Injin da Dan Adam a Cikin Zagaye: An Fahimce Shi Kuskure

**Lazaros Toumanidis**

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Wannan littafin jagora yana nazarin daya daga cikin mahimman yankuna — kuma mafi fahimtar kuskure — na fasahar wucin-gadi ta zamani: shigar da dan adam cikin zagayen koyo. Daga lakabi na bayanai da koyon aiki zuwa RLHF da koyo daga misalai, hukuncin dan adam yana da daraja marar misali.

Littafin yana nufin masu bincike, injiniyoyi, da dalibai wadanda suke son fahimtar lokacin, dalilin, da yadda dole ne dan adam ya shiga cikin zagayen koyo.

:::{admonition} Bugu na Hausa :class: note Wannan shi ne fassarar Hausa. Asalin turanci shi ne bugun farko na tunani. Wasu babi-babi na fasaha na iya zama har yanzu cikin ci gaba. :::

---

## Tsarin Littafi

**Kashi na Farko — Tushe da Ginshi□ ai** — yana kafa tsarin tunani na HITL ML da matakan sarrafa kai.

**Kashi na Biyu — Dabarun Asali** — yana rufe lakabi na bayanai, koyon aiki, da koyon injin mai mu'amala.

**Kashi na Uku — Koyo daga Ra'ayin Dan Adam** — yana rufe RLHF, koyo daga misalai da kwatancen.

**Kashi na Hudu — Aikace-aikace** — yana nazarin misalai na NLP, gane hoto ta kwamfuta, da lafiya.

**Kashi na Biyar — Tsarin Aiki da Aiwatarwa** — yana rufe kayan aiki, dandamali, da ma'aunai na tsarin HITL.

**Kashi na Shida — Da'a da Makomar Fasaha** — yana magana kan adalci, nuna bambanci, da makomar huldar dan adam da algorithm.

# Sanarwar Asali

*An gabatar a matsayin wani □ angare na rashin cikawa na bu□ atun digiri na Doctor of Philosophy.*

---

Ina bayar da sanarwa a nan cewa aikin da aka gabatar a wannan littafi na ne, sai inda ba shi ba ne.

Wa□ annan ra'ayoyin a cikin wa□ annan shafukan an samar da su tare da taimakon: masu bincike da suka wallafa takardunsu kuma wani lokaci suka ajiye su a bayan biyan ku□ i; masu lakabi wa□ anda aikinsu ya sa samfurin ya yiwu amma sunayensu ba su bayyana a wani bangare na godiya ba; masu gudummawar bu□ a□ □ en tushe wa□ anda suka rubuta dakunan karatu da aka yi amfani da su a duk misalin lambar; da rubutaccen rubutu na wayewar dan adam, an ha□ a shi kuma an takaitashi ta manyan samfurin harshe wa□ anda ba su da ra'ayi kan kowane □ in su.

Kurakurai nawa ne. Ra'ayoyin sun kasance na ha□ in gwiwa.

Ina tabbatarwa cewa ba a gabatar da wannan aiki don wani digiri ko cancanta ta □ wararru, kyautar ilimi, tallafi, □ addamarwar matayar aiki, ko wata saTa'a ta LinkedIn ba, sai dai yadda aka bayyana.

Ina □ ara tabbatarwa cewa ba a ha□ a wani bangare na kwatancen SOTA (Matsayin Zamani) a wannan aiki ba. Masu karatu ba za su sami tebur da ke jera hanyoyi 47 a gwaji 12 ba, inda hanyar da aka ba da shawara ta cimma ingantattun fa'idodi masu ma'ana a 9 daga cikin su (†p $< 0.05$). Wannan ba ne saboda ba za a iya gina irin wannan tebur ba, amma saboda gina shi zai kasance daga batu.

Batu shi ne: dan adam yana cikin zagaye. Koyaushe ya kasance. Wannan littafi yana □ o□ arin fahimtar abin da hakan ke nufi.

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*Sanya hannu: Dan Adam a Cikin Zagaye*

*Ranar: Duk lokacin da kake karanta wannan*

*Cibiya: Babu. Wannan shi ne batu.*

---

**Game da taken kari** :class: note

*An fahimce shi kuskure* yana nufin filin, matsayin dan adam a ciki, da mai karatu — duka lokaci guda. Idan kana tsammanin yana nufin kai kawai, ka iya zama daidai. Idan kana tsammanin yana nufin injuna kawai, kana tabbatar da taken kari.

# Takaitaccen Bayani

**Koyon Injin da Dan Adam a Cikin Zagaye: An Fahimce Shi Kuskure**

Labarin da ya yi rinjaye game da koyon injin yana sanya dan adam a matsayin mai shiga na wucin gadi — yana nan lokacin tara bayanai da horar da samfuri, sannan a hankali ake cireshi yayin da atomatik yana inganta. Wannan littafi yana jayayya cewa wannan labarin ba daidai ba ne, koyaushe bai daidai ba, kuma fahimtar *dalilin da ya sa* ba daidai ba ne na □ aya daga cikin mafi amfanin abubuwa da mai aikin koyon injin zai iya yi.

Koyon Injin da Dan Adam a Cikin Zagaye (HITL) shine ilimin da ke mai da shiga dan adam a cikin tsarin koyo da gangan, tsari, da kuma ci gaba. Maimakon □ aukar ra'ayin dan adam a matsayin ku□ in da za a rage, HITL yana □ aukarsa a matsayin alama da za a fahimta: inda ya fito, abin da yake □ oye, yadda za a tattarashi yadda ya kamata, da yadda za a ha□ a shi cikin aminci a cikin samfurin da ke □ abi'anta kamar yadda masu zanen su da masu amfani da su ke nufin gaske.

Wannan littafi yana ba da cikakken magance filin a yankuna shida. Kashi na Farko yana kafa ginshi□ ai da rarrabuwa ta huldar dan adam da injin. Kashi na Biyu ya □ unshi dabarun gargajiya: lakabi da bayanai, koyon aiki, da koyon injin mai mu'amala. Kashi na Uku ya magance tsarin zamani na daidaitawa: koyon □ arfafawa daga ra'ayin dan adam (RLHF), koyo daga misalai, da koyon fifiko. Kashi na Hudu yana amfani da wa□ annan dabaru ga sarrafa harshen yau da kullum, gane hoto ta kwamfuta, da kiwon lafiya. Kashi na Biyar ya □ unshi tsarin aiki da ababen more rayuwar HITL a sikeli: dandamali na lakabi, sarrafa ingancin tarin aiki, da hanyoyin kimantawa. Kashi na Shida yana duba da'a da makomar filin.

Kowane babi ya ha□ a da ma'anoni na yau da kullum, magance lissafi inda ya dace, da misalan lambar da za a iya gudanar da su a cikin Python.

Wannan littafi an yi niyya ga masu aikin da ke gina tsarin ML, masu bincike da ke nazarin su, □ alibai da ke koyo game da su, da kuma kowa wanda ya ta□ a mamaki wane ya sanya wa bayanai lakabin da ya horar da samfurin da ya yanke hukuncin da ya shafi rayuwarsa.

Amsa, a duk lokuta, ita ce: dan adam. Galibi ana biya shi kasa. Da wuya a yabe shi. Koyaushe yana cikin zagaye.

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**Kalmomi masu mahimmanci:** koyon injin da dan adam a cikin zagaye, koyon aiki, labelin bayanai, koyon □ arfafawa daga ra'ayin dan adam, RLHF, koyon fifiko, tarin aiki, koyon injin mai mu'amala, daidaitar AI, AI mai alhakin

**Rarrabuwar batu:** Koyon Injin · Huldar Dan Adam da Kwamfuta · Fasahar Wucin Gadi · Kimiyyar Bayanai

**Yawan kalmomi:** Ya isa.

# Godiya

Kowane littafi game da koyon injin da dan adam a cikin zagaye ya kamata ya fara da mutanen da ke cikin zagayensa.

**Masu lakabi.** Wannan littafi, a wani □ angare, game da ku ne — game da aikinku, hukumcinku, da kuma gaskiyar cewa sunayenku ba su bayyana a wani katin bayanai, ko wata takardar godiya ta takarda, ko wata ganewa ta jarida game da samfurin da aikinku ya sa ya yiwu ba. Filin yana farawa da □ aukar wannan nauyin. Yana da dogon hanya.

**Masu gudummawar bu□ a□ □ en tushe** wa□ anda dakunan karatunsu ake amfani da su a duk misalin lambar: al'ummar NumPy (tun 2006), masu taimaka wa scikit-learn (tun 2007), □ ungiyar PyTorch, al'ummar Hugging Face, da masu kula da yawa wa□ anda ke gyara kurakurai a karshen mako ba tare da lada ko karramawa ba. □ ididdige kimiyya an gina shi akan wannan aiki mai □ oye.

**Masu bincike** wa□ anda suka wallafa aikinsu openly — a cikin wallafe-wallafen arXiv, wuraren bu□ a□ □ en dama, da wuraren ajiya na GitHub — sun sa yana yiwuwa ga masu aiki su koyo ba tare da dama ta cibiya ba. Da masu bincike wa□ anda suka duba takardun cikin hankali da gaskiya, wata nau'i ta lakabi ta □ wararru da ilimi ta yadda ya sa ya yi imanin ba aiki ba ne.

**Malamai** wa□ anda suka sa ilimi ya samu: masu rubutu a shafin yanar gizo, masu rubuta koyarwa, masu amsa tambayoyi a Stack Overflow, masu gabatarwa a taro wa□ anda suka aiko da abubuwan gabatarwarsu. Koyo ba ya faruwa a ke□ e, kuma ba ya faruwa ba tare da mutane da ke son bayyana abubuwa ga ba□ i ba.

**Masu sukar.** Masu bincike wa□ anda suka nuna cewa bayanan bayanai suna □ oye son kai, cewa halaye na masu lakabi yana tsara □ abi'ar samfuri, cewa mutanen da ke cikin zagaye galibi sune mafi rauni daga cikin mahalarta tsarin. Wannan filin ya fi alheri saboda □ in ku na tambayoyin da ke sa rashin jin da□ i.

---

> *Akwai sanannun sani — abubuwan da muka sani cewa mun san su. Akwai sanannun rashin sani — abubuwan da muka sani ba mu san su ba. Akwai rashin sanannun rashin sani — abubuwan da ba mu san cewa ba mu san su ba.*
>
> *Sa'an nan akwai abubuwan da kowa a filin ya san su amma ba sa shiga cikin takardun sosai.*
>
> *Wannan littafi, a wani □ angare, □ o□ ari ne na nau'in □ arshe.*

---

*Idan ka ba da gudummawa ga wannan filin kuma ba ka ga sunanka a nan ba: an gode ka. Zagaye ya ha□ a da kai.*

# Menene Koyon Injin da Dan Adam a Cikin Zagaye?

> *Injin da zai iya koyo daga kwarewa… amma kawai idan ka ba shi kwarewar da ta dace. – Fassara daga Alan Turing*

## Sauye-Sauyen Atomatik

Kowane sabon zagaye na atomatik yana haifar da bu□ atu sabon kan kulawar dan adam. Lokacin da kamfanonin jiragen sama suka gabatar da atomatik tuka, direbobi suka zama masu sa ido — ba su da alhakin sarrafa daki da daki amma suna da aikin mafi wahala na sanin *lokacin* da za su □ auki mulki. Lokacin da kasuwa suka gabatar da biyan ku□ i kai tsaye, sun gano cewa tsarin yana bu□ atar kulawar dan adam fiye da masu kudi na gargajiya, ba □ asa ba. Kuma lokacin da koyon injin ya fara yanke hukunci a sikeli — a likitanci, ku□ i, aiki, da sarrafa abun ciki — ya haifar da bu□ atu mai yawa, ci gaba, na hukuncin dan adam.

Wannan shine **sauye-sauyen atomatik** {cite}`bainbridge1983ironies`: da yake tsarin atomatik ya zama mai □ arfi, sakamakun gazawarsa suna da muhimmanci fiye da kima, kuma saboda haka kulawar □ an adam mai □ arfi ta zama dole fiye da kima. Koyon injin ba banda ba ne.

Koyon Injin da Dan Adam a Cikin Zagaye (HITL ML) shine filin da ya □ auki wannan sauye-sauye da mahimmanci kuma ya gina shi cikin zanen tsarin daga farko. Maimakon □ aukar shiga dan adam a matsayin gini na wucin gadi da za a □ arshe a cire, HITL ML yana □ aukar huldar dan adam da injin a matsayin *fasali* — tushen alama, gyarawa, da jagora wanda zai iya sa samfurin ya yi daidai fiye da kima, ya fi dacewa da □ imar dan adam, kuma ya fi aminci.

---

## Ma'anar Koyon Injin da Dan Adam a Cikin Zagaye

Babu wata ma'ana guda □ aya da aka amince da ita na HITL ML, kuma ana amfani da kalmar ta hanyoyi da yawa masu juna cikin wallafe-wallafen. Don wannan littafi muna □ aukar ma'ana faffadan amma daidai:

:::{admonition} Ma'ana :class: important

**Koyon Injin da Dan Adam a Cikin Zagaye** yana nufin duk tsarin ML ko hanyar da a cikinta ra'ayin dan adam ake ha□ awa cikin tsarin koyo ta hanyar *da gangan, tsari, da kuma ci gaba* — ba kawai a lokacin □ ir□ ira bayanan da aka yi lakabi ba, amma a duk lokacin horarwa, kimantawa, da aikawa.

:::

Wannan ma'ana tana da manyan rukunai uku:

**Da gangan.** HITL ba tasiri na dan adam na ha□ ari ba ne (misali, gaskiyar cewa bayanin horarwa asali dan adam ne ya □ ir□ ira shi). Yana nufin tsarin da aka tsara da gangan don neman, ha□ awa, da amfana da ra'ayin dan adam.

**Tsari.** Ra'ayin yana da nau'i da aka bayyana — lakabi, gyarawa, hukuncin fifiko, misali — da kuma matsayi da aka bayyana a cikin algorithm □ in koyo.

**Ci gaba.** Zagayen ra'ayi yana ci gaba akan lokaci, yana ba da damar tsarin ya inganta yayin da ya samu sabbin yanayi, ya yi kuskure, kuma ya kar□ i jagoran dan adam.

Wannan ma'ana ta ha□ a da tsarin lakabi na gargajiya, koyon aiki, ML mai mu'amala, RLHF, da koyon kwaikwayo. Yana ke□ ance tarawa ta wucin gadi da koyon supervised na layi guda kawai (ko da yake iyaka ba ta bayyana ba, kamar yadda za mu tattauna a Babi na 2).

---

## Tarihin Gajere

### Tsarin □ wararru da Injiniyan Ilimi (1960s–1980s)

Tsarin AI na farko kusan gaba □ aya dan adam ne cikin zagaye: injiniyoyin ilimi sun zauna tare da □ wararrun yanki na watanni, suna yin lissafin □ a'idoji cikin hankali a cikin tsarin □ wararru kamar MYCIN da DENDRAL. Kowane yanki na ilimi dan adam ne ya bayar da shi a fili. Injin ya kasance mai aiwatarwa; dan adam ya kasance oracle.

Wa□ annan tsarin sun yi aiki da kyau sosai a yankuna masu kunkuntar amma sun kasance masu laushi — ba su iya fa□ a□ awa bayan □ a'idodinsu da aka yi da hannu ba kuma sun kasance masu tsada don kulawa.

### Juyin Lissafi (1980s–2000s)

Sauyin zuwa koyon injin na lissafi a cikin 1980s da 1990s ya canza yanayin shiga dan adam. Maimakon yin rubutu na ilimi a matsayin □ a'idoji, dan adam yanzu yana ba da *misalai* — bayanan da aka yi lakabi wa□ anda suke ba da damar samfuri su fitar da salon. Matsayin dan adam ya zama na mai lakabi: yin lakabi ga takardun karatu, alamanta hotuna, rubuta magana.

Wannan babban mataki ne gaba, amma ya haifar da wani □ angare da ke toshe yanayi sabon: **bayanan da aka yi lakabi suna da tsada**. Masu bincike sun fara tambaya yadda za su samu mafi amfanin amfani da □ o□ arin lakabi na dan adam. Wannan tambayar ta haifi **koyon aiki**, wanda aka tsara shi a farkon 1990s {cite}`cohn1994improving`.

### Zamanin Koyon Zurfin (2010s–yanzu)

Juyin koyon zurfin ya haifar da sabon tsari: samfurin masu □ iliyan □ iliyan na sinadarai wa□ anda za su iya koyon ayyuka masu rikitarwa daga bayanai — amma suna bu□ atar bayanan da aka yi lakabi mafi girma kwatankwacin su. ImageNet (hoto miliyan 14 da aka yi lakabi) da ayyukan lakabi masu sikeli wa□ anda suka zo bayan sun nuna iko da kuma ku□ in sikeli.

A lokaci □ aya, aikawa ML a sikeli ya fallasa matsaloli sabon: samfurin masu daidai a matsakaici amma ba daidai ba a tsari don □ ungiyoyi musamman, wa□ anda suke □ ir□ ira gaskiya da amincewar kuskure, wa□ anda ke inganta wakilcin aunawa maimakon □ imar dan adam. Wa□ annan gazawar sun □ arfafa sabon nau'in shiga dan adam: ba kawai lakabi ba, amma *daidaitawa* — aikin sa samfurin ya □ abi'anta ta hanyoyin da dan adam ke nufin gaske.

Mafi bayyanar bayyanar wannan aikin HITL mai mai da hankali kan daidaitawa shine **Koyon □ arfafawa daga Ra'ayin Dan Adam (RLHF)** {cite}`christiano2017deep`, wanda ya zama ginshi□ in tsarin kamar InstructGPT {cite}`ouyang2022training` da iyawar bin umarni na samfurin harshe na zamani.

---

## Me Yasa HITL? Hujjar Hukuncin Dan Adam

Mene ne ya sa hukuncin dan adam yana da □ ima sosai don ha□ awa cikin tsarin koyon injin? Siffofi da yawa:

### 1. Hankali na Yau da Kullum da Ilimin Duniya

Dan adam yana kawo ilimin baya mai yawa ga kowane aiki. Lokacin da likitan hoto ya yi wa hoto X lakabi, tana jan abubuwa daga shekarun horo, fahimtar jikin mutum, da ilimin da aka □ oye game da yadda cututtuka ke kama — ilimi mai wahala wajen □ ayyadewa ko samu daga bayanai kawai.

### 2. Gine-ginen Ma'ana

Lakabin yana da ma'ana domin dan adam yana fahimtar abin da suke nufi. Ajin "cat" a ImageNet yana nufin ra'ayi mai laushi wanda dan adam ke gane shi ta asali amma babu wata ma'anar yau da kullum da ta cika canzawa. Samfurin yana koyon fa□ a□ awar lakabi (wa□ anne hotuna ake yin mu'amala da shi) amma ba zai iya koyon ra'ayin kansa ba, yana haifar da gazawar a lokuta na iyaka wanda kowane dan adam zai yi da kyau.

### 3. Daidaitan □ ima

Dan adam yana da fifiko, □ ima, da hukunci na da'a wanda samfurin ML ba za su iya samo daga bayanai kawai ba. Ko rubutu "mai taimako" ko "mai cutarwa" ba tambaya ta kimiyya kawai ba ne — yana dogara akan □ imar dan adam wanda ke bambanta a cikin mutane dabandaban da mahallin. HITL shine babbar hanyar da za a iya sadar da wa□ annan □ imomi ga tsarin ML.

### 4. Iya Daidaitawa

Hukuncin dan adam zai iya daidaitawa ga sabbin yanayi ba tare da sake horarwa ba. Samfuri da aka horar akan bayanai na tarihi na iya gazawa da tsanaki lokacin da duniya ta canza; dan adam na iya gane sauyi kuma ya amsa da kyau.

### 5. Alhaki

A yankunan masu mahimmanci — likitanci, shari'a, ku□ i — yanke shawara suna bu□ atar alhakin ga dan adam. Tsarin HITL yana sa wannan alhaki ya yiwu ta hanyar sanya dan adam a matsayin da zai iya fahimtar, tabbatar da, da □ etare □ abi'ar injin.

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## Zagayen Ra'ayi

Tsarin tsakiyar HITL ML zagaye ne na ra'ayi tsakanin samfuri da □ aya ko fiye da dan adam:

```
+----------------------------------------------+
|                                              |
|   Samfuri yana yanke hukunci / roƙo          |
|   ----------------------------------->       |
|                                  Dan Adam    |
|   Dan Adam yana ba da ra'ayi     <-------- |
|   ----------------------------------         |
|                                              |
|   Samfuri yana sabuntawa kan ra'ayi         |
|                                              |
+----------------------------------------------+
```

Yanayin ra'ayi yana bambanta sosai a cikin tsarin HITL:

| Nau'in ra'ayi | Misali | Tsarin farko |
|---|---|---|
| Lakabi aji | "Wannan imel faci ne" | Koyon supervised |
| Gyarawa | "Wuri ya kamata ya zama ORG, ba PER ba" | Koyon aiki / mu'amala |
| Fifiko | "Amsar A ta fi ta B" | RLHF / jeri |
| Misali | Nuna robot yadda za ya kama abu | Koyon kwaikwayo |
| Harshen yau da kullum | "Kasance mai ta□ aita a amoshin ka" | Daidaitar umarni |
| Alama da □ oye | Mai amfani ya danna / bai danna ba | Ra'ayin da □ oye |

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## Abin da HITL Ba Shi Ba

Ya cancanta mu daidaita abin da ke *waje* da ma'anarmu.

**HITL ba daidai yake da aikawa da dan adam a cikin zagaye ba** (wani lokaci ana kiransa "dan adam a saman zagaye"), inda dan adam ke sa ido kan yanke shawara ta atomatik kuma zai iya □ etare amma ba suna ciyar da gyarawa zuwa horarwa ba. Za mu tattauna wannan bambance a Babi na 2.

**HITL ba □ aramar kulawa kawai ba ne.** Tsarin lakabi na shirye-shirye kamar Snorkel yana amfani da ayyukan lakabi (galibi □ a'idoji da dan adam ya rubuta) don □ ir□ ira lakabin da ke da □ ura a sikeli. Wannan wata nau'i ce ta shigar dan adam mai tsari, amma ra'ayin ba sake zagaye ba ne ta hanyar da HITL galibi ke nuna shi.

**HITL ba kawai amfani da bayanai da aka yi lakabi ba ne.** Kowane samfuri na koyon supervised yana amfani da bayanai da dan adam ya yi lakabi. HITL yana nufin musamman tsarin inda ra'ayin dan adam shine *wuri mai aiki, sake zagaye* na tsarin koyo.

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## Tattalin Arzikin Ra'ayin Dan Adam

Ra'ayin dan adam yana da □ ima amma yana da tsada. Lakabi ta hoto na likitanci na iya kashe daga goma zuwa □ aruruwa dalar Amurka kowace hoto lokacin da □ wararren ne ya yi, dangane da fanin kuma rikitarwar aikin {cite}`monarch2021human`. Lakabin masu aiki a yanar gizo kamar Amazon Mechanical Turk na iya kashe $0.01–$0.10 kowace item {cite}`hara2018data` a

inganci mafi □ asa. Kalubalen asali na HITL ML shine **yin mafi amfanin kowane naushi na ra'ayin dan adam**.

Wannan yana haifar da tambaya ta ha□ awa da ke gudana ta cikin mafi yawan wannan littafi:

:::{admonition} Tambayar Tsakiyar HITL ML :class: tip

*Da wani tsarin kasafin ku□ i na lokaci da kulawar dan adam, ta yaya za mu yanke shawara abin da za mu tambaye dan adam, lokacin da za mu tambaya, da yadda za mu ha□ a amoshin su cikin horarwa ta samfuri?*

:::

Amsar wannan tambayar tana dogara kan yanki, nau'in ra'ayi, ku□ in lakabi, ha□ arin kuskure, da halin yanzu na samfuri — shi ya sa HITL ML ilimi mai arziki kuma har yanzu yana ha□ aka.

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## Duba a Kan Littafi

Sauran wannan littafi yana da tsari kamar haka. **Kashi na Biyu** ya □ unshi ginshi□ ai uku na gargajiya na HITL: lakabi, koyon aiki, da ML mai mu'amala. **Kashi na Uku** ya magance tsarin zamani na koyo daga ra'ayi — RLHF, koyon kwaikwayo, da koyon fifiko — wa□ anda suka zama tsakiya ga AI na zamani. **Kashi na Hudu** yana duba HITL ta fuskantoki na yanki musamman. **Kashi na Biyar** yana □ aukar hangen mai aiki akan dandamali, tarin aiki, da kimantawa. **Kashi na Shida** yana magance da'a da yin duba gaba.

> **See also:** Don duba mai mai da hankali kan mai aiki na filin, duba {cite}`monarch2021human`. Don takarda asali ta koyon aiki da ta fara yawancin magance yau da kullum na HITL, duba {cite}`settles2009active`.

# Rarrabuwa ta Huldar Dan Adam da Injin

Fa□ i kalmomin yau da kullum shine shara□ i don tunani mai bayyana. Ana amfani da kalmar "dan adam a cikin zagaye" cikin sassauci a aiki — wani lokaci yana nufin dan adam yana yin lakabi akan bayanan horarwa, wani lokaci cewa dan adam zai iya □ etare yanke shawarar samfuri, kuma wani lokaci cewa dan adam yana kula da tsarin koyo a lokaci na gaske. Wa□ annan abubuwa ne masu mahimmanci daban-daban.

Wannan babi yana zana shimfidar huldar dan adam da injin a ML, yana ba da kalmomi na ra'ayi da ake amfani da su a sauran littafi.

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## Matakin Atomatik

Mafi mahimmancin bambance shine yadda dan adam ke shiga cikin yanke shawarar tsarin sosai. Tsarin da aka sani daga ka'idar atomatik {cite}`sheridan1992telerobotics` yana bambance matakai goma, amma don manufofin ML rukunai uku suna kama canjin muhimmanci:

### Dan Adam a Cikin Zagaye (HITL)

Dan adam shine *mai shiga aiki a cikin tsarin koyo*. Yanke shawara — ko a □ alla, yanke shawara masu muhimmanci — yana bu□ atar shigar dan adam kafin a □ arshe su. Tsarin ba zai iya aiki ba tare da shiga dan adam da ke ci gaba.

*Misalai:* Tsarin koyon aiki da ke tambaya likita kafin □ ara wani yanayi zuwa bayanin horarwa. Mai lakabi da ke yin lakabi misalai wa□ anda nan da nan ake amfani da su don sabunta samfuri. Mai sanya wa lakabi RLHF da ke kwatanta sakamakon samfuri.

### Dan Adam a Saman Zagaye (HOTL)

Tsarin yana aiki ta atomatik amma dan adam yana sa ido kuma zai iya shiga tsakani. Dan adam shine *mai sa ido*, ba mai shiga ba. Ra'ayi na iya ko ba zai iya koma cikin horarwa ba.

*Misalai:* Tsarin sarrafa abun ciki da ke yin alamar rubutu ta atomatik; mai duba dan adam yana bitar □ angare kuma yana gyara yanke shawara. Atomatik tuka da ke tuka jirgin amma yana fa□ akar da matukin jirgi game da sabon abu.

### Dan Adam a Ciki Umarni (HIC)

Dan adam yana yanke duk shawarwari; tsarin yana ba da *shawarwari ko bayanai* amma ba shi da atomatik. Wannan mafi rauni ne na aikawa ML.

*Misalai:* Tsarin tallafi na ganewar cutar da ke nuna wa likita □ iyadar iya yiwuwar samfuri amma yana barin yanke hukuncin □ arshe gaba □ aya ga likita.

> **Wane matakin ya dace?** :class: tip
>
> Matakin atomatik da ya dace yana dogara akan ku□ in kurakurai, amincin samfuri, □ warewar dan adam da yake samuwa, da □ untatawa na latency na aiki. Wa□ annan abubuwa suna canzawa yayin da samfuri ya nuna — mafi yawan tsarin suna fara HITL kuma suna □ aura zuwa HOTL yayin da amincewar ke □ aruwa.

```
Bayanai mentah --> Sarrafa --> Fasali --> Horarwa --> Kimantawa --> Aikawa
    ^                ^                      ^              ^              ^
    |                |                      |              |              |
 Tattara       Lakabi                   Koyon          Gwaji        Sa ido
 ra'ayi        & sanya                  aiki           ra'ayi       & gyarawa
               lakabi
```

| Matakin | Matsayin dan adam | Babi |
|---|---|---|
| Tattara | Yanke shawara abin da za a tattara; tsarin samfuri | 3, 4 |
| Lakabi | Bayar da lakabin, tsari, metadata | 3, 13 |
| Horarwa | Tambayoyin koyon aiki; ra'ayin yanar gizo | 4, 5, 6 |
| Kimantawa | Kimantawar dan adam na sakamakon samfuri | 14 |
| Aikawa | Sa ido, sarrafa ke□ antawa, gyarawa | 12, 14 |

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## Shiga Mai Aiki da Wanda Ba Na Aiki Ba

A cikin HITL *mai aiki*, tsarin yana za□ ar wa□ anne bayanai za a gabatar ga dan adam — yana tambaya da dabarun lissafi. A cikin HITL *wanda ba na aiki ba*, dan adam yana ba da ra'ayi akan duk bayanai da suka isa (misali, jerin lakabi da aka sanya jeri jeri).

Shiga mai aiki yana da inganci fiye da kima saboda ra'ayin yana gudana inda zai fi inganta samfuri. Shiga wanda ba na aiki ba yana da sau□ i don aiwatarwa da sarrafa.

Bambance na ala□ a shine tsakanin ra'ayi **jeri** da **yanar gizo**:

- **Jeri:** Dan adam yana yin lakabi ga babban rukunin misalai; samfuri yana sake horarwa. Maimaita.
- **Yanar gizo (cigaban yanayi):** Ra'ayin dan adam yana isa cikin ci gaba; samfuri yana sabunta a hankali.

Aikin jeri shine al'ada a masana'anta (ayyukan lakabi da aka biyo bayan gudu na horarwa). Aikin yanar gizo ya fi yanayi ga tsarin mai mu'amala da yanayin koyon □ arfafawa.

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## Mai Sanya Lakabi □ aya da Masu Yawa

Yawancin gabatarwa na yau da kullum na HITL suna □ auka mai sanya lakabi □ aya, mai daidaito. A aiki, lakabi yana ha□ ar da mutane da yawa, kuma hukumcin su yana bambanta.

**Ha□ awa** yana ha□ a lakabi da yawa cikin lakabi □ aya, galibi ta za□ in rinjaye ko tsarin lissafi (Babi na 13).

**Rashin jituwa a matsayin alama** — wasu masu bincike suna jayayya cewa rashin jituwa na masu sanya lakabi bayanai ne masu □ ima wa□ anda ba za a runtuma zuwa lakabi □ aya na "zinariya" ba. Hanyoyin da ke la'akari da hangen nesa ga NLP, misali, suna kiyaye lakabi da yawa a matsayin lakabin laushi wa□ anda ke nuna rikitarwan gaskiya na bayanai {cite}`uma2021learning`.

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## Tsarin Ha□ e

Za mu iya wakilta duk tsarin HITL da biyar-biyar:

$$\text{HITL config} = (\text{level}, \text{modality}, \text{stage}, \text{frequency}, \text{annotators})$$

inda:

- **matakin** $\in$ {HITL, HOTL, HIC}
- **yanayi** $\in$ {lakabi, gyarawa, misali, fifiko, NL, da □ oye}
- **matakin aiki** $\in$ {tattara, lakabi, horarwa, kimantawa, aikawa}
- **yawan lokaci** $\in$ {jeri, yanar gizo}
- **masu sanya lakabi** $\in \mathbb{N}^+$ (yawan masu sanya lakabi kowace item)

Wannan rarrabuwa tana ba mu damar kwatanta tsarin HITL iri-iri kan axes □ aya kuma tunani game da musayar tsakanin su. Sauran littafi yana huda zuwa sel musamman na wannan sararin.

> **See also:** Don bincike na yadda yanke shawara na lakabi ke shafar □ abi'ar samfuri a □ asa, duba {cite}`bender2021stochastic`. Don bita na tsarin ML mai mu'amala, duba {cite}`amershi2014power`.

# Bayanan Bayanai da Lakabi

Lakabi bayanai shine mafi yawaitar nau'in shiga dan adam a ML. Kafin samfuri ya iya koyo, dole ne wani ya gaya masa abin da amsar daidai take — kuma wannan wani galibi dan adam ne. Wannan babi ya □ unshi ka'idar da aikin lakabi: abin da ke sa lakabi ya yi wuya, yadda za a tsara ayyukan lakabi, yadda za a auna inganci, da yadda za a magance rashin jituwa.

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## Nau'in Lakabi

Ayyukan lakabi suna bambanta sosai a cikin tsarinsu, wahalarsu, da ku□ insu. Manyan nau'in sun ha□ a da:

### Rarrabuwa

Mai lakabi yana sanya kowane misali zuwa □ aya daga cikin $K$ rukunan da aka riga aka □ ayyadewa. Wannan mafi sau□ in aikin lakabi ne ta kwamitin tunani, amma □ ayyadewa tsarin rukunan mai kyau (*rarrabuwa*) na iya zama mai ban mamaki wahala.

**Rarrabuwa biyu** (shin wannan hoto cat ne?) shine mafi sau□ in yanayi. **Nau'i da yawa** (wane nau'in dabba shine wannan?) yana bu□ atar masu lakabi su za□ i za□ i □ aya daga jeri. **Lakabi nau'i da yawa** (wa□ anne batutuwa wannan labarin ya □ unshi?) yana ba da damar lakabin lokaci guda da yawa.

### Lakabi Jeri

Kowane alamar cikin jeri tana kar□ ar lakabi. Gane Sunan Wuri (NER) shine misali na gargajiya — masu lakabi suna alamta span na rubutu a matsayin MUTUM, □ UNGIYA, WURI, dkk. Lakabi galibi ana yi shi ta amfani da tsarin BIO (Farko-Ciki-Waje) ko BIOES:

```
B-ORG    O          B-ORG    O    O    O
```

### Span da Lakabi Dangantaka

Bayan yin lakabi ga alamomi □ aya □ aya, masu lakabi na iya bu□ atar: - Gano span (maganganu masu alamu da yawa) kuma sanya nau'i - Alama *dangantaka* tsakanin span ("Apple" SAMU "Shazam") - Sanya lakabi ga sar□ o□ in coreference (duk ambaton wuri □ aya)

Wa□ annan ayyuka suna da wahala ta kwamitin tunani kuma suna da yarjejeniya □ asa da masu sanya lakabi.

### Akwatin Iyaka da Gano Abubuwa

Masu lakabi suna zana murabba'i a kusa da abubuwa a cikin hotuna kuma suna sanya lakabi aji ga kowane akwati. Daidaito na gida yana da muhimmanci: akwati mai □ an□ anta yawancin muhalli; akwati mai girma yana ha□ a baya. Kayan aiki na zamani na lakabi suna □ ididdige ha□ uwa-akan-□ ungiya (IoU) tare da lakabi na tunani don yin alamar matsalolin inganci.

### Rarrabuwa

Lakabi matakin pixel: kowane pixel yana sanyawa aji (rarrabuwa ta ma'ana) ko zuwa misali na abin musamman (rarrabuwa misali). Rarrabuwa mai inganci na daga cikin mafi tsada nau'in lakabi, tare da ku□ i daga goma zuwa sama da dala □ ari kowace hoto don yanayi masu rikitarwa dangane da yanki da tallafin kayan aiki.

### Rubutu da Fassara

Sauti □ rubutu (bayanin horarwa ASR), rubutu mai hannu □ rubutu (bayanan OCR), ko harshen tushe □ harshen manufa (bayanan MT). Wa□ annan ayyuka suna bu□ atar □ warewa ta harshe kuma ba za a iya yi da aminci ta masu sanya lakabi da ba su samu horo ba.

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## Jagoran Lakabi

Mafi muhimmin abin da ke □ ayyade ingancin lakabi shine ingancin **jagoran lakabi**: umarni na rubutu da masu lakabi ke bi.

Jagoran mai kyau: - Ya bayyana manufar aikin kuma ya bayyana *dalilin da ya sa* lakabi yana da muhimmanci - Ya ba da ma'ana mai bayyana ga kowane rukuni tare da misalai masu kyau da mara kyau - Ya magance lokuta na iyaka na yau da kullum da lokuta masu wuya a fili - Ya bayyana abin da za a yi lokacin da ba a tabbata ba (misali, alamar "tsallake" da za□ in tilas) - Ya ha□ a misalai masu aiki na cikakkiyar lakabi

Jagoran mai muni ya dogara akan masu lakabi don "amfani da hankali na yau da kullum" don lokuta na iyaka — wanda ke haifar da yanke shawara masu karo da juna wa□ anda ke raunana ingancin samfuri da kara rashin jituwa tsakanin masu sanya lakabi.

> **Ha□ aka jagoran yana sake zagaye** :class: note
>
> Kada ka yi tsammanin rubuta jagoran cikakke a farko. Gudanar da □ aramin zagayen lakabi na gwaji, bincika rashin jituwa, kuma sabunta jagoran. Maimaita. Jagoran da aka ha□ aka da kyau galibi yana ta cikin zagaye na gyarawa 3–5 kafin su kwantar.

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## Auna Ingancin Lakabi: Yarjejeniyar Tsakanin Masu Sanya Lakabi

Lokacin da masu sanya lakabi da yawa suka yi lakabi ga bayanai □ aya, ana iya auna yarjejeniyarsu. Yarjejeniya mai □ ari tana ba da shawara aiki an □ ayyadewa da kyau kuma masu sanya lakabi sun fahimce shi. Yarjejeniya □ asa tana ba da shawara rikitarwa a cikin aiki, jagoran, ko bayanai kanta.

### Kappa na Cohen

Don masu sanya lakabi biyu da ke yin lakabi ga bayanai zuwa $K$ rukunan, **kappa na Cohen** {cite}`cohen1960coefficient` yana gyara yarjejeniyar da aka ga don damari:

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

inda $P_o$ shine □ angare yarjejeniya da aka ga kuma $P_e$ shine iya yiwuwar yarjejeniya ta damari (an □ ididdige daga rarrabawar lakabi na iyaka).

$\kappa = 1$ yana nufin cikakkiyar yarjejeniya; $\kappa = 0$ yana nufin yarjejeniya ba fi damari ba; $\kappa < 0$ yana nufin rashin jituwa na tsari.

| $\kappa$ iyaka | Fassara |
|---|---|
| $< 0$ | □ asa da damari |
| $0.0 - 0.20$ | □ arami |
| $0.21 - 0.40$ | Mai gaskiya |
| $0.41 - 0.60$ | Matsakaici |
| $0.61 - 0.80$ | Muhimmanci |
| $0.81 - 1.00$ | Kusan cikakke |

### Kappa na Fleiss

Yana fa□ a□ a kappa na Cohen zuwa $M > 2$ masu sanya lakabi. Kowane mai sanya lakabi yana yin lakabi ga kowane item ba tare da wani ba; tsari yana ha□ a daga cikin masu sanya lakabi:

$$\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$$

inda $\bar{P}$ shine matsakaicin yarjejeniyar ma'anar ma'anar juna a duk ma'anar masu sanya lakabi, kuma $\bar{P}_e$ shine yarjejeniyar da ake tsammani □ ar□ ashin sanyawa da damari.

### Alpha na Krippendorff

Mafi yawan metric □ in gaba □ aya, yana tallafa wa duk yawan masu sanya lakabi, duk nau'in sikelin (suna, tsari, tazara, rabo), da bayanai da bata {cite}`krippendorff2011computing`:

$$\alpha = 1 - \frac{D_o}{D_e}$$

inda $D_o$ shine rashin jituwa da aka ga kuma $D_e$ shine rashin jituwa da ake tsammani. Alpha na Krippendorff galibi shine mafi so a aikin ilimi saboda sassaucinsa.

```
import numpy as np
from sklearn.metrics import cohen_kappa_score

# Simulate two annotators labeling 200 items into 3 categories
rng = np.random.default_rng(0)
true_labels = rng.integers(0, 3, size=200)
```

```
# Annotator 1: mostly agrees with ground truth
ann1 = true_labels.copy()
flip_mask = rng.random(200) < 0.15
ann1[flip_mask] = rng.integers(0, 3, size=flip_mask.sum())

# Annotator 2: less consistent
ann2 = true_labels.copy()
flip_mask2 = rng.random(200) < 0.30
ann2[flip_mask2] = rng.integers(0, 3, size=flip_mask2.sum())

kappa_12 = cohen_kappa_score(ann1, ann2)
kappa_1true = cohen_kappa_score(ann1, true_labels)
kappa_2true = cohen_kappa_score(ann2, true_labels)

print(f"Cohen's κ (ann1 vs ann2):   {kappa_12:.3f}")
print(f"Cohen's κ (ann1 vs truth):  {kappa_1true:.3f}")
print(f"Cohen's κ (ann2 vs truth):  {kappa_2true:.3f}")
```

---

## Sarrafa Rashin Jituwa

Lokacin da masu sanya lakabi suka yi rashin jituwa, akwai dabarun da yawa:

### Za□ in Rinjaye

Ana □ aukar lakabi mafi yawa a matsayin ma'aunin zinariya. Sau□ i da □ arfi lokacin da yawan masu sanya lakabi kowace item ba shi ba. Gazawa lokacin da □ aramin □ ungiya na masu sanya lakabi sun fi daidai ta tsari.

### Za□ in Nauyi

Ana nauyata masu sanya lakabi ta daidai ta □ ididdigan su (an samo daga yarjejeniya tare da ma'aunin zinariya ko sauran masu sanya lakabi). Masu sanya lakabi mafi daidai suna da tasiri fiye da kima.

### Lakabin Laushi

Maimakon runtuma lakabi zuwa □ aya, kiyaye rarrabawa. Idan masu sanya lakabi 3 daga cikin 5 sun ce "mai kyau" kuma 2 sun ce "neutral," wakilci wannan a matsayin $(p_{\text{pos}}, p_{\text{neutral}}, p_{\text{neg}}) = (0.6, 0.4, 0.0)$. Horarwa akan lakabin laushi yana inganta □ ididdigan iya yiwuwa.

### Shari'a

Babban mai sanya lakabi ko □ wararren yanki yana shari'a rashin jituwa. Ma'aunin zinariya amma yana da tsada; galibi ana kiyaye shi don yankuna masu ha□ ari.

### Samfurin Lissafi

Hanyoyin mafi inganci suna □ iyana iyawar mai sanya lakabi ta yiwuwar. Tsarin **Dawid-Skene** {cite}`dawid1979maximum` yana □ ididdiga matrices na rikicewar mai sanya lakabi da ma'aunin □ arshe na item a lokaci □ aya ta EM. Duba Babi na 13 don cikakkiyar bayanai.

---

## □ urar Lakabi da Tasirinta

Lakabi na gaske yana da □ ura. Tasirin □ urar lakabi akan horarwar samfuri yana dogara kan nau'in □ ura:

- □ **ura ta damari** (lakabin da aka juye ta damari) yana raunana aiki amma samfurin suna da juriya mai ban mamaki ga matakan matsakaici (har ~20% ga ayyuka da yawa).
- □ **ura ta tsari/makiya** (lakabin da ke kuskure koyaushe a salon musamman) yana da damashi fiye da kima kuma yana da wahala gano.
- □ **ura ta yanayin ajin** (kuskure mafi yiwuwa ga aji musamman) yana nuna iyakar yanke shawara na samfuri.

□ a'idar amfani ta aiki: tare da $n$ misalin horarwa da □ angare $\epsilon$ na lakabin lalace, aiki na samfuri yana raguwa kusan kamar dai kana da $(1 - 2\epsilon)^2 n$ misalan tsabta {cite}`natarajan2013learning`. Don $\epsilon = 0.2$, wannan daidai yake da rasa kashi 36% na bayananku.

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import cross_val_score

rng = np.random.default_rng(42)
X, y = make_classification(n_samples=2000, n_features=20, random_state=42)

noise_levels = np.linspace(0, 0.45, 15)
mean_accs = []

for eps in noise_levels:
    y_noisy = y.copy()
    flip = rng.random(len(y)) < eps
    y_noisy[flip] = 1 - y_noisy[flip]
    scores = cross_val_score(LogisticRegression(max_iter=500), X, y_noisy, cv=5)
    mean_accs.append(scores.mean())

plt.figure(figsize=(7, 4))
plt.plot(noise_levels, mean_accs, 'o-', color='#2b3a8f', linewidth=2)
plt.xlabel("Label noise rate (ε)", fontsize=12)
plt.ylabel("Cross-validated accuracy", fontsize=12)
plt.title("Effect of Label Noise on Model Performance", fontsize=13)
plt.axvline(0.2, color='#e05c5c', linestyle='--', alpha=0.7, label='20% noise')
plt.legend()
plt.tight_layout()
plt.savefig('label_noise_effect.png', dpi=150)
plt.show()
print(f"\nAccuracy at 0% noise:  {mean_accs[0]:.3f}")
print(f"Accuracy at 20% noise: {mean_accs[round(0.2 / 0.45 * 14)]:.3f}")
print(f"Accuracy at 40% noise: {mean_accs[-2]:.3f}")
```

---

## Ku□ in Lakabi da Gudu

Fahimtar tattalin arzikin lakabi yana da mahimmanci don tsare-tsaren aiki.

| Nau'in aiki | Gudu na yau da kullum | Ku□ i kowace item (□ wararru) |
|---|---|---|
| Rarrabuwa hoto biyu | 200–500/hr | $0.02–0.10 |
| NER (rubutu gajere) | 50–150 items/hr | $0.10–0.50 |
| Fitar da dangantaka | 20–60 items/hr | $0.30–1.50 |
| Rarrabuwa hoto likitanci | 5–30 items/hr | $10–100 |
| Lakabi bidiyo | 5–20 min bidiyo/hr | $20–200 |

Wa□ annan alkaluma □ iyasin girma ne na kusan kuma suna bambanta sosai ta □ warewa da ake bu□ ata, ingancin kayan aiki na lakabi, bayyanar jagoran, da kwarewa na mai sanya lakabi. Ya kamata a □ auke su a matsayin misali, ba umarnin □ a'ida ba.

> **See also:** Za□ u□ □ ukan kayan aiki na lakabi an □ unshi su a Babi na 12. Samfurin lissafi don lakabi na taron aiki (Dawid-Skene, MACE) an □ unshi su a Babi na 13.

# Koyon Aiki

Bayanan da aka yi lakabi suna da tsada. Babban ra'ayin koyon aiki shine cewa *ba duk misalin da ba a yi lakabi ba yana da bayanai □ aya ba* — samfuri zai iya inganta da sauri idan ya samu za□ ar wa□ anne misalai za su tambaya game da su. Maimakon yin lakabi ga bayanai ta damari, tsarin koyon aiki yana tambaya oracle (galibi mai sanya lakabi □ an adam) akan misalai mafi yiwuwar inganta samfuri.

Wannan babi ya □ unshi ka'idar da aikin koyon aiki: dabarun tambaya, tsarin samfuri, ma'aunin dakatar, da abubuwan da ke da amfani ga aikawa na gaske.

---

## Tsarin Koyon Aiki

Tsarin **koyon aiki bisa tafkin** na yau da kullum ya □ unshi:

- **Rukunin da aka yi lakabi** $\mathcal{L} = \{(x_i, y_i)\}_{i=1}^{n}$ — □ arami a farko
- **Tafkin da ba a yi lakabi ba** $\mathcal{U} = \{x_j\}_{j=1}^{m}$ — galibi mafi girma fiye da $\mathcal{L}$
- **Oracle** $\mathcal{O}$ wanda zai iya mayar da $y = \mathcal{O}(x)$ don duk $x$ da aka tambaya
- **Dabarun tambaya** $\phi$ wanda ke za□ ar tambayar gaba $x^* = \phi(\mathcal{L}, \mathcal{U}, f_\theta)$

Zagayen koyon aiki:

```
1. Fara: L = ƙaramin saitin farko da aka yi lakabi, U = tafkin da ba a yi lakabi ba
2. Horar: f_θ ← train(L)
3. Tambaya: x∗ = argmax φ(x; f_θ) akan x ∈ U
4. Lakabi: y∗ = O(x∗)
5. Sabunta: L ← L ∪ {(x∗, y∗)}, U ← U \ {x∗}
→ Maimaita daga 2 har sai kasafin kuɗi ya ƙare
```

Manufar ita ce kaiwa ga ingancin samfuri da aka nufa ta amfani da ka□ an ka□ an tambayoyin oracle iya.

---

## Tushen Ka'ida

Tambayar yanayi ita ce: nawa koyon aiki zai iya taimaka? A mafi kyau, koyon aiki na iya cimma *raguwa mai tsanaki* a rikitarwan lakabi — kaiwa kuskure $\epsilon$ tare da $O(\log(1/\epsilon))$ lakabin maimakon $O(1/\epsilon)$ da ake bu□ ata ta koyon wucin gadi, a □ alla a yanayi mai yiwuwa tare da dabarun tambaya mai kyau {cite}`settles2009active`.

A aiki, tabbaci ya fi wuya samu. **Koyon aiki na agnostic** {cite}`balcan2006agnostic` yana nuna cewa adana lakabi yana yiwuwa ko da lokacin da ra'ayin manufa ba cikin ajin hasashe ba, amma

adana yana dogara sosai kan coefficient na rashin jituwa — auna yadda rukunin hasashe masu yiwuwa ke raguwa yayin da bayanai ke tara.

Tasirin da ke da amfani na aiki: fa'idar koyon aiki ta fi girma lokacin da **iyakar yanke shawara yana sau□ i kuma mai □ ullewa** (don haka tambayoyin rashin tabbas da sauri suna kawar da hasashe mara daidai), kuma mafi □ an□ anta lokacin da ajin hasashe yana da girma ko iyakar ta rikitarwa.

---

## Dabarun Tambaya

### Samfurin Rashin Tabbas

Mafi sau□ in kuma mafi amfani na dabarun: tambaya misali wanda samfuri ke *mafi rashin tabbas* game da shi {cite}`lewis1994sequential`.

**Mafi □ arancin amincewar** yana tambaya misali wanda samfuri ke mafi □ arancin amincewar a cikin hasashen farko:

$$x^* = \arg\max_{x \in \mathcal{U}} \left(1 - P_\theta(\hat{y} \mid x)\right)$$

**Samfurin iyaka** yana la'akari da tazara tsakanin manyan iya yiwuwar biyu da aka hasashe:

$$x^* = \arg\min_{x \in \mathcal{U}} \left(P_\theta(\hat{y}_1 \mid x) - P_\theta(\hat{y}_2 \mid x)\right)$$

**Samfurin entropy** yana amfani da cikakkiyar rarrabawar da aka hasashe:

$$x^* = \arg\max_{x \in \mathcal{U}} \left(-\sum_{k=1}^{K} P_\theta(y_k \mid x) \log P_\theta(y_k \mid x)\right)$$

Samfurin entropy shine mafi □ a'ida daga cikin uku — yana la'akari da duk aji — kuma galibi yana inganta sauran akan matsaloli nau'i da yawa.

### Tambaya ta Kwamitin (QbC)

Horar da **kwamitin** na $C$ samfurin (ta amfani da bagging, farawa daban-daban, ko tsarin daban-daban). Tambaya misali wanda kwamitin ke rashin jituwa game da shi sosai:

$$x^* = \arg\max_{x \in \mathcal{U}} \ \text{disagreement}(\{f_c(x)\}_{c=1}^{C})$$

Ana iya auna rashin jituwa a matsayin **entropy na za□ i** (entropy akan za□ in rinjaye na kwamitin) ko **KL divergence** daga rarrabawar yarjejeniya.

QbC yana ba da □ iyasin rashin tabbas mafi kyau fiye da samfuri □ aya amma yana bu□ atar horar da samfurin da yawa, wanda yana da tsada ta □ ididdiga.

### Canjin Samfuri da Ake Tsammani

Tambaya misali wanda zai haifar da mafi girman canji ga samfuri na yanzu idan aka yi lakabi ga shi. Don samfurin bisa gradient, wannan ya dace da misali mai mafi girman iya □ arfin gradient da ake tsammani {cite}`settles2008analysis`:

$$x^* = \arg\max_{x \in \mathcal{U}} \sum_{y \in \mathcal{Y}} P_\theta(y \mid x) \, \|\nabla_\theta \mathcal{L}(f_\theta(x), y)\|$$

Wannan dabarun yana da □ arfin ka'ida amma yana bu□ atar □ ididdige gradient don kowane □ an takara, yana mai da shi mai tsada don manyan samfurin.

### Hanyoyin Core-Set / Geometrical

Dabarun bisa rashin tabbas na iya kasancewa **masu nuna son zuciya ga wa□ anda ke waje**: misali na musamman na iya zama mara tabbas sosai amma ba ya wakiltar rarrabawar bayanai. Hanyoyin Core-set suna magance wannan ta neman samfurin iri-iri da ke rufe sararin fasali.

Algorithm **k-center greedy** {cite}`sener2018active` yana nemo mafi □ arami na aya wa□ anda suka sa kowane aya da ba a yi lakabi ba yana cikin $\delta$ na aya □ aya da aka tambaya akalla:

$$x^* = \arg\max_{x \in \mathcal{U}} \min_{x' \in \mathcal{L}} d(x, x')$$

watau, tambaya aya mafi nisa daga duk aya da aka yi lakabi a yanzu. Wannan yana □ arfafa saitin lakabi mai fa□ a□ awa.

### BADGE

**Koyon Aiki na Jeri ta Embeddings na Gradient Iri-Iri** {cite}`ash2020deep` yana ha□ a rashin tabbas da iri-iri: yana za□ ar jeri na misalai wa□ anda embeddings na gradient (game da lakabi da aka hasashe) duka suna da girma (rashin tabbas) da iri-iri (suna rufe yankuna daban-daban na sararin gradient). Wannan yana □ aya daga cikin dabarun zamani mafi gasa.

---

## □ iyasin Rashin Tabbas don Samfurin Zurfin

Dabarun da ke sama suna □ auka damar □ wanso iya yiwuwar □ ididdigan da aka daidaita daga samfuri. Don sau□ in samfurin (Logistic regression, masu rarraba softmax), wannan yana da sau□ i. Don cibiyoyin jijiya mai zurfi, samun □ iyasin rashin tabbas mai aminci yana bu□ atar fasaha □ ari.

### Nau'in Rashin Tabbas Biyu

Bayan Kendall da Gal {cite}`kendall2017uncertainties`, muna bambanta:

**Rashin tabbas aleatoric** (rashin tabbas na bayanai): hayaniya da ke cikin lura wa□ anda ba za a rage ta hanyar tara bayani fiye da ba. Hoto mara bayyana yana da rashin tabbas na aleatoric — babu □ ari na horarwa daga rarrabawar □ aya zai sa samfuri ya fi amincewar akan shi.

**Rashin tabbas epistemic** (rashin tabbas na samfuri): rashin tabbas saboda iyakancin bayanin horarwa ko samfuri da bai ga misalai makamancin haka ba. Rashin tabbas na epistemic *zai iya*

ragewa ta hanyar yin lakabi ga bayani fiye da kima — kuma saboda haka shine adadin da ya dace don za□ in tambaya na koyon aiki.

Don koyon aiki, muna son tambaya misalai masu rashin tabbas epistemic mai □ ari, ba rashin tabbas aleatoric mai □ ari ba. Tambaya misali mai rikitarwa ta asali yana bata □ o□ arin oracle: babu lakabi da za su bayar ya kasance daidai sosai.

### MC Dropout

Hanyar da ke da amfani ga □ iyasin rashin tabbas epistemic don cibiyoyin jijiya ita ce **MC Dropout** {cite}`gal2016dropout`: yi amfani da dropout a lokacin inference kuma gudanar da $T$ na gaba. Bambancin a cikin hasashe shine □ iyasin rashin tabbas epistemic.

```
import numpy as np
import torch
import torch.nn as nn

torch.manual_seed(42)
rng = np.random.default_rng(42)

class MCDropoutNet(nn.Module):
    def __init__(self, input_dim=20, hidden=64, output_dim=2, p_drop=0.3):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden), nn.ReLU(), nn.Dropout(p_drop),
            nn.Linear(hidden, hidden),    nn.ReLU(), nn.Dropout(p_drop),
            nn.Linear(hidden, output_dim)
        )

    def forward(self, x):
        return self.net(x)

def mc_uncertainty(model, x, T=30):
    """
    Run T stochastic forward passes with dropout active.
    Returns mean prediction and epistemic uncertainty (predictive variance).
    """
    model.train()  # keep dropout active during inference
    with torch.no_grad():
        preds = torch.stack([
            torch.softmax(model(x), dim=-1) for _ in range(T)
        ])  # shape: (T, N, C)
    mean_pred = preds.mean(0)
    # Epistemic uncertainty: mean of variances across passes
    epistemic = preds.var(0).sum(-1)
    # Aleatoric uncertainty: entropy of mean prediction
    aleatoric = -(mean_pred * (mean_pred + 1e-9).log()).sum(-1)
    return mean_pred, epistemic, aleatoric

# Quick demonstration
model = MCDropoutNet(input_dim=20, output_dim=2)
# In-distribution example (simulated)
x_familiar   = torch.randn(1, 20) * 0.5
```

```
# Out-of-distribution example (far from training distribution)
x_unfamiliar = torch.randn(1, 20) * 3.0

for name, x in [("In-distribution ", x_familiar), ("Out-of-distribution", x_unfamiliar)]:
    _, ep, al = mc_uncertainty(model, x)
    print(f"{name} | epistemic: {ep.item():.4f} | aleatoric: {al.item():.4f}")
```

A cikin cibiyar jijiya da ba a horar da ita a sama, misalai biyu suna nuna rashin tabbas makamancin haka. Bayan horarwa, misali da ke waje da rarrabawar zai nuna rashin tabbas epistemic mai □ ari — samfuri bai koyi zana daidai don shigarwa nesa daga rarrabawar horarwa ba.

### Ha□ in Gwiwa Mai Zurfi

Horar da $M$ na samfurin da aka farawa ba tare da juna ba kuma matsakaici hasashensu yana ba da □ iyasin rashin tabbas mafi sau□ i kuma galibi mafi aminci fiye da MC Dropout {cite}`lakshminarayanan2017simple`. Rashin jituwa tsakanin mambobin ha□ in gwiwa shine alaman rashin tabbas epistemic.

Don koyon aiki a sikeli, MC Dropout da ha□ in gwiwa mai zurfi duka suna □ ara nauyi kwatankwacin $T$ ko $M$ hanyoyin gaba. A aiki, $T = 10$–$30$ don MC Dropout ko $M = 5$ mambobin ha□ in gwiwa galibi ya isa wajen jera misalai ta rashin tabbas epistemic, ko da dai □ imomi na cikakke ba su da daidaitawa.

---

## Zagayen Koyon Aiki Cikakke

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from copy import deepcopy

rng = np.random.default_rng(42)

# Generate dataset
X, y = make_classification(
    n_samples=2000, n_features=20, n_informative=10,
    n_classes=3, n_clusters_per_class=1, random_state=42
)
X_train, y_train = X[:1500], y[:1500]
X_test,  y_test  = X[1500:], y[1500:]

def entropy_query(model, X_pool):
    """Return index of most uncertain sample (entropy)."""
    probs = model.predict_proba(X_pool)
    ent = -np.sum(probs * np.log(probs + 1e-9), axis=1)
    return np.argmax(ent)

def random_query(X_pool):
    """Random baseline."""
    return rng.integers(0, len(X_pool))
```

```
def run_active_learning(strategy='entropy', n_initial=30, n_queries=120, query_batch=5):
    labeled_idx = list(rng.choice(len(X_train), n_initial, replace=False))
    unlabeled_idx = [i for i in range(len(X_train)) if i not in labeled_idx]
    accs = []

    for step in range(n_queries // query_batch):
        model = LogisticRegression(max_iter=500, C=1.0)
        model.fit(X_train[labeled_idx], y_train[labeled_idx])
        accs.append(accuracy_score(y_test, model.predict(X_test)))

        # Query
        X_pool = X_train[unlabeled_idx]
        for _ in range(query_batch):
            if strategy == 'entropy':
                q = entropy_query(model, X_pool)
            else:
                q = random_query(X_pool)
            labeled_idx.append(unlabeled_idx.pop(q))
            X_pool = X_train[unlabeled_idx]

    return np.array(accs)

labels_used = np.arange(1, 25) * 5 + 30  # label counts at each step

accs_active = run_active_learning(strategy='entropy')
accs_random = run_active_learning(strategy='random')

plt.figure(figsize=(8, 5))
plt.plot(labels_used, accs_active, 'o-', label='Entropy sampling', color='#2b3a8f', linewidth
plt.plot(labels_used, accs_random, 's--', label='Random baseline',  color='#e05c5c', linewidt
plt.xlabel("Number of labeled examples", fontsize=12)
plt.ylabel("Test accuracy", fontsize=12)
plt.title("Active Learning vs. Random Sampling", fontsize=13)
plt.legend(fontsize=11)
plt.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('active_learning_curve.png', dpi=150)
plt.show()

print(f"Active learning reaches {accs_active[-1]:.3f} accuracy")
print(f"Random sampling reaches {accs_random[-1]:.3f} accuracy")
print(f"Active learning saves ~{int((accs_random.tolist().index(min(accs_random, key=lambda a
```

---

## Matsalar Farawa Mai Sanyi

Koyon aiki yana bu□ atar samfuri da aka horar don □ ididdiga aya da ba a yi lakabi ba — amma a farko, kana da ka□ an (ko ba shi) misalin da aka yi lakabi. Wannan shine **matsalar farawa mai sanyi**.

Mafita masu amfani:

1. **Farawa ta damari:** Yi lakabi ga □ aramin saitin farko ta damari (misalai 20–100) kafin fara koyon aiki.
2. **Farawa bisa clustering:** Yi amfani da k-means akan tafkin da ba a yi lakabi ba; yi lakabi ga misali □ aya daga kowane cluster. Wannan yana tabbatar da iri-iri a cikin saitin farko da aka yi lakabi.
3. **Za□ i bisa embedding:** Yi amfani da encoder da aka horar a baya don □ oye misalai; za□ i iri-iri ta core-set.

Don mafi yawan ayyuka, □ an □ aramin lakabin farko ta damari galibi ya isa don fara koyon aiki; adadin daidai yana dogara kan daidaito na aji, girman fasali, da rikitarwan samfuri.

---

## Koyon Aiki na Yanayin Jeri

A aiki, masu sanya lakabi suna aiki a jeri — yana da □ arancin inganci don horar da kuma aikawa da sabon samfuri bayan kowane lakabi □ aya. **Koyon aiki na yanayin jeri** yana za□ ar saitin $b$ misalai don yi lakabi a lokaci □ aya.

Za□ in mafi rashin tabbas $b$ misalai kawai yana haifar da **tsattsauran maimaitawa**: misalin mara tabbas sosai suna tara tare (misali, misalai kusa da iyakar yanke shawara a yanki □ aya). Dabarun jeri mafi kyau suna inganta duka rashin tabbas *da* iri-iri a cikin jeri.

**Tsarin Determinantal Point Process (DPPs)** suna ba da hanyar □ a'ida don samfuri na jeri iri-iri: suna bayyana rarrabawa akan □ ananan saiti wa□ anda ke hukunta abubuwa makamancin haka. Iya yiwuwar □ aramin saiti $S$ □ ar□ ashin DPP ya dace da $\det(L_S)$ inda $L$ shine matrix na kernel da ke □ oye kamancin.

---

## Ma'aunin Dakatar

Yaushe koyon aiki ya kamata ya tsaya? Ma'aunin yau da kullum:

- **Kasafin ku□ i ya □ are:** Mafi sau□ i — tsaya lokacin da kasafin ku□ in lakabi ya □ are.
- **Kushewar aiki:** Tsaya lokacin da daidaito na samfuri akan saitin tabbatarwa da aka □ aure bai inganta fiye da $\delta$ don $k$ zagaye na jere ba.
- **Iyakar amincewar:** Tsaya lokacin da ka□ an misalin da ba a yi lakabi ba suna da rashin tabbas a saman iyaka.
- **Raguwar □ arin girma mai yiwuwa:** □ iyasta □ arin mafi yawa daga lakabi □ ari; tsaya lokacin da wannan ya fa□ i □ asa da iyaka {cite}`bloodgood2009method`.

---

## Lokacin da Koyon Aiki ke Aiki (da Lokacin da Ba Aiki Ba)

Koyon aiki galibi yana aiki lokacin da: - Lakabi yana da tsada kuma tafkin da ba a yi lakabi ba yana da girma - Bayanai yana da tsari bayyananne samfuri zai iya amfani da shi don gano misalai masu bayanai - Ajin samfuri ya dace da aiki

Koyon aiki yana yin mara kyau lokacin da: - Samfuri na farko ya yi mara kyau sosai (farawa mai sanyi) kuma ba zai iya jera misalai da ma'ana ba - Dabarun tambaya yana za□ ar wa□ anda

ke waje ko misalin mislabeled (juriya na hayaniya yana da muhimmanci) - Rarrabawar bayanai tana canzawa tsakanin tafkin da ba a yi lakabi ba da rarrabawar gwaji

Damuwa ta aiki ta muhimmanci shine **rashin daidaito na rarrabawa**: koyon aiki galibi yana tambaya misalai kusa da iyakar yanke shawara, yana □ ir□ ira saitin da aka yi lakabi mai nuna son zuciya wanda ba zai iya wakilta rarrabawar gwaji da kyau ba. Wannan na iya haifar da iyakar yanke shawara da aka horar da kyau amma □ ididdigan iya yiwuwar mara daidai.

> **See also:** Bita na asali shine {cite}`settles2009active`. Tushe na ka'ida (rikitarwan lakabi, iyakar agnostic): {cite}`balcan2006agnostic`. Don koyon aiki na musamman ga koyon zurfin, duba {cite}`ash2020deep` (BADGE) da {cite}`sener2018active` (core-set). Don kimantawa mai sukar lokacin da koyon aiki ke taimaka gaske, duba {cite}`lowell2019practical`. Akan rashin tabbas aleatoric da epistemic don samfurin mai zurfi, duba {cite}`kendall2017uncertainties`; don ha□ in gwiwa mai zurfi a matsayin □ iyasin rashin tabbas, duba {cite}`lakshminarayanan2017simple`; don MC Dropout a matsayin Bayesian inference na kusan, duba {cite}`gal2016dropout`.

# Koyon Injin Mai Mu'amala

Koyon aiki yana tambaya tambaya mai mai da hankali: da wani kasafin ku□ i, wa□ anne misalai ya kamata a yi lakabi? Koyon Injin Mai Mu'amala (IML) yana tambaya tambaya mafi faffada: yadda za a tsara *hulda* gaba □ aya tsakanin □ an adam da tsarin koyo don kasancewa mai amfani □ warai, mai dadi, da daidai?

IML ya bambanta ta **nan da nan** da **kai tsaye** na zagayen ra'ayin dan adam da samfuri. Inda ML na gargajiya ya □ unshi □ an adam ya mika bayanai kuma ya jira horarwa ta □ are, IML yana ba da damar □ an adam su lura da □ abi'ar samfuri, ba da ra'ayi, kuma su ga samfuri ya amsa — galibi a cikin da□ i□ a ka□ an.

---

## Ka'idojin Koyon Injin Mai Mu'amala

Amershi et al. {cite}`amershi2014power` suna gano halaye uku na bayyana na IML:

**1. Ra'ayin Mai Sauri:** Samfuri yana sabunta da sauri sosai har □ an adam ya iya gane tasirin ra'ayinsu. A iyaka, sabuntawar samfuri yana faruwa a lokaci na gaske.

**2. Sarrafa Kai Tsaye:** □ an adam yana hulda da samfuri ta bayanan ko ta hasashen samfuri — ba ta fayilolin tsarin ko daidaita abubuwan sama ba.

**3. Gyarawa Mai Sake Zagaye:** Tsari yana mai sake zagaye gaske: aikin □ an adam na gaba an sanar da shi ta □ abi'ar samfuri ta yanzu, wanda ya kasance ta ayyukan □ an adam na baya.

Wannan yana □ ir□ ira **zagayen mabiya ha□ in gwiwa** mai □ auri: duka □ an adam da samfuri suna canzawa akan lokaci don amsa wa juna. □ an adam yana koyon abin da samfuri ya fahimta; samfuri yana koyon abin da □ an adam ya damu da shi.

---

## Kwatancen da Koyon Aiki

IML da koyon aiki suna ha□ awa sosai amma ba □ aya ba ne:

| Dukiya | Koyon Aiki | ML Mai Mu'amala |
|---|---|---|
| Tambayar farko | Me za a yi lakabi? | Yadda za a yi mu'amala? |
| Latency na ra'ayi | Na iya kasancewa jeri (kwanaki) | Galibi lokaci na gaske ko kusan |
| Yawan sabuntar samfuri | Kowace zagaye (jeri) | Kowace mu'amala (yanar gizo) |
| □ arfin □ an adam | Yana amsa tambayoyin samfuri | Zai iya koyar da samfuri da gangan |

| Dukiya | Koyon Aiki | ML Mai Mu'amala |
|---|---|---|
| Zanen mai amfani | Damuwa ta biyu | Damuwa ta tsakiya |
| Nauyin kwamitin tunani na □ an adam | Ba a □ iyana ta fili ba | An la'akari da fili |

A cikin koyon aiki, injin yana kula da mu'amala. A cikin IML, □ an adam na iya □ aukar shugabanci kuma — yana ba da misalai, gyarawa, ko ra'ayi akan duk wane gefen □ abi'ar samfuri wanda ya zama matsala.

---

## Mu'amala ta Shugabanci Hadaye

Tsarin **shugabanci hadaye** yana ba da damar duka □ an adam da injin su □ auki shugabanci a lokuta daban-daban {cite}`allen1999mixed`. Tsarin da injin ya kula da shi gaba □ aya yana tambaya tambayoyi kuma □ an adam yana amsa. Tsarin da □ an adam ya kula da shi gaba □ aya yana barin □ an adam ya yanke duk komai. Tsarin shugabanci hadaye yana daidaita duka biyun.

A aiki, mafi kyau na tsarin IML suna ha□ a: - **Shugabancin injin:** "Ba ni da tabbas game da wa□ annan misalai — za ka iya yin lakabi ga su?" - **Shugabancin □ an adam:** "Na lura samfuri koyaushe yana kuskure game da wannan rukuni — bari in ba da misalai fiye da kima" - **Tabbatarwa:** Samfuri yana fito da fahimtarsa ta yanzu; □ an adam yana tabbatar ko yin gyarawa

Tsarin IML mai kyau yana mai da fahimtar samfuri ta yanzu a bayyane kuma za a iya gyara shi. Wannan shine bu□ atar **fahimtar**: □ an adam kawai zai iya kula da samfuri da suke fahimta, a □ alla kusan kusan.

---

## Abubuwan Dan Adam a IML

IML yana kawo abubuwan dan adam — nauyin kwamitin tunani, gajiya, daidaito, da amana — kai tsaye cikin zagayen koyo. Zanen IML mara kyau yana haifar da:

**Gajiyar lakabi:** □ an adam suna yanke shawara da sauri, mara kyau yayin da zaman ke tsawaita. Kurakurai da ke shiga bayanin horarwa.

**Nuna son zuciya na anchoring:** □ an adam suna dogara sosai akan shawarwarin samfuri na yanzu. Idan mai amfani da hannu ya cika hasashen samfuri, masu sanya lakabi sun fi □ arancin gyara shi ko da lokacin da bai daidai ba — tushen tsari na □ urar lakabi wanda ke ha□ awa cikin zagaye na lakabi {cite}`geva2019annotator`. Pre-annotation na iya hanzarta gudu {cite}`lingren2014evaluating` yayin da a lokaci □ aya yana rage yawan da masu sanya lakabi ke kama kurakuran samfuri; wa□ annan tasiri biyu dole ne a aunata su da juna a zanen mai amfani na IML.

**Rashin daidaitan amana:** □ an adam ko suna amana sosai (suna yarda da sakamakon samfuri mara daidai) ko ba sa amana (suna watsi da shawarwari daidai). Salon biyu suna rage □ imar ha□ in gwiwar dan adam da samfuri.

**Daidaito na zama:** □ an adam na iya yanke shawara daban-daban akan misali □ aya a lokuta daban-daban, musamman bayan zaman mai tsawo. Duba daidaito (sake gabatarwa da misalai na baya) zai iya gano da gyara wannan.

Zanen IML mai kyau yana rage wa□ annan matsaloli ta za□ u□ □ ukan mai amfani: gabatar da amincewar samfuri a fili, bazata tsarin gabatarwa, iyakanta tsawon zaman, da gina cikin duba daidaito.

---

## Nau'in Ra'ayin IML a Aiki

### Ra'ayin Matakin Misali

□ an adam yana ba da lakabi ko gyarawa don misali musamman. Wannan mafi yawaitar nau'in ne kuma yana kai tsaye dacewa da koyon supervised.

### Ra'ayin Matakin Fasali

□ an adam yana nuna wa□ anne fasali suke da dacewa ko mara dacewa. "Samfuri ya kamata ya kula da kalmomi 'gaggawa' da 'lokaci □ arshe' don wannan rukuni." Wannan ya fi bayyananne fiye da ra'ayin matakin misali kuma na iya zama mai inganci fiye da kima don ayyuka musamman.

Tsarin **TFIDF Interactive** da makamancin su suna ba da damar masu sanya lakabi su haskaka kalmomi masu dacewa a cikin takardun rubutu. Wa□ annan fastoci ana canza su zuwa □ untatawa ko □ arin kulawa akan kulawun samfuri.

### Ra'ayin Matakin Samfuri

□ an adam yana kai tsaye gyara □ abi'ar samfuri akan ajin shigarwa: "Duk lokacin da shigarwa ta □ unshi [X], sakamakon ya kamata ya zama [Y]." Wannan yana yin daidai da □ a'idojin logic ko □ untatawa a cikin hanyoyi kamar Posterior Regularization {cite}`ganchev2010posterior` ko koyon bisa □ untatawa.

---

## Nazarin Yanayi: Google Teachable Machine

Teachable Machine shine tsarin IML na yanar gizo mai sau□ in samun damar wanda ke ba da damar masu amfani marasa fasaha su horar da masu rarraba hotuna a cikin mai bincike. Mai amfani:

1. Yana □ aukar misalai na kowane aji ta amfani da kameran su
2. Yana horar da samfuri tare da danna □ aya (daidaita MobileNet a cikin mai bincike)
3. Nan da nan yana ganin hasashen samfuri akan bidiyo mai rai
4. Yana □ ara misalai fiye da kima don aji inda samfuri ba daidai ba

Wannan yana nuna zagayen IML na asali: ba da misalai □ lura da samfuri □ gano gazawa □ ba da misalai mafi mai nufin fiye da kima. Ra'ayin lokaci na gaske (sakamakon samfuri yana sabunta a lokaci na gaske, galibi a yawan firam mai mu'amala akan kayan aikin zamani) yana sa zagayen ha□ in gwiwa ya zama kai tsaye da hankali.

---

## Aiwatar da Sau□ in Zagayen IML

```python
import numpy as np
from sklearn.linear_model import SGDClassifier
from sklearn.preprocessing import StandardScaler
import warnings
warnings.filterwarnings('ignore')

class SimpleIMLSystem:
    """
    Minimal IML system that allows online feedback and displays
    model state after each annotation.
    """

    def __init__(self, n_features=10, n_classes=2):
        self.model = SGDClassifier(loss='log_loss', max_iter=1, warm_start=True,
                                   random_state=42)
        self.scaler = StandardScaler()
        self.X_seen = []
        self.y_seen = []
        self.n_classes = n_classes
        self.initialized = False

    def update(self, x, y_true):
        """Receive a single labeled example and update the model."""
        self.X_seen.append(x)
        self.y_seen.append(y_true)

        if len(self.X_seen) >= 2 * self.n_classes:
            X_arr = np.array(self.X_seen)
            y_arr = np.array(self.y_seen)
            X_scaled = self.scaler.fit_transform(X_arr)
            self.model.partial_fit(X_scaled[-1:], y_arr[-1:],
                                   classes=list(range(self.n_classes)))
            self.initialized = True

        return self

    def predict_with_confidence(self, x):
        """Predict label and return confidence."""
        if not self.initialized:
            return None, 0.0
        x_scaled = self.scaler.transform(x.reshape(1, -1))
        probs = self.model.predict_proba(x_scaled)[0]
        return self.model.predict(x_scaled)[0], probs.max()

    def current_accuracy(self, X_val, y_val):
        if not self.initialized:
            return None
        X_scaled = self.scaler.transform(X_val)
        return (self.model.predict(X_scaled) == y_val).mean()
```

```
# Simulate an IML session
rng = np.random.default_rng(42)
X_all, y_all = np.random.default_rng(0).random((500, 10)), np.random.default_rng(0).integers(
X_val, y_val = X_all[400:], y_all[400:]

system = SimpleIMLSystem(n_features=10, n_classes=2)
accs, confidences = [], []

print("Step | Labels | Accuracy | Example confidence")
print("-" * 50)

for step in range(100):
    x, y = X_all[step], y_all[step]
    pred, conf = system.predict_with_confidence(x)
    system.update(x, y)

    if (step + 1) % 10 == 0:
        acc = system.current_accuracy(X_val, y_val)
        if acc is not None:
            accs.append(acc)
            print(f"  {step+1:3d} |  {step+1:4d}  |  {acc:.3f}   | {conf:.3f}")
```

---

## Jarabawar Kaka

Wata hanyar da ke da amfani don kimanta zanen mai amfani na IML — kuma zanen tsarin HITL gaba □ aya — shine abin da za mu kira **Jarabawar Kaka** (asalin kiraye da aka gabatar a nan a matsayin □ untatawa na zane, ba a matsayin nuni ga aikin baya ba):

> *Mace da aka haifa a 1930 ya kamata ta iya amfani da wannan na'ura ta murya, kuma idan ta ji damuwa, ta ragewa mai kyau zuwa mai amfani da ha□ in ma□ alli ko mai amfani da rubutu.*

Jarabawa ba ta farko game da isa ba, ko da yake haka ne ma. Ita game da **zane don gogayya** ne. Idan tsari yana bu□ atar □ irar cibiyar jijiya, zagayen horarwa, ko rarrabawar iya yiwuwa don amfani da shi yadda ya dace, ya gaza Jarabawar Kaka. □ an adam a cikin zagaye ya kamata ya iya shiga ba tare da fahimtar gefen injin na zagaye ba.

Abubuwa masu amfani don zanen IML suna da □ wan□ wasa:

**Makoma ta farko ta murya:** Yanayin mu'amala na farko ya kamata ya zama harshen yau da kullum ko motsi — ba masu juyawa na sinadarai ko iyakar tabbatarwa ba. □ wararru na iya son masu juyawa; kowa ya kamata ya iya cewa "ba daidai ba ne."

**Lalacewa mai kyau:** Lokacin da yanayin mai amfani da shi ya fifita ya gaza ko ya haifar da damuwa, tsari ya kamata ya ba da wata hanya — ba allon fanko ko sa□ on kuskure ba. Mai amfani da shi wanda ba zai iya hulda ba ba zai iya koyar da shi ba.

**Halin samfuri mai bayyana:** Ya kamata a ga fahimtar yanzu ta samfuri a cikin kalmomin dan adam. Ba "amincewar: 0.73" amma "Na tabbata sosai wannan [X] ne, amma na ga misalai kamar wannan suka je hanyoyi biyu." Ya kamata rashin tabbas ya kasance a cikin harshe wanda

ke gayyata gyarawa.

**Juriya ga rikitarwa:** Mai amfani da shekara 93 da injiniyan ML mai shekara 23 za su yi mu'amala daban-daban da tsari □ aya. Jarabawar Kaka na tambaya ko tsari zai iya kar□ a duka biyun — ba ta gano shekarun mai amfani ba, amma ta tsara mu'amala wa□ anda ke aiki akan iyakar □ warewar da jin da□ in.

Jarabawa tana □ aukar mahimmanci musamman yayin da tsarin ML ke motsi daga kayan aiki na bincike zuwa ababen more rayuwa na yau da kullum. Tsarin tallafi na hoto likitanci da likitan hoto ke amfani da shi, mai rarraba takardun doka da ma'aikata ke amfani da shi, tsarin ra'ayin ilimi da malamai ke amfani da shi — kowannensu yana ha□ a dan adam a cikin zagaye wa□ anda ba su yi rajista don kasancewa mai horar da AI ba. Zane don su ba kyautar wani ba ne; shi ne manufar.

:::{admonition} Ka'idar Zane :class: tip Jarabawar Kaka □ untatawa ce ta zane, ba rukunin da ake nufin ba. Tsarin da suke wucewa sun fi □ arfi ga iri-iri na mai amfani, sun fi sadaukarwa ga tazara na □ warewar, kuma suna da gaskiya fiye da kima game da abin da suke bu□ ata daga dan adam a cikin zagaye. Idan tsari yana bu□ atar bayani kafin amfani, yana ro□ on □ an adam ya yi aiki □ ari. Ya kamata a tabbatar da wannan aiki ta fa'ida kwatankwacin shi. :::

---

## IML da Samfurin Ginshi□ ai

ML mai mu'amala na zamani yana dogara fiye da kima akan **samfurin ginshi□ ai da aka horar a baya** {cite}`bommasani2021opportunities` a matsayin tushe. Maimakon horarwa daga farko, masu amfani suna daidaita babban samfuri da aka horar a baya tare da □ aramin adadin misalai na mu'amala. Wannan zai iya rage yawan misalai da ake bu□ ata don kaiwa aiki mai amfani sosai — a yanayi mafi kyau, ka□ an kamar misalai 5–50 maimakon dubbai, dangane da yadda embeddings na baya ya dace da aikin manufa {cite}`bommasani2021opportunities`.

Dabarun da ke ba da damar wannan sun ha□ a: - **Tambayar ka□ an-ka□ an:** Bayar da misalai a cikin tagar mahallin LLM - **Daidaita adapter:** Sabunta □ ananan modules na adapter yayin daskarewa da tushe - **Daidaita sinadarai mai inganci (PEFT):** LoRA, daidaita prefix, da makamancin hanyoyin da ke ba da damar sabuntar da sauri, mara albarkatu

Samfurin ginshi□ ai suna canza yanayin IML: □ an adam ba su □ ara koyar da samfuri daga farko amma *kula da* samfuri wanda ya riga ya san abubuwa da yawa. Kalubale yana canzawa daga "yadda za a ba da misalai masu isa" zuwa "yadda za a □ ayyadewa ainihin abin da muke son ya bambanta daga abin da samfuri ya riga ya yi."

> **See also:** Bita daga {cite}`amershi2014power` har yanzu shine mafi kyau na ka'idojin IML. Don tsarin shugabanci hadaye musamman, duba {cite}`allen1999mixed`. Don tasirin anchoring a cikin lakabi, duba {cite}`geva2019annotator` da {cite}`lingren2014evaluating`.

# Koyon Ta Karfafawa daga Ra'ayin Dan Adam

Babu wata dabara da ta fi kawo HITL ML cikin babbar hanya fiye da Koyon Ta Karfafawa daga Ra'ayin Dan Adam (RLHF). Shi ne hanyar da ke bayan InstructGPT {cite}`ouyang2022training` da wani □ angare na farko na tsarin bin umarni a cikin manyan samfurin harshe na zamani da yawa {cite}`stiennon2020learning`. Fahimtar RLHF — ba kawai a matsayin mai jita-jita don bi ba, amma a matsayin hanyar □ a'ida ga daidaitawa — yana da muhimmanci ga kowa da ke aiki a AI na zamani.

---

## Matsalar Daidaitawa

Manyan samfurin harshe (LLMs) da aka horar kawai akan hasashen alamar gaba suna inganta don manufar wakili: yi hasashe a abin da rubutu ke zuwa gaba a rukunin rubutu na dan adam. Wannan manufa ta ala□ a da, amma ta bambanta da, abin da muke nufin gaske: amsoshi masu taimako, daidai, amintacce, kuma dacewa da □ imar dan adam.

Rashin daidaito tsakanin manufar horarwa da □ abi'ar da ake so ana kiransa **matsalar daidaitawa** {cite}`russell2019human`. Musamman, samfuri na harshe da aka horar akan rubutu na yanar gizo yana koyon: - Samar da ci gaba mai kama da gaskiya (wanda na iya kasancewa ba daidai ba ta gaskiya) - Nuna nuna son zuciya da cutarwa da ke cikin bayanin horarwa - Kasancewa mai sau□ in guduwa ko mai yaudarar lokacin da wannan shine abin da ya biyo bayan tambayar ta lissafi

RLHF yana magance daidaitawa ta sanya fifikon dan adam *wani □ angare na manufar inganta*.

---

## Tsarin RLHF

RLHF yana gudana a matakai uku:

```
Mataki na 1: Daidaita Supervised (SFT)
  --> Tattara bayanin misali (dan adam yana rubuta amsoshi mafi kyau)
  --> Daidaita LLM na tushe akan misalai

Mataki na 2: Horarwa ta Samfurin Lada
  --> Tattara fifiko biyu biyu (dan adam yana kimanta A da B)
  --> Horar da samfurin lada R(x, y) don hasashen fifikon dan adam

Mataki na 3: Daidaita RL
```

```
--> Daidaita LLM ta amfani da PPO/RL don inganta R(x, y)
--> Hukuncin KL yana hana tafiya mai nisa daga samfurin SFT
```

### Mataki na 1: Daidaita Supervised

Farawa daga samfuri na tushe da aka horar a baya $\pi_0$, muna tattara saitin bayanai na (tambaya, mafi kyau amsa) biyu, da masu kwangila na dan adam wa□ anda suka bi jagoran cikakku. Ana daidaita samfuri akan wa□ annan misalai ta amfani da cross-entropy na yau da kullum:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x,y)\sim\mathcal{D}_{\text{demo}}}\left[\log \pi_\theta(y \mid x)\right]$$

Samfurin SFT $\pi_{\text{SFT}}$ farawa mafi kyau ce don RLHF fiye da samfuri na tushe mentah.

### Mataki na 2: Horarwa ta Samfurin Lada

Don saitin tambayoyi $\{x_i\}$, muna □ ir□ ira $K$ amsoshi kowace tambaya ta amfani da $\pi_{\text{SFT}}$ kuma mun gabatar da su ga masu sanya lakabi dan adam a matsayin kwatancen biyu biyu: "Wane amsa ya fi, A ko B?"

Ana horar da samfurin lada $r_\phi$ don hasashen wa□ annan fifiko. □ ar□ ashin tsarin **Bradley-Terry** (Babi na 8), iya yiwuwar cewa amsa $y_w$ an fi so da $y_l$ ita ce:

$$P(y_w \succ y_l \mid x) = \sigma\left(r_\phi(x, y_w) - r_\phi(x, y_l)\right)$$

Ana horar da samfurin lada don rage asarar jeri biyu biyu:

$$\mathcal{L}_{\text{RM}}(\phi) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}_{\text{pref}}}\left[\log \sigma\left(r_\phi(x, y_w) - r_\phi(x, y_l)\right)\right]$$

Ana farawa da samfurin lada daga samfurin SFT tare da kai scalar wanda ya maye gurbin saman ta □ arshe.

### Mataki na 3: Daidaita RL tare da PPO

Tare da samfurin lada da aka horar, muna iya amfani da koyon □ arfafawa don daidaita LLM. Kowane tambaya $x$ yanayi ne; kowane amsa $y$ hanya ce ta za□ u□ □ ukan alamar; kuma ladan shine $r_\phi(x, y)$.

Manufar inganta ta ha□ a da **hukunci na KL divergence** don hana samfuri daga tafiya mai nisa daga tushen SFT (wanda zai haifar da hacking lada {cite}`krakovna2020specification,gao2023scaling`):

$$\max_\theta \mathbb{E}_{x\sim\mathcal{D}, y\sim\pi_\theta(\cdot|x)}\left[r_\phi(x, y) - \beta \cdot \text{KL}\left[\pi_\theta(\cdot \mid x) \| \pi_{\text{SFT}}(\cdot \mid x)\right]\right]$$

Sinadarin $\beta$ yana sarrafa □ arfin hukuncin KL. □ aramin $\beta$ yana ba da damar inganta fiye da kima amma yana hadari hacking lada; babban $\beta$ yana sanya samfuri kusa da SFT amma yana iyakance riba ta daidaitawa.

**Inganta Manufar Kusan (PPO)** {cite}`schulman2017proximal` shine algorithm na yau da kullum don wannan mataki, an za□ a don daidaitarsa dangane da hanyoyin gradient na manufa mentah.

---

## Nuni na Sau□ i na RLHF

Tsarin cikakke na RLHF yana bu□ atar ababen more rayuwar mafi sikeli. Misali mai zuwa yana nuna ra'ayoyin muhimmanci a □ aramin sikeli.

```
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from torch.nn import functional as F

torch.manual_seed(42)
rng = np.random.default_rng(42)

# ----------------------------------------------
# Toy setup: responses are 4-dimensional vectors
# "Quality" is known analytically (sum of positive values)
# We simulate a reward model learning this from pairwise feedback
# ----------------------------------------------

class RewardModel(nn.Module):
    def __init__(self, d=4, hidden=32):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(d, hidden), nn.ReLU(),
            nn.Linear(hidden, hidden), nn.ReLU(),
            nn.Linear(hidden, 1)
        )
    def forward(self, x):
        return self.net(x).squeeze(-1)

def true_quality(x):
    """The hidden ground-truth reward function."""
    return x.sum(dim=-1) + 0.5 * (x ** 2).mean(dim=-1)

# Generate pairwise preference data
N_PAIRS = 500
X1 = torch.randn(N_PAIRS, 4)
X2 = torch.randn(N_PAIRS, 4)
q1, q2 = true_quality(X1), true_quality(X2)
# Human prefers X1 when q1 > q2 (with some noise)
noise = torch.randn(N_PAIRS) * 0.5
preferred_1 = ((q1 - q2 + noise) > 0).float()

# Train reward model
rm = RewardModel(d=4, hidden=32)
optimizer = optim.Adam(rm.parameters(), lr=3e-3)

losses = []
for epoch in range(200):
    r1 = rm(X1)
    r2 = rm(X2)
```

```
    # Bradley-Terry loss
    logit = r1 - r2
    loss = F.binary_cross_entropy_with_logits(logit, preferred_1)
    optimizer.zero_grad(); loss.backward(); optimizer.step()
    losses.append(loss.item())

# Evaluate: does the reward model agree with true quality?
X_eval = torch.randn(1000, 4)
with torch.no_grad():
    r_pred = rm(X_eval).numpy()
    r_true = true_quality(X_eval).numpy()

corr = np.corrcoef(r_pred, r_true)[0, 1]
print(f"Reward model correlation with true quality: {corr:.4f}")
print(f"Final training loss: {losses[-1]:.4f}")

import matplotlib.pyplot as plt
plt.figure(figsize=(8, 3))
plt.subplot(1, 2, 1)
plt.plot(losses, color='#2b3a8f', linewidth=1.5)
plt.xlabel("Epoch"); plt.ylabel("Pairwise loss")
plt.title("Reward Model Training")

plt.subplot(1, 2, 2)
plt.scatter(r_true[:200], r_pred[:200], alpha=0.4, s=15, color='#2b3a8f')
plt.xlabel("True quality"); plt.ylabel("Predicted reward")
plt.title(f"Reward Model vs. Truth (r={corr:.3f})")
plt.tight_layout()
plt.savefig('reward_model.png', dpi=150)
plt.show()
```

---

## Kalubale a RLHF

### Hacking Lada

Gazawar muhimmanci: manufar yana nemo hanyoyin samun lada mai □ ari daga samfurin lada wa□ anda ba su dace da □ abi'ar mai kyau gaske ba. Misali, LLM na iya koyon samar da amsoshi masu yabo ko masu ha□ i□ a (wa□ anda masu sanya lakabi galibi suna □ ididdiga mafi □ ari) maimakon daidai.

Hacking lada yana yiwuwa fiye da kima lokacin da: - Ana horar da samfurin lada akan bayanin fifiko □ asa da isa - Ana ba da damar manufa ta tafiya mai nisa daga tushen SFT (□ aramin $\beta$) - Rarrabawar samfurin lada tana canzawa a lokacin horarwar PPO

**Dabarun rage:** Hukunci KL, horarwar samfurin lada mai sake zagaye, kimantawa iri-iri, □ untatawa na AI na tsarin mulki.

### Nuna Son Zuciya na Mai Kimantawa

Masu sanya lakabi dan adam suna da nuna son zuciya ta tsari. Suna fifiko da amsoshi mafi tsawo (nuna son zuciya ga yawa), rubutu mai kama amincewar (nuna son zuciya ga amince-

war), da amsoshi wa□ anda suka yarda da imaninsu na baya. Wa□ annan nuna son zuciya suna yada cikin samfurin lada.

Gazawar sycophancy da aka fi sani na samfurin RLHF — inda samfuri ke gaya wa masu amfani abin da suke son ji maimakon abin da ya kasance gaskiya — a wani □ angare sakamakon fifikon mai kimantawa don amsoshi masu jituwa ne.

### Kulawa Mai Sikeli

Don ayyuka masu rikitarwa, dan adam ba zai iya kimanta da aminci wane amsa AI ya fi daidai ba. Mai sanya lakabi da ke kwatanta shaidun lissafi mafi tsawo biyu ko aiwatarwa biyu na lambar na iya kawai za□ ar mai □ arin karatu, ba tare da la'akari da daidaito ba. **Kulawa mai sikeli** ita ce matsalar bincike bu□ a□ □ e na tsara hanyoyin kimantawa wa□ anda suke zama amintacce yayin da rikitarwar aiki ke □ aruwa {cite}`bowman2022measuring`.

---

## AI na Tsarin Mulki (RLAIF)

**AI na Tsarin Mulki** {cite}`bai2022constitutional`, da aka ha□ aka a Anthropic, yana rage dogaro akan masu sanya lakabi dan adam ta amfani da AI kansa don □ ir□ ira lakabin fifiko mai jagora ta saitin □ a'idoji (wani "tsarin mulki"). Tsari:

1. □ ir□ ira amsoshi ga tambayoyin masu iya cutar
2. Yi amfani da mai sukar AI don kimanta amsoshi □ ar□ ashin □ a'idojin tsarin mulki
3. Gyara amsoshi mai jagora ta ra'ayin AI (RLAIF — RL daga Ra'ayin AI)
4. Horar da samfurin lada akan fifikon da AI ya □ ir□ ira
5. Daidaita tare da RLHF ta amfani da wannan samfurin lada

RLAIF zai iya □ ir□ ira bayanin fifiko a sikeli mafi girma fiye da sanya lakabi dan adam, kuma yana ba da damar sarrafa □ an □ arami akan □ imomi da aka □ oye a cikin samfurin lada.

> **See also:** Takarda ta asali ta InstructGPT {cite}`ouyang2022training` tana bayyana amfani na farko na mafi sikeli na RLHF ga LLMs. Aikin RLHF na asali don RL mai zurfi shine {cite}`christiano2017deep`. An bayyana PPO a cikin {cite}`schulman2017proximal`. AI na tsarin mulki daga {cite}`bai2022constitutional`.

# Koyo daga Misalai

Lokacin da aiki ya yi wuya □ ayyadewa amma sau□ i nuna, na iya zama mai inganci fiye da kima don koyar da misali maimakon bayyana ta □ a'ida. □ wararren dan adam yana nuna hannun robot yadda za ya kama abu; shiryen programmer tare da IDE □ insu yana ba da jeri na gyarawa daidai; □ wararren chess yana yin wasa. **Koyo daga misalai** yana fitar da manufa daga irin wannan bayanan □ abi'a, yana kauce da bu□ atar ayyukan lada da aka yi da hannu ko bayanan aikin a fili.

---

## Kwaikwayon □ abi'a

Mafi sau□ in hanya shine **kwaikwayon □ abi'a (BC)**: □ auki misalin a matsayin bayanai supervised kuma koyi zawa daga yanayi zuwa ayyuka.

Da saitin bayanai na nau'in yanayi-aiki $\mathcal{D} = \{(s_i, a_i)\}$ daga mai nuna □ wararru, muna daidaita manufa $\pi_\theta(a \mid s)$ ta hanyar rage negative log-likelihood:

$$\mathcal{L}_{\text{BC}}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{(s,a)\in\mathcal{D}} \log \pi_\theta(a \mid s)$$

Wannan daidai koyon supervised na yau da kullum da ake amfani da shi ga bayanai mai jeri ne.

### Matsalar Canjin Covariate

BC yana da raunin asali: **canjin rarrabawa** tsakanin horarwa da aikawa. Misalai na □ wararru suna rufe yanayin da □ wararru ke ziyarta. Amma a lokacin aikawa, manufa da aka koyo na iya yanke shawara daban-daban ka□ an, yana jagorancinta zuwa yanayin da □ wararru ba su ta□ a ziyartar — yanayin da manufa ba shi da kulawa kuma na iya gazawa mara kyau.

Muhimmanci, kurakurai suna **ha□ awa**: sauye-sauye □ arami yana haifar da yanayi mara sani, inda aiki mara daidai ka□ an yana haifar da yanayi mara sani fiye da kima, da sauransu. Aiki yana ragewa a matsayin $O(T^2\epsilon)$ inda $T$ shine tsawon □ ullun da $\epsilon$ shine yawan kuskure a kowane mataki — mafi muni sosai fiye da raguwar $O(T\epsilon)$ na manufa ta oracle {cite}`ross2010efficient`.

> **Misali: Tuka Kai Tsaye** :class: note
>
> Samfurin kwaikwayon □ abi'a don adana hanya da aka horar akan bayanin tuka dan adam yana aiki da kyau akan hanyoyi madaidaiciya (yanayi kusa da rarrabawar horarwa). Amma lokacin da ya kauce ka□ an — yanayi wanda babu mai tuka dan

adam da zai kasancewa domin sun riga sun gyara — ba shi da bayanai don kula da shi kuma na iya tuka waje da hanya.

```
Algorithm DAgger:
  Initialize: D <- {} (empty dataset)
  Train initial policy pi_1 on M expert demonstrations

  for iteration i = 1, 2, ..., N:
    1. Run pi_i in the environment to collect states {s_1, ..., s_t}
    2. Query expert for actions on each visited state: a_t = pi*(s_t)
    3. Aggregate: D <- D u {(s_1, a_1), ..., (s_t, a_t)}
    4. Train pi_{i+1} by supervised learning on D
```

DAgger yana cimma nadama ta $O(T\epsilon)$ — □ aya da manufa ta oracle — saboda rarrabawar horarwa ta kusanya don dacewa da rarrabawar aikawa.

Bu□ ata ta muhimmanci ita ce □ wararru za a iya tambayar su akan duk yanayi, gami da yanayin da □ wararru ba za su ta□ a ziyarta su ta halitta ba. Wannan yana yiwuwa a kwaikwayo (ro□ a □ wararru don gyara robot daga tsarin da ba na yau da kullum ba) amma na iya zama kalubale ko mara lafiya a cikin tsarin jiki na gaske.

---

## Koyon □ arfafawa Mai Juyawa

Wani lokaci ana fi fahimtar □ abi'ar □ wararru ba a matsayin jeri na ayyuka don kwaikwayo amma a matsayin sakamakon inganta ayyukan lada da ba a sani ba. **Koyon □ arfafawa Mai Juyawa (IRL)** {cite}`ng2000algorithms` yana dawo da wannan ayyukan lada da ke □ oye daga misalai.

Da misalai $\tau = \{(s_1, a_1), \ldots, (s_T, a_T)\}$, IRL yana nemo ayyukan lada $R(s, a)$ inda manufar □ wararru ta fi a □ ar□ ashin $R$.

Burgeshi na IRL akan BC: idan muka dawo da ayyukan lada na gaskiya, za mu iya sake inganta shi a sabbin muhalli, tare da motsin daban-daban, ko tare da masu tsara aiki ingantattun — yana ya□ uwa fiye da yanayin da aka nuna sosai.

### MaxEnt IRL

**MaxEnt IRL** {cite}`ziebart2008maximum` yana warware matsalar rikitarwa na IRL (akwai ayyukan lada da yawa masu dacewa da kowane saitin misalai) ta hanyar za□ ar ayyukan lada wanda, yayin da yake dacewa da □ abi'ar da aka nuna, yana haifar da rarrabawar akan hanyoyi masu *entropy mafi girma*. An rarraba hanyoyi a matsayin:

$$P(\tau \mid R) \propto \exp\left(\sum_t R(s_t, a_t)\right)$$

Manufar koyo tana dacewa da tsammani na fasali na □ wararru da aka ga $\mu_E = \mathbb{E}_{\tau\sim\pi^*}[\phi(\tau)]$ ga tsammani na fasali na samfuri $\mu_\theta = \mathbb{E}_{\tau\sim\pi_\theta}[\phi(\tau)]$.

---

## GAIL: Koyon Kwaikwayo ta Adawa ta Generative

**GAIL** {cite}`ho2016generative` yana tsallake koyon ayyukan lada gaba □ aya, ta amfani da tsarin kama GAN don kai tsaye dacewa da rarrabawar yanayi-aiki ta □ wararru.

Ana horar da mai nuna bambanci $D_\psi$ don bambanta nau'i na yanayi-aiki na □ wararru $(s, a) \sim \pi^*$ daga nau'i na manufa $(s, a) \sim \pi_\theta$:

$$\mathcal{L}_D = -\mathbb{E}_{\pi^*}[\log D_\psi(s, a)] - \mathbb{E}_{\pi_\theta}[\log(1 - D_\psi(s, a))]$$

Ana horar da mai □ ir□ ira (manufar $\pi_\theta$) don yaudarar mai nuna bambanci — watau, don samar da nau'i na yanayi-aiki wa□ anda suke kama da na □ wararru. Sakamakon mai nuna bambanci $\log D_\psi(s, a)$ yana zama alaman lada don manufar.

GAIL yana cimma aikin matakin □ wararru akan gwajin sarrafa ci gaba da □ arancin misalai fiye da BC, kuma yana fa□ a□ awa mafi kyau fiye da MaxEnt IRL a muhalli masu rikitarwa.

---

## Kwaikwayon □ abi'a a NLP: Misali na Aiki

```python
import numpy as np
import torch
import torch.nn as nn
from torch.utils.data import DataLoader, TensorDataset

torch.manual_seed(42)

# ----------------------------------------------
# Toy NLP task: rewriting sentences to be more formal
# We simulate this as a simple sequence transformation
# In practice: fine-tuning a seq2seq model on expert rewrites
# ----------------------------------------------

class SimpleSeq2Seq(nn.Module):
    def __init__(self, vocab_size=100, embed_dim=32, hidden_dim=64):
        super().__init__()
        self.embed = nn.Embedding(vocab_size, embed_dim, padding_idx=0)
        self.encoder = nn.GRU(embed_dim, hidden_dim, batch_first=True)
        self.decoder = nn.GRU(embed_dim, hidden_dim, batch_first=True)
        self.proj = nn.Linear(hidden_dim, vocab_size)
        self.hidden_dim = hidden_dim

    def forward(self, src, tgt):
        src_emb = self.embed(src)
        _, hidden = self.encoder(src_emb)
        tgt_emb = self.embed(tgt)
        out, _ = self.decoder(tgt_emb, hidden)
        return self.proj(out)

# Generate synthetic demonstration data
VOCAB = 100
```

```
rng = np.random.default_rng(42)
N, SEQ_LEN = 1000, 12

src_seqs = torch.tensor(rng.integers(1, VOCAB, (N, SEQ_LEN)), dtype=torch.long)
# "Expert" transformation: shift tokens by 1 (toy formalization)
tgt_seqs = torch.clamp(src_seqs + 1, 1, VOCAB - 1)
tgt_in  = torch.cat([torch.ones(N, 1, dtype=torch.long), tgt_seqs[:, :-1]], dim=1)

dataset = TensorDataset(src_seqs, tgt_in, tgt_seqs)
loader = DataLoader(dataset, batch_size=64, shuffle=True)

model = SimpleSeq2Seq(vocab_size=VOCAB)
optimizer = torch.optim.Adam(model.parameters(), lr=1e-3)
criterion = nn.CrossEntropyLoss(ignore_index=0)

# Behavioral cloning training
train_losses = []
for epoch in range(20):
    epoch_loss = 0
    for src, tgt_i, tgt_o in loader:
        logits = model(src, tgt_i)
        loss = criterion(logits.reshape(-1, VOCAB), tgt_o.reshape(-1))
        optimizer.zero_grad(); loss.backward(); optimizer.step()
        epoch_loss += loss.item()
    train_losses.append(epoch_loss / len(loader))

print(f"Initial loss: {train_losses[0]:.3f}")
print(f"Final loss:   {train_losses[-1]:.3f}")

# Evaluate: check token accuracy on held-out examples
model.eval()
with torch.no_grad():
    src_test = src_seqs[-100:]
    tgt_test_in = tgt_in[-100:]
    tgt_test_out = tgt_seqs[-100:]
    logits = model(src_test, tgt_test_in)
    preds = logits.argmax(dim=-1)
    acc = (preds == tgt_test_out).float().mean().item()
    print(f"Token accuracy on held-out set: {acc:.3f}")
```

---

## Kwatancen Hanyoyin Koyon Kwaikwayo

| Hanya | Yana bu□ atar lada? | An tambaya □ wararru akan yanar gizo? | Yana ya□ uwa zuwa sabon motsi? | Rikitarwa |
|---|---|---|---|---|
| Kwaikwayon □ abi'a | A'a | A'a | Mai muni (canjin rarrabawa) | □ arami |
| DAgger | A'a | Eh | Matsakaici | Matsakaici |

| Hanya | Yana bu□ atar lada? | An tambaya □ wararru akan yanar gizo? | Yana ya□ uwa zuwa sabon motsi? | Rikitarwa |
|---|---|---|---|---|
| MaxEnt IRL | Yana dawo da shi | A'a | Mai kyau | Babba |
| GAIL | A'a | A'a | Mai kyau | Babba |

---

## Aikace-aikace

**Robots.** Koyar da robots don sarrafa abubuwa, kewayawa muhalli, ko yin ayyukan gida. Ana tattara misalai na jiki ta teleoperation ko koyar da kinesthetic.

**Tuka kai tsaye.** Tsarin motoci na farko da ke tuka da kansu kamar ALVINN {cite}`pomerleau1989alvinn` da NVIDIA DAVE sun dogara sosai akan kwaikwayon □ abi'a daga bayanin tuka dan adam.

**AI na wasa.** Koyon kwaikwayo akan wasannin dan adam yana dawo da wakilan kafin daidaita RL. AlphaStar ya horar akan kwaikwayon dan adam kafin RL; wannan hanya ta zama yau da kullum lokacin da misalai matakin dan adam suke samuwa.

□ **ir□ ira lambar.** Daidaita samfuri na harshe akan misalai mafi kyau na lambar (GitHub Copilot, Codex) wata nau'i ce ta kwaikwayon □ abi'a.

**Tallafi na yanke shawara na asibiti.** Koyo daga jeri na yanke shawara na likita □ wararru don tsarin ma'aunan rikitarwa.

> **See also:** Bincike na asali na BC/DAgger yana a {cite}`ross2011reduction`. MaxEnt IRL daga {cite}`ziebart2008maximum`. GAIL daga {cite}`ho2016generative`. Don bita cikakke na koyon kwaikwayo, duba {cite}`osa2018algorithmic`.

# Koyo daga Kwatancen da Jeri

Ro□ on dan adam su sanya kyautatuwar inganci ta cikakke ga sakamakon yana da wuya. Mene ne ingancin lissafi na wannan rubutu, akan sikelin 1 zuwa 10? Tambayar ba ta bayyananne: dan adam yana rasa sikeli na ciki mai daidaito, kuma maki □ insu suna shafawa sosai ta anchoring, mahallin, da gajiya.

Ro□ on dan adam su *kwatanta* sakamakon biyu yana da sau□ i sosai: wane rubutu ya fi, A ko B? Hukunci na kwatancen sun fi daidaito, sun fi aminci, kuma suna ta□ a fifikon dan adam kai tsaye fiye da □ ididdigan cikakke. Wannan babi ya □ unshi tushen lissafi da aikace-aikacen da ke da amfani na koyo daga kwatancen da jeri.

---

## Dalilin da Ya Sa Kwatancen Sun Fi □ ididdigan

### Asalin Kimiyyar Ilimin □ an Adam

Fifikon hukunci na kwatancen yana da tarihi mai tsawo a cikin kimanta yanayi. Dokar Hukunci na Kwatancen ta Thurstone {cite}`thurstone1927law` (1927) ta nuna cewa ko da lokacin da dan adam yana da hukunci na cikakke mara daidaito, hukuncin su na dangi yana bin doka mai yiwuwa mai daidaito. Kwatancen suna amfani da gaskiyar cewa dan adam sun fi □ warewa a *fahimtar dangi* fiye da daidaita cikakke.

### Ingancin Lissafi

Kowane kwatancen biyu biyu yana ba da bayanai game da *matsayin dangi* na abubuwan biyu akan sikelin inganci. Tare da $K$ abubuwa, $K-1$ kwatancen suna iya jera duk abubuwa; kawai $O(\log K)$ kwatancen ake bu□ ata don nemo item mafi □ ari. □ ididdigan cikakke galibi yana bu□ atar hukunci fiye da kima don cimma daidaito □ aya.

### Girman Sikeli

Don samfurin generative, yawan sakamakon daban-daban yana da tasiri mara iyaka. □ ididdiga sakamakon ta cikakke yana bu□ atar kafa sikeli mai ha□ in gwiwa a duk sakamakon; kwatanta sakamakon yana bu□ atar kawai hukunci na gida, na dangi wa□ anda suka daidaita da juna ta halitta.

---

## Tsarin Bradley-Terry

Tsarin iya yiwuwa mai rinjaye don kwatancen biyu biyu shine **tsarin Bradley-Terry (BT)** {cite}`bradley1952rank`. Kowane item $i$ yana da maki inganci da ke □ oye $\alpha_i \in \mathbb{R}$. Iya yiwuwar

cewa item $i$ an fi so da item $j$ a cikin kwatancen kai tsaye shine:

$$P(i \succ j) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}} = \sigma(\alpha_i - \alpha_j)$$

inda $\sigma$ shine logistic sigmoid. Wannan daidai yake da □ auka cewa ingancin da ake gane na item $i$ shine $\alpha_i + \epsilon$ inda $\epsilon$ shine ma'aunin hayaniya na logistic.

### Iya Gane

Tsarin BT yana iya gane har zuwa fassara: idan $\alpha$ mafita ce, haka ma $\alpha + c$ don duk yawan $c$. Al'ada ta yau da kullum ita ce □ ayyade maki □ aya (misali, $\alpha_0 = 0$) ko □ untatawa $\sum_i \alpha_i = 0$. Ana iya gane maki kawai lokacin da **graph na kwatancen** (nodes = abubuwa, edges = nau'i da aka ga) **yana da ha□ i** — idan graph yana da sassan da ba sa ha□ i, ala□ an maki a cikin sassan ba su da ma'ana.

### □ iyasin Sinadari

Da saitin bayanai na kwatancen biyu biyu $\mathcal{D} = \{(i, j, y_{ij})\}$ inda $y_{ij} = 1$ idan $i$ an fi so da $j$, log-likelihood shine:

$$\mathcal{L}(\alpha) = \sum_{(i,j,y_{ij}) \in \mathcal{D}} \left[ y_{ij} \log \sigma(\alpha_i - \alpha_j) + (1 - y_{ij}) \log \sigma(\alpha_j - \alpha_i) \right]$$

Wannan ayyuka mai gwiwa ne na $\alpha$ kuma za a iya inganta shi ta hanyar ha□ aka gradient ko hanyar Newton.

```
import numpy as np
from scipy.optimize import minimize
from scipy.special import expit  # sigmoid

rng = np.random.default_rng(42)

# ------------------------------------------------
# Simulate Bradley-Terry: 8 items with true quality scores
# Generate pairwise comparisons and recover the scores
# ------------------------------------------------

N_ITEMS = 8
true_alpha = rng.normal(0, 1, N_ITEMS)
print(f"True quality ranking: {np.argsort(true_alpha)[::-1]}")

# Generate comparisons: every pair compared 5 times
comparisons = []
for i in range(N_ITEMS):
    for j in range(i + 1, N_ITEMS):
        for _ in range(5):
            p_i_wins = expit(true_alpha[i] - true_alpha[j])
            winner = i if rng.random() < p_i_wins else j
            loser  = j if winner == i else i
            comparisons.append((winner, loser))
```

```
print(f"Total comparisons: {len(comparisons)}")

def neg_log_likelihood(alpha, comparisons):
    """Bradley-Terry negative log-likelihood."""
    alpha = np.array(alpha)
    loss = 0.0
    for winner, loser in comparisons:
        log_prob = np.log(expit(alpha[winner] - alpha[loser]) + 1e-10)
        loss -= log_prob
    return loss

def neg_grad(alpha, comparisons):
    alpha = np.array(alpha)
    grad = np.zeros(len(alpha))
    for winner, loser in comparisons:
        p = expit(alpha[winner] - alpha[loser])
        grad[winner] -= (1 - p)
        grad[loser]  -= (-p)
    return grad

# Fix alpha[0] = 0 to resolve scale ambiguity
result = minimize(
    lambda a: neg_log_likelihood(np.concatenate([[0.0], a]), comparisons),
    x0=np.zeros(N_ITEMS - 1),
    jac=lambda a: neg_grad(np.concatenate([[0.0], a]), comparisons)[1:],
    method='L-BFGS-B'
)
est_alpha = np.concatenate([[0.0], result.x])

# Compare true vs estimated ranking
true_rank = np.argsort(true_alpha)[::-1]
est_rank  = np.argsort(est_alpha)[::-1]

print(f"\nTrue ranking (item indices): {true_rank}")
print(f"Estimated ranking:           {est_rank}")
rank_corr = np.corrcoef(true_alpha, est_alpha)[0, 1]
print(f"Correlation with true scores: {rank_corr:.4f}")
```

---

## Tsarin Thurstone

Tsarin Thurstone {cite}`thurstone1927law` yana da ala□ a da Bradley-Terry amma yana amfani da hayaniya ta Gaussian maimakon logistic:

$$P(i \succ j) = \Phi\left(\frac{\alpha_i - \alpha_j}{\sqrt{2}\sigma}\right)$$

inda $\Phi$ shine CDF na yau da kullum. Lokacin da $\sigma = 1/\sqrt{2}$, wannan ya zama daidai da BT tare da sau□ a□ an bambancin sukeli. A aiki, samfurin biyu suna ba da sakamakon kusan □ aya.

---

## Ha□ a Jeri

Lokacin da kowane mai sanya lakabi ya ba da cikakken jeri na $K$ abubuwa (maimakon kwatancen biyu biyu), matsala ita ce **ha□ a jeri**: ha□ a jerin jeri da yawa zuwa jeri na yarjejeniya.

□ **ididdigan Borda:** Kowane item yana kar□ ar maki daidai da yawan abubuwa da aka jera □ asansa a cikin jerin kowane mai sanya lakabi. Ana ha□ a maki daga cikin masu sanya lakabi. Sau□ i da □ arfi.

**Kemeny–Young:** Nemo jeri wanda ke rage jimla na rashin jituwa biyu biyu (nisan Kendall tau) tare da jerin kowane mai sanya lakabi. Wannan NP-hard ne don babban $K$ amma mai yiwuwa don □ aramin saitin.

**RankNet / ListNet:** Hanyoyin cibiyar jijiya wa□ anda ke koyon ayyukan □ ididdiga daga jerin jeri, yana ba da damar ya□ uwa ga abubuwa da ba a ga su ba.

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## Makiyayan Biyu Biyu

A cikin koyon fifiko **yanar gizo**, abubuwa suna isa cikin yanayi kuma dole ne mu yanke shawara wa□ anne nau'i za a kwatanta, daidaita bincike (koyon game da abubuwa da ba a sani ba) da amfani (gabatar da abubuwa masu inganci). Wannan shine matsalar **makiyan biyu biyu** {cite}`yue2009interactively`.

Algorithm □ in muhimmanci: - **Doubler:** Yana kiyaye item jagora; yana □ alubalantar shi da abokan hamayya ta damari - **RUCB (Iyakar Amincewar Mai Dangi):** Yana □ ididdige tazarar amincewar salon UCB don iya yiwuwar kowane item ya fi kowane item □ in na biyu - **MergeRank:** Yana ha□ a kwatancen salon gasar da UCB

Ana amfani da makiyayan biyu biyu a tsarin shawarwarin yanar gizo (wane labari za a nuna gaba, da ra'ayin da ke □ oye) da neman fifiko mai mu'amala don tattara bayanin RLHF.

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## Koyon Fifiko don Samfurin Harshe

A mahallin RLHF (Babi na 6), ana amfani da tsarin Bradley-Terry don horar da samfurin lada. Wani bambancin muhimmanci shine **Inganta Fifiko Kai Tsaye (DPO)** {cite}`rafailov2023direct`, wanda ke nuna cewa za a iya inganta manufar RLHF kai tsaye daga bayanin fifiko ba tare da horar da samfurin lada daban ba:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right]$$

DPO yana da sau□ i fiye da cikakken RLHF (babu zagayen horarwa na PPO, babu samfurin lada) yayin da cimma sakamakon kwatankwacin ko mafi kyau akan gwaji da yawa {cite}`rafailov2023direct`. Ya zama hanya mai fa□ in amfani maimakon RLHF bisa PPO don daidaitar bin umarni, ko da yake hanyoyin biyu har yanzu suna cikin amfani mai aiki kuma □ arfinsu na dangi yana dogara akan aiki.

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## Tattara Bayanin Fifiko Mai Inganci

Ingancin bayanin fifiko yana □ ayyade ingancin samfurin lada. Abubuwan da ke da muhimmanci:

**Iri-iri na tambaya.** Bayanin fifiko ya kamata ya rufe cikakkiyar rarrabawar tambayoyin da samfuri zai fuskanta a aikawa. Gi□ □ in rufe yana haifar da □ abi'ar samfurin lada mara aminci a wa□ ancan yankuna.

**Iri-iri na amsa.** Kwatanta amsoshi biyu makamancin haka yana ba da bayanai ka□ an. Amsoshin da aka kwatanta ya kamata su bambanta sosai don masu sanya lakabi su sami fifiko bayyananne.

**Yarjejeniya na masu sanya lakabi.** Yarjejeniya □ asa da yawa tsakanin masu sanya lakabi yana ba da shawara ma'aunin kwatancen ba a bayyananne. Auna yarjejeniya (kappa na Cohen) kuma gyara jagoran lokacin da ta kasance □ asa da iyakar da za a iya yarda.

**Daidaita □ididdigan.** Masu sanya lakabi ya kamata su fahimci *dalilin da ya sa* amsa □ aya ta fi: taimako, daidaito, aminci, salo? Ayyuka wa□ anda ke ha□ a ma'aunai da yawa galibi suna samar da fifiko mara daidaito. Galibi ya fi kyau don tattara fifiko akan kowane ma'auni daban.

> **See also:** Tsarin Bradley-Terry: {cite}`bradley1952rank`. Thurstone: {cite}`thurstone1927law`. Makiyayan biyu biyu: {cite}`yue2009interactively`. Inganta Fifiko Kai Tsaye (DPO): {cite}`rafailov2023direct`.

# HITL a Sarrafa Harshen Yau da Kullum

Harshen yau da kullum shine yanki inda HITL ML ya sami tasiri mafi girma a iya magana — kuma inda mafi zurfinsa na kwamitin tunani suke fitar da kansu. Harshe yana da ma'ana ta zamantakewa: ma'anarsa an gina ta ta hanyar al'umman dan adam, pragmatics □ insa yana dogara akan mahallin da niyyar, kuma ingancin sa kawai ake iya □ ididdiga ta masu karatu dan adam. Amma wannan kuma yana nufin cewa lakabi na NLP ba kawai tsari ne na tattara lura ba. Tsari ne na *gina* rukunan.

---

## Matsalar □ ir□ ira a Lakabi na NLP

A cikin hoto likitanci, akwai gaskiya ta □ asa: ciwon daji yana nan ko ba shi ba. Lakabi na likita na iya kasancewa mara tabbas, amma yana □ o□ arin bin wani abu na gaske. Lakabi na NLP galibi yana da bambance da yawa. Lokacin da mai sanya lakabi ya alamta tweet a matsayin "mai guba," babu kwayoyin guba a cikin tweet wanda muke □ o□ arin gano. **Lakabi yana □ ir□ ira rukuni.**

Wannan yana da sakamako mai zurfi wa□ anda galibi ba a fahimta su sosai:

□ **ungiyar aiki na lakabi tana bayyana al'amari.** Schema ta lakabi don "magana mara kyau" tana □ oye hankali da duk wanda ya tsara shi da duk wanda ya yi amfani da shi. □ ungiya na masu sanya lakabi da ke magana Turanci daga bango na dimokra'iyya □ aya, suna aiki □ ar□ ashin jagoran da □ ungiyar manufofi ta kamfani ta rubuta, tana samar da saitin bayanai wanda ke nuna fahimtar wannan □ ungiyar ga laifin — ba wani ma'aunin dan adam na duniya ba. Samfurin da aka horar akan irin wa□ annan bayanai za su nuna iyakar da ke □ oye □ aya.

**Jagoran su ne ka'ida, ko sun yarda ko ba su yarda ba.** Kowane schema na lakabi yana yin da'awar akan ontology. Yanke shawara cewa "ba da manufar" da "izgili" ajin □ aya ne da'awar ka'ida ce, ba sauki na neutral ba. Yanke shawara don yin lakabi ga "fushi" a matsayin ajin □ aya maimakon bambanta "fushi mai adalci" daga "fushi mai □ iyayya" yana ruguje bambance wanda na iya yin muhimmanci ga aikin □ asa. Wa□ annan yanke shawara ana yi su □ ar□ ashin matsincin samarwa kuma da wuya a sake duba su.

**Rashin daidaito na lakabi akan lokaci.** Harshe na zamantakewa yana ha□ aka. Samfurin guba da aka horar a 2018 zai rarrabe abun ciki daga 2024 ba daidai ba saboda ba saboda ba a horar da shi da isa ba amma saboda ma'anar zamantakewa na wasu kalmomi ta canza. Lakabi na NLP ba ana bitar rarrabawa mai tsaye ba — ana bitar iyaka mai motsi wanda aikin lakabi yana taimaka □ ir□ ira shi a wani □ angare.

:::{admonition} Matsalar Kayan Aikin Lakabi :class: important

Geva et al. {cite}`geva2019annotator` sun nuna cewa saitin bayanai na NLI (Natural Language Inference) sun □ unshi kayan aikin tsari da aka gabatar ta tsarin lakabi kansa. Masu sanya lakabi da aka ro□ a su rubuta hasashe na "□ aukar" don wani yanki da aka ba su galibi suna

amfani da salon tsari musamman; masu sanya lakabi suna rubuta hasashe na "sabani" suna amfani da wasu. Samfurin yana koyon rarrabawa bisa wa□ annan kayan aikin maimakon ala□ ar ma'ana da aka nufa — suna warware aikin lakabi, ba aikin NLP na □ asa ba.

Wannan ba sakaci ba ne. Sakamakon da ba a kaurace shi ba ne na sanya dan adam su □ ir□ ira misalai don dacewa da lakabi. Tsarin HITL yana saka alaman tsari wanda ba a ta□ a nufa shi a cikin bayanai ba. :::

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## Lakabi na Rarrabuwa Rubutu

Mafi sau□ in aikin lakabi na NLP shine sanya rukuni ga takarda rubutu. Binciken ji, rarrabuwa ta batu, gano niyya, da tace faci duk su ne ayyukan rarrabuwa.

**Kalubale musamman ga rarrabuwa rubutu:**

*Son zuciya.* Rukunan kamar "mai guba" ko "mai laifi" yana da son zuciya ta asali kuma yana bambanta cikin mahallin al'adu, bango na mai sanya lakabi, da mahallin dandamali. A duk masu sanya lakabi, fahimtar laifi suna bambanta sosai ta halaye na dimokra'iyya — gaskiya mai kai tsaye ga adalci {cite}`blodgett2020language`.

*Rikitarwan lakabi.* Takardun da yawa suna cikin rukunan da yawa ko suna kan iyaka. Bita da ke da 60% mai kyau da 40% mai muni yana rikitarwa gaske, ba mislabeled ba. Tilasta lakabi □ aya yana watsi da bayanai na gaske (duba Babi na 13 akan lakabin laushi da rashin jituwa na mai sanya lakabi).

*Girma na lakabi.* Samfurin ji 2-aji na iya isa don sa ido na kafofin yada labarai na zamantakewa; sikelin maki 7 na iya zama dole don binciken ra'ayin kaya. Girman da ya dace yana dogara akan aikin □ asa, amma galibi ana □ ayyade shi kafin lakabi — yanke shawara mai muhimmanci da aka yi tare da bayanai □ asa da isa.

---

## Gane Sunan Wuri

Lakabi na NER yana bu□ atar gano span na rubutu da sanya nau'in wuri (MUTUM, □ UNGIYA, WURI, dkk.). Lakabi ya fi rikitarwa fiye da rarrabuwa takarda saboda dalilai da yawa:

**Iyakar span suna rikitarwa.** A "CEO Tim Cook na Apple ya sanar da…", shin wuri na MUTUM ya rufe "Tim Cook" ko "CEO Tim Cook na Apple"? Jagoran dole ne su magance wa□ annan lokuta a fili, kuma yarjejeniya tsakanin masu sanya lakabi akan span koyaushe ta kasance □ asa fiye da akan nau'i.

**Sanya nau'i yana bu□ atar ilimin duniya.** "Apple" ORG ne a mahallin □ aya, KAYAN AIKI a wani, kuma ba ko □ aya ba a "fitilar apple." Masu sanya lakabi suna bu□ atar isa ga ilimin yanki don yin sanya nau'i daidai.

**Wuri mai juna juna.** "The University of California, Berkeley" tana □ unshi □ UNGIYA mai juna juna a ciki WURI. Tsarin BIO na yau da kullum ba zai iya wakilta wuri mai juna juna ba; tsarin mafi rikitarwa (misali, BIOES, ko tsarin bisa graph) ake bu□ ata.

**Ingancin lakabi.** Pre-annotation tare da tsarin NER na tushe yana hanzarta lakabi sosai ta ba da damar masu sanya lakabi su gyara hasashe maimakon yin lakabi daga farko. A wani bincike

na NER asibiti, an ga □ aruwa ta gudu ta 30–60% {cite}`lingren2014evaluating`; girman irin wa□ annan riba yana dogara sosai kan ingancin samfuri na tushe da yanki.

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## Fitar da Dangantaka da Lakabi na Ma'ana

Bayan gano wuri, fitar da dangantaka yana bu□ atar yin lakabi ga *ala□ a* tsakanin wurin:

- Masu sanya lakabi dole ne su fahimci wuri biyu da kuma mai hasashe da ke ha□ a su
- Nau'in dangantaka yana da bambance na ma'ana masu rikitarwa (YA_DAUKI_AIKI vs. AN_DAUKE_YA vs. AN_KAFA_YA)
- Misalai mara kyau (nau'i na wuri ba shi da dangantaka) sun fi yawa fiye da masu kyau

**Yarjejeniya tsakanin masu sanya lakabi don fitar da dangantaka** galibi ta kasance □ asa fiye da ayyukan rarrabuwa. Don schemas masu bayyana, □ imomi na $\kappa$ a cikin iyaka 0.65–0.80 galibi an ruwaito {cite}`pustejovsky2012natural`; don ala□ ar ma'ana masu rikitarwa (dalili na al'amari, ala□ ar lokaci), yarjejeniya na iya fa□ uwa □ asa da haka, dangane da zanen schema da horar da mai sanya lakabi.

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## Gyarawa Bayan Fassara ta Injin (MTPE)

Gyarawa bayan fassara ta injin wata nau'i ce ta HITL NLP mai girma. Mai fassara dan adam yana gyara sakamakon tsarin MT maimakon fassara daga farko:

**Gyarawa haske (LPE):** Kawai ana gyara kurakurai masu mahimmanci. Ya dace lokacin da bu□ atun ingancin fassara sun kasance matsakaici.

**Gyarawa cikakke (FPE):** Ana gyara sakamakon zuwa ingancin bugu. Sakamakon da aka gyara galibi yana zama bayanin horarwa don inganta MT □ ari — zagayen gyarawa na gaske na dan adam a cikin zagaye.

**HTER (Ma'aunin Gyarawa na Fassara ta Dan Adam):** Metric □ in da ke auna nisan gyarawa tsakanin sakamakon MT da fassarar da aka gyara, an daidaita ta tsawon jumla {cite}`graham2015accurate`. A matsayin □ a'idar mai aiki kusan, ana □ aukar HTER □ asa da kusan 0.3 a matsayin sakamakon MT mai kyau; sama da 0.5, fassara daga farko na iya zama mafi sauri — ko da yake wa□ annan iyakar suna bambanta ta yanki da nau'in harshe.

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## AI na Tattaunawa da Lakabi na Zance

Sanya lakabi ga zance yana gabatar da tsarin lokaci:

- **Lakabi matakin juyawa:** yin lakabi ga kowane juyawa (niyya, ji, inganci)
- **Lakabi matakin zance:** kimanta daidaito na gaba □ aya, nasarar aiki, gamsuwar mai amfani
- **Lakabi hanya-hanya na mu'amala:** gano lokuta na gazawa musamman a cikin tattaunawa

HITL yana da muhimmanci musamman a cikin tattaunawa saboda gazawar samfuri galibi suna da laushi da ha□ awa: kuskuren gaskiya a juyawa na 3 na iya zama ba a bayyana shi ba har

juyawa na 7. Masu sanya lakabi dan adam da ke bin tattaunawa suna iya gano wa□ annan salon gazawar dogon lokaci wa□ anda ma'aunai ta atomatik gaba □ aya suke rasa.

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## Sanya Lakabi na Shirye-shirye da Kulawa □ asa

Lokacin da bayanan da aka yi lakabi suka yi □ arami, **kulawa** □ **asa** yana ba da damar amfani da ayyukan sanya lakabi da yawa masu □ ura, masu ha□ awa don □ ir□ ira lakabin iya yiwuwar a sikeli. **Snorkel** {cite}`ratner2017snorkel` shine tsarin canonical:

```
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import f1_score

rng = np.random.default_rng(42)

# ---------------------------------------------------------
# Toy weak supervision: sentiment classification
# 5 labeling functions (LFs) with different coverage/accuracy
# Label: +1 (positive), -1 (negative), 0 (abstain)
# ---------------------------------------------------------

N = 1000
true_labels = rng.choice([-1, 1], size=N)
X_features = np.column_stack([
    true_labels * 0.8 + rng.normal(0, 0.5, N),
    rng.normal(0, 1, N),
    rng.normal(0, 1, N),
])

def make_lf(accuracy, coverage, seed):
    rng_lf = np.random.default_rng(seed)
    votes = np.zeros(N, dtype=int)
    active = rng_lf.random(N) < coverage
    correct = rng_lf.random(N) < accuracy
    votes[active & correct]  = true_labels[active & correct]
    votes[active & ~correct] = -true_labels[active & ~correct]
    return votes

LFs = np.column_stack([
    make_lf(accuracy=0.85, coverage=0.60, seed=1),
    make_lf(accuracy=0.78, coverage=0.45, seed=2),
    make_lf(accuracy=0.70, coverage=0.80, seed=3),
    make_lf(accuracy=0.90, coverage=0.30, seed=4),
    make_lf(accuracy=0.60, coverage=0.90, seed=5),
])

def majority_vote(LF_matrix):
    labels = []
    for i in range(LF_matrix.shape[0]):
        votes = LF_matrix[i][LF_matrix[i] != 0]
```

```
        labels.append(0 if len(votes) == 0 else int(np.sign(votes.sum())))
    return np.array(labels)

mv_labels = majority_vote(LFs)
covered = mv_labels != 0

print(f"Coverage:                    {covered.mean():.1%}")
print(f"MV accuracy (covered):       {(mv_labels[covered] == true_labels[covered]).mean():.3f

X_train, y_train = X_features[covered], mv_labels[covered]
X_test,  y_test  = X_features[~covered], true_labels[~covered]

if len(X_train) > 10 and len(X_test) > 10:
    clf = LogisticRegression().fit(X_train, y_train)
    preds = clf.predict(X_test)
    print(f"F1 on uncovered test set:    {f1_score(y_test, preds):.3f}")
```

---

## RLHF don Samfurin Harshe: Gaskiyar Lakabi

Babi na 6 ya □ unshi RLHF ta fasaha. Daga hangen NLP, aikin lakabi ya fi wuya fiye da yadda ya bayyana daga waje.

**Abin da ake ro□ a masu sanya lakabi da gaske su yi** — kwatanta sakamakon samfuri guda biyu kuma nuna wanda ya fi "kyau" — yana kama da sau□ i. A aiki, "kyau" wani tsari ne mara bayyananne wanda masu sanya lakabi ke warware ta amfani da hanyoyi na sirri. Wasu suna bai wa fluency muhimmanci sosai; wasu suna bai wa daidaito na gaskiya; wasu suna hukunta yawa. Ba tare da jagoran mai □ arfi ba, saitin bayanin fifiko da ya samu yana nuna ba □ imar dan adam a sah ba amma dabarun warwarewa musamman na □ ungiyar aiki na lakabi da aka yi hayar.

**Abubuwan muhimmanci na lakabi su ne:**

- *Taimako:* Shin amsar ta amsa tambayar da kyau? Ta bayyananne, ta ba da bayanai sosai, kuma ta dace sosai?
- *Gaskiya:* Shin amsar ta kasance daidai ta gaskiya? Wannan yana bu□ atar masu kimantawa su sami ilimin yanki — bu□ ata wanda ke □ ir□ ira matsalolin inganci mai girma a sikeli, tun da masu sanya lakabi na gaba □ aya ba za su iya tabbatar da da'awa na □ wararru ba.
- *Rashin cutar:* Shin amsar ta kaurace daga abun ciki mai guba, nuna son zuciya, mai cutarwa, ko mara dacewa? Wa□ annan hukunci suna bu□ atar jagoran cikakku saboda "mai cutarwa" yana dogara sosai akan mahallin kuma yana canzawa cikin al'adu, lokaci, da mahallin dandamali.
- *Daidaitar □ ididdigan:* Shin amsar ta bayyana rashin tabbas da ya dace? Masu sanya lakabi suna fifiko ta tsari don amsoshi wa□ anda suke kama amincewar, wanda ke □ ir□ ira alaman lada a kan tawali'u na epistemological da ya dace.

Mu'amalar tsakanin ma'aunai tana da rikitarwa: amsar mafi taimako ga tambaya mai ha□ ari na iya kasancewa mafi cutar. Jagoran dole ne su □ ayyadewa yadda za a musayar ma'aunai masu gasa — kuma wa□ annan musayar a zahiri yanke shawara na manufofi ne, ba yanke shawara na lakabi ba. □ ungiyar aiki na lakabi tana yanke manufofi.

**Girman sikeli yana tattara tasirin halaye na dimokra'iyya.** RLHF don manyan samfurin yana

ha□ a da □ ungiyar aiki na lakabi da ba su da girma (□ aruruwa zuwa dubbai ka□ an) da ke yanke shawara miliyan □ aruruwa □ asa. Halaye na al'adu da dimokra'iyya na wannan □ ungiya suna yada zuwa cikin □ abi'ar samfuri a sikeli ta hanyar da ba za ta faru ba idan lakabi ya kasance mafi yadi. Wannan yana □ aya daga cikin matsalolin tsari mafi □ arancin tattaunawa a cikin tsarin RLHF na yanzu.

---

## Zagayen Lakabi–Samfuri

A NLP fiye da kowane yanki, tsarin lakabi da ha□ aka samfuri suna rikita tsarin akan lokaci:

1. Ana daidaita masu sanya lakabi ta amfani da sakamakon samfuri da ake nan a matsayin tunani (galibi da □ oye).
2. Samfurin lada yana koyon abin da lakabi galibi suka fi so.
3. Ana daidaita mai □ ir□ ira don samar da sakamakon da ke samun lada mai □ ari.
4. Wa□ annan sakamakon suna shafawa abin da ke kama "mai kyau" a zagaye na lakabi masu zuwa.

Wannan zagaye ba mara lafiya ta asali ba ne — shi ne abin da ke ba da damar RLHF ya □ usanci — amma yana nufin cewa □ abi'ar samfuri tana tsarawa ta iyaka mai motsi wanda tsarin lakabi kansa yana taimaka motsa. Bambance abin da samfuri ya koyo saboda yana nuna fifikon dan adam daga abin da ya koyo saboda ya koyi dacewa da heuristics na masu sanya lakabi yana da wuya ta gwaji.

Babu mafita mai tsabta. Aiki mafi kyau na yanzu yana sa ido don canzawa ta amfani da hukunci na fifiko na □ aure da aka tattara a tazara na yau da kullum, da kuma □ aukar sigar jagoran lakabi a matsayin canji na gwaji.

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## Kimantawa ta Samfurin NLP na Generative

Ba kamar samfurin rarrabuwa tare da metric □ in daidaito bayyananne, kimantawar ingancin □ ir□ ira yana bu□ atar hukuncin dan adam:

| Hanyar kimantawa | Bayani | Ku□ i |
|---|---|---|
| Kimantawa kai tsaye (DA) | Masu sanya lakabi suna □ ididdiga inganci akan sikeli | Matsakaici |
| Hukunci na kwatancen | Masu sanya lakabi suna kwatanta sakamakon biyu | □ arami |
| MQM (Ma'aunai Inganci Mai Yawan Girma) | Taxonomy na kuskure mai tsari | Babba |
| Fifikon RLHF | Lakabin fifiko da ake amfani da su don horarwa | Matsakaici |
| LLM a matsayin al□ ali | LLM yana □ ididdiga sakamakon (yana dacewa matsakaici da dan adam) | □ arami sosai |
| BERTScore, BLEU | Ma'aunai ta atomatik (ala□ a mara daidai da hukuncin dan adam) | □ arami sosai |

Ma'aunai ta atomatik (BLEU don MT, ROUGE don takaitawa) suna da arha amma yana dacewa da □ arancin daidaito tare da hukunci na inganci na dan adam {cite}`reiter2018structured`.

Hanyoyin LLM-a-matsayin-al□ ali suna nuna yarjejeniya matsakaici tare da masu sanya lakabi dan adam {cite}`zheng2023judging` kuma ana amfani da su fiye da kima don sake zagaye mai sauri, amma ba ya kamata su maye gurbin kimantawar dan adam don kima na □ arshe. Don yanke shawara tare da ha□ ari na gaske, kimantawar dan adam har yanzu yana dole.

> **See also:** Kulawa □ asa ta Snorkel: {cite}`ratner2017snorkel`. Jagoran lakabi na NLP: {cite}`pustejovsky2012natural`. Kayan aikin lakabi a NLI: {cite}`geva2019annotator`. Nuna son zuciya na mai sanya lakabi da saitin bayanai na NLP: {cite}`blodgett2020language`. Don kimantawa ta samfurin generative: {cite}`reiter2018structured`.

# HITL a Gane Hoto ta Kwamfuta

Gane hoto ta kwamfuta yana ba da wasu mafi bayyana na □ an adam na HITL ML. Gasar ImageNet, wanda aka gina akan hotuna miliyan 14 da dan adam ya yi lakabi, ya □ addamar da zamanin koyon zurfin. Lakabi na hoto likitanci ta likitan hoto yana □ arfafa AI na gano cutar. Motoci masu tuka kan su suna dogara akan firam da dan adam ya yi lakabi da miliyoyi don koyon gane duniya.

Abin da ke da sau□ in rasa: wa□ annan ba kawai lokuta ne na dan adam da ke ba da gaskiyar □ asa ba. Su ne lokuta ne na dan adam da ke □ ir□ ira saitin bayanai wa□ anda ke □ oye za□ u□ □ ukan fahimta, al'adu, da aiki musamman — za□ u□ □ uka wa□ anda ke bayyana kawai daga baya, lokacin da samfurin suka gazawa ta hanyoyi masu iya hasashe.

---

## Yadda Za□ u□ □ ukan Lakabi ke Zama Nuna Son Zuciya na Samfuri

Tsarin yau da kullum yana □ aukar lakabi a matsayin tattara bayanai: dan adam suna lura da duniya kuma suna rubuta abin da suka gani. Tsarin mafi daidai shine cewa lakabi shine *zanen saitin bayanai*: dan adam suna yanke shawara wa□ anne rukunan za a yi amfani da su, inda za a zana iyaka, wa□ anne lokuta na iyaka za a ha□ a, da yadda za a warware rikitarwa — kuma duk wa□ annan yanke shawara suna tsara abin da samfuri da aka horar zai gane.

### Yanayin ImageNet

ImageNet {cite}`russakovsky2015imagenet` shine saitin bayanai mafi muhimmanci a cikin tarihin gane hoto ta kwamfuta. Saita lakabin sa yana samo asali daga synsets na WordNet, da farko an za□ a don kasancewa da yawa da kuma bambancin ma'ana. Sakamako da yawa na wannan za□ i na zane suka bayyana shekaru baya:

- **Rukunan mutum sun □ oye ala□ a na halaye na dimokra'iyya.** Sigar farko na lakabin synset na ImageNet don mutum sun ha□ a da da yawa wa□ anda yanzu za a □ auke su mara darajar ko nuna son zuciya, suna nuna duka tushen WordNet na tarihi da yanke shawara da □ oye na □ ungiyar aiki na lakabi game da wa□ anne lakabin za a yi amfani da su ga hotuna guda guda {cite}`yang2020towards`. Lakabin da aka yi amfani da su ga hotuna na mutum sun □ oye ala□ a ta jinsi, jima'i, da aji wa□ anda kai tsaye suka yada zuwa cikin embeddings na samfuri.

- **Taxonomy ta jinsin-jinsin tsari mai kyau, na kusan duk sauran.** ImageNet yana iya bambanta karnukan nau'in 120 amma yana ruguje bambancin mai yawa a cikin kayan aiki, motoci, abinci, da daki-daki zuwa rukunan □ aya. Wannan sakamakon bin tsarin WordNet ne, ba za□ in da gangan game da abin da ke da muhimmanci ba. Samfurin da aka horar akan ImageNet suna nuna daidaito mara daidaito □ aya.

- **Ma'aunan gani na Yamma, mai magana da Turanci.** An tattara hotuna da farko daga Flickr da binciken Intanet ta amfani da tambayoyi da Turanci. Rarrabawar da ta samu tana nuna sosai zuwa yanayin gani da abubuwan al'adu na □ asashe masu arziki, masu magana da Turanci.

Babu □ ayan wa□ annan kurakurai ba ne. Yanke shawara ne na zanen lakabi da aka yi da sauri a sikeli, galibi ta mutane wa□ anda ba su yi tsammanin yadda saitin bayanai za a yi amfani da shi ba. Darasin ba shine cewa ya kamata a gina ImageNet daban ba (ko da yake ya kamata), amma **zanen lakabi zanen samfuri ne**, kuma ya kamata a □ auke shi da kulawa □ aya.

:::{admonition} Schema na lakabi ka'idar duniya ce :class: note

Kowane taxonomy na lakabi yana yin da'awa game da bambance wa□ anda ke da muhimmanci. Za□ ar rarrabewa "mota" daga "motar da aka yi watsi da wuri biyu" yayin da suka dunkule duk wayoyin hannu zuwa aji □ aya da'awar ka'ida ce game da wa□ anne bambance ke da ma'ana ta ma'ana. Za□ ar yin lakabi ga "mutum" a matsayin aji □ aya ba tare da la'akari da hali, sutura, ko aiki ba wata da'awar ka'ida ce. Samfurin da aka horar akan wa□ annan schemas za su yi bambancen □ aya, kuma ba fiye da haka — ba za su fa□ a□ a fiye da rukunan da aka horar su don bambanta ba. :::

---

## Lakabi na Rarrabuwa Hoto

Mafi sau□ in aikin lakabi na CV shine sanya lakabin □ aya ko fiye ga hoto gaba □ aya.

**Hierarchy na lakabi.** Lakabi "kare" yana yaro na "dabba" a cikin hierarchy na ma'ana. Samfurin da aka horar akan taxonomies mai saman iya ba su fa□ a□ a da kyau a cikin matakin abstraction. ImageNet yana amfani da hierarchy bisa synset wanda ke ba da damar kimantawa a matakai da yawa na musamman.

**Rikitarwa ta lakabi nau'i da yawa.** Yanayin titin na iya □ unshi mota, mutum, keke, da hasken zirga-zirga a lokaci □ aya. Yanke shawara wa□ anne lakabin za a ha□ a yana bu□ atar jagoran bayyananne game da iyakar dacewa.

**Rarrabawar dogon wutsiya.** Saitin bayanai na hotuna na yanayi suna bin doka mai iko: rukunan ka□ an suna da yawa sosai; mafi yawansu suna da □ arami. Koyon aiki yana da □ ima musamman don rukunan dogon wutsiya inda samfurin ta damari yana ba da ka□ an misalai.

---

## Gano Abubuwa: Lakabi na Akwatin Iyaka

Gano abubuwa yana bu□ atar masu sanya lakabi su zana akwatuna daidai da axis a kusa da misali na kowane ajin abubuwa. Wannan yana gabatar da bu□ atun daidaito na geometrical da lokuta na iyaka masu yawa.

**Ma'aunai na ingancin lakabi:**

*IoU (Ha□ uwa-akan-□ ungiya)* yana auna ha□ uwa tsakanin akwati da aka sanya lakabi da akwati na tunani:

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

IoU $\geq$ 0.5 shine iyaka ta yau da kullum don gano "daidai" a PASCAL VOC; COCO {cite}`lin2014microsoft` yana amfani da iyaka da yawa daga 0.5 zuwa 0.95, wanda yana da bu□ atu fiye da kima kuma yana ba da bayanai fiye da kima.

**Lokuta na iyaka na lakabi wa□ anda dole ne a warware su a cikin jagoran:** - Abubuwa masu toshe: yin lakabi ga □ angare da ake gani ko kimanta fa□ in cikakke? - Abubuwa masu yanke (a wani □ angare waje da firam): ha□ a ko cire? - Yankuna masu cunkoson jama'a: yi amfani da lakabi na musamman "fama da cunkoson jama'a" ko yin lakabi ga misali □ aya □ aya?

Kowane □ ayan wa□ annan yanke shawara yana canza abin da "gano daidai" ke nufi — kuma saboda haka yana canza abin da samfuri aka horar don yi.

```python
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as patches

def compute_iou(boxA, boxB):
    """Compute IoU between two boxes [x1, y1, x2, y2]."""
    xA = max(boxA[0], boxB[0])
    yA = max(boxA[1], boxB[1])
    xB = min(boxA[2], boxB[2])
    yB = min(boxA[3], boxB[3])
    inter_area = max(0, xB - xA) * max(0, yB - yA)
    boxA_area = (boxA[2] - boxA[0]) * (boxA[3] - boxA[1])
    boxB_area = (boxB[2] - boxB[0]) * (boxB[3] - boxB[1])
    union_area = boxA_area + boxB_area - inter_area
    return inter_area / (union_area + 1e-6)

ref_box  = [1.0, 1.0, 4.0, 4.0]
ann1_box = [1.1, 0.9, 4.1, 4.2]   # close
ann2_box = [0.5, 0.5, 3.5, 3.8]   # less precise

print(f"IoU (ann1 vs ref):  {compute_iou(ann1_box, ref_box):.3f}")
print(f"IoU (ann2 vs ref):  {compute_iou(ann2_box, ref_box):.3f}")
print(f"IoU (ann1 vs ann2): {compute_iou(ann1_box, ann2_box):.3f}")

fig, ax = plt.subplots(figsize=(5, 5))
ax.set_xlim(0, 5); ax.set_ylim(0, 5); ax.set_aspect('equal')
ax.add_patch(patches.Rectangle(
    (ref_box[0], ref_box[1]), ref_box[2]-ref_box[0], ref_box[3]-ref_box[1],
    linewidth=2.5, edgecolor='#2b3a8f', facecolor='none', label='Reference'))
ax.add_patch(patches.Rectangle(
    (ann1_box[0], ann1_box[1]), ann1_box[2]-ann1_box[0], ann1_box[3]-ann1_box[1],
    linewidth=2, edgecolor='#0d9e8e', facecolor='none', linestyle='--',
    label=f'Ann1 (IoU={compute_iou(ann1_box, ref_box):.2f})'))
ax.add_patch(patches.Rectangle(
    (ann2_box[0], ann2_box[1]), ann2_box[2]-ann2_box[0], ann2_box[3]-ann2_box[1],
    linewidth=2, edgecolor='#e05c5c', facecolor='none', linestyle=':',
    label=f'Ann2 (IoU={compute_iou(ann2_box, ref_box):.2f})'))
ax.legend(fontsize=10)
ax.set_title("Bounding Box Agreement (IoU)", fontsize=12)
plt.tight_layout()
plt.savefig('bbox_iou.png', dpi=150)
```

```
plt.show()
```

---

## Rarrabuwa ta Ma'ana da Misali

Lakabi na rarrabuwa yana bu□ atar sanya ajin lakabi ga kowane pixel a cikin hoto — daga cikin mafi tsada nau'in lakabi.

**Rarrabuwa ta ma'ana:** Kowane pixel yana cikin ajin ma'ana (hanya, sararin sama, mutum, mota). Duk pixel na ajin □ aya suna raba lakabi □ aya, ba tare da la'akari da wane abu na musamman suke cikin shi ba.

**Rarrabuwa misali:** Kowane misali na abu na musamman yana kar□ ar lakabi na musamman. Fama da cunkoson jama'a na mutum 20 yana zama tufafi 20 daban-daban.

**Rarrabuwa Panoptic:** Yana ha□ a duka biyun: rukunan "abu" (abubuwa masu iya □ idaya) suna da tufafi misali; rukunan "abubuwa" (hanya, sararin sama) suna da tufafi ma'ana.

**Lakabi mai tallafi na SAM:** Samfurin Segment Anything Meta {cite}`kirillov2023segment` yana □ ir□ ira tufafin rarrabuwa mai inganci daga danna kan □ aya. Masu sanya lakabi suna danna tsakiyar abu; SAM yana ba da shawara tufafi; mai sanya lakabi yana yarda ko gyara. Marubutan SAM sun ruwaito hanzartar da injin lakabi na kusan 6.5× akan sanya lakabi bisa polygon; riba suna bambanta cikin nau'in yanayi da kayan aiki na lakabi.

SAM yana wakilta canjin mafi faffada: **matsayin mai sanya lakabi yana canzawa daga zane zuwa duba da tabbatarwa**. Wannan yana da abubuwa masu amfani don ingancin lakabi fiye da sauri. Lokacin da masu sanya lakabi suna zane, hankalinsu yana shiga a duk tsari. Lokacin da masu sanya lakabi suna duba kuma suna danna "yarda," akwai shaidar cewa suna rasa kuraku-rai cikin sau□ in fiye da kima — sigar nuna son zuciya na atomatik musamman ga mahallin lakabi.

---

## Koyon Aiki don Gane Hoto ta Kwamfuta

Koyon aiki yana da □ ima musamman a CV saboda: 1. Hotuna suna da yawan girma kuma suna da arziki ta fasali — embeddings daga samfurin da aka horar a baya suna □ aukar alaman □ arfi don □ iyasin rashin tabbas 2. Tafkin mara lakabi yana da arha (hotuna, firam na bidiyo) 3. Lakabi (musamman rarrabuwa) yana da tsada kuma ba za a iya cunkusar da shi da sau□ in don yankuna musamman ba

```
import numpy as np
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)

X, y = make_classification(n_samples=3000, n_features=50, n_informative=25,
                           n_classes=5, n_clusters_per_class=2, random_state=42)
X_train, y_train = X[:2500], y[:2500]
X_test,  y_test  = X[2500:], y[2500:]
```

```
def margin_uncertainty(model, X_pool):
    probs = model.predict_proba(X_pool)
    sorted_p = np.sort(probs, axis=1)
    return sorted_p[:, -2] - sorted_p[:, -1]  # most negative = most uncertain

n_init = 50
results = {'active': [], 'random': []}
label_counts = list(range(50, 401, 30))

for strategy in ['active', 'random']:
    labeled = list(rng.choice(len(X_train), n_init, replace=False))
    unlabeled = [i for i in range(len(X_train)) if i not in labeled]

    for target in label_counts:
        while len(labeled) < target and unlabeled:
            if strategy == 'active' and len(labeled) >= 10:
                model_temp = LogisticRegression(max_iter=300).fit(
                    X_train[labeled], y_train[labeled])
                margins = margin_uncertainty(model_temp, X_train[unlabeled])
                idx = int(np.argmin(margins))
            else:
                idx = rng.integers(0, len(unlabeled))
            labeled.append(unlabeled.pop(idx))

        clf = LogisticRegression(max_iter=300).fit(X_train[labeled], y_train[labeled])
        results[strategy].append(accuracy_score(y_test, clf.predict(X_test)))

plt.figure(figsize=(7, 4))
plt.plot(label_counts, results['active'], 'o-', color='#2b3a8f',
         linewidth=2, label='Active (margin sampling)')
plt.plot(label_counts, results['random'], 's--', color='#e05c5c',
         linewidth=2, label='Random baseline')
plt.xlabel("Labeled training images", fontsize=12)
plt.ylabel("Test accuracy", fontsize=12)
plt.title("Active Learning for 5-Class Image Classification", fontsize=13)
plt.legend(fontsize=11); plt.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('cv_active_learning.png', dpi=150)
plt.show()
```

---

## Koyon Semi-Supervised tare da Jagoran Dan Adam

Yawan bayanai na gani da ba a yi lakabi ba da yake samuwa yana mai da koyon semi-supervised yana da jan hankali musamman don CV.

**Horarwa-kan-kansa / pseudo-labeling:** Horar da samfuri akan bayanai da aka yi lakabi; yi amfani da hasashe masu amincewar □ ari akan bayanai da ba a yi lakabi ba a matsayin pseudo-lakabin; sake horarwa. Tambayar zanen da ke da muhimmanci ita ce iyakar amincewar. Iyaka □ arami yana kawo misalai fiye da kima amma yana gabatar da hayaniya; iyaka mai □ ari yana

barin mafi yawan tafkin da ba a yi lakabi ba ba a yi amfani da shi ba. Shiga dan adam zai iya kula da wannan iyaka — masu sanya lakabi suna bitar samfuri na pseudo-lakabin misalai don daidaita shi.

**FixMatch da daidaito ta ha□ aka:** Wa□ annan hanyoyin suna horar da samfurin don samar da hasashe daidai □ ar□ ashin ha□ aka. Babban ra'ayin HITL: ana ba da shawarwari ta dan adam ba kawai don lakabin amma don **zanen ha□ aka** — wa□ anne daidaito ya kamata samfuri ya koyo? Samfuri don hoto likitanci ya kamata ya kasance daidai da jujjuyawa da haske amma ba da sukeli ba; samfuri don gano rubutu bai kamata a mai da shi daidai da canzawar hangen nesa ba. Wa□ annan za□ u□ □ uka musamman na yanki suna bu□ atar □ warewa ta dan adam, kuma kuskuren su yana raunana koyon semi-supervised sosai.

**Koyon aiki na semi-supervised mai aiki:** Ha□ in gwiwa mafi inganci: koyon aiki yana tattara lakabin dan adam inda rashin tabbas na samfuri ya fi girma; horarwa-kan-kansa yana yin lakabi ta atomatik ga wutsiyar amincewar □ ari. □ o□ arin dan adam yana tattara inda yake da □ ima fiye da kima, kuma samfuri yana □ aukar sauran ta auto-labeling. A kowane zagaye, bitar dan adam na samfuri ta damari na pseudo-lakabin yana ba da duba inganci ba tare da bu□ atar duba cikakke ba.

---

## Lakabi na Bidiyo

Lakabi na bidiyo yana ninka kalubalen lakabi na hoto ta lokaci:

- **Bin sawu:** Dole ne a gano abubuwa cikin firam. Masu sanya lakabi suna yin lakabi ga firam muhimmanci; algorithms na bin sawu suna interpolate tsakanin su. Gazawar bin sawu — toshe, sake shiga, motsi mai sauri — suna bu□ atar sake lakabi na dan adam a yawan fiye da bin sawu na daidai.
- **Daidaito na lokaci:** Ya kamata iyaka da aka zana a firam $t$ su kasance daidai a sararin duniya da firam $t + 1$. Lakabi mara daidaito alama ce ta horarwa da ke gaya wa samfuri cewa abubuwa suna tsalle ba da tsari ba — wata nau'i ta □ urar lakabi musamman mai cutarwa don samfurin gano.
- **Girman sikeli:** Bidiyo na awa 1 a 30fps shine firam 108,000. Lakabi cikakke ba mai yiwuwa ba ne; dole ne a tsara dabarun samfuri da hankali don tabbatarwa cewa al'amurra da ba a cika ba (lokuta na iyaka, kusan gazawa, yanayin gazawa) ba a cire su ta tsari ba.

Kayan aikin lakabi na bidiyo na zamani suna tallafa wa **bin sawu mai wayo** wanda ke yada lakabi cikin firam kuma yana yin alamar firam inda amincewar bin sawu ta fa□ i □ asa da iyaka, yana kira mai sanya lakabi don sake duba. Wannan amfani kai tsaye ne na ra'ayin koyon aiki ga tsarin lakabi kansa: kayan aikin yana tambaya mai sanya lakabi ainihin inda interpolation □ insa ya kasance mara tabbas.

**Matsalar al'amari da ba a cika ba a cikin tsarin atomatik.** Don aikace-aikace inda sakamakun al'amurra mara cika suka lalace — tuka kai tsaye, kewayar UAV — rarrabawar firam da ake gani a aiki na yau da kullum yana dacewa mara kyau sosai tare da rarrabawar firam wa□ anda ke da muhimmanci sosai. Saitin bayanai da aka gina ta samfurin firam na tuka ta daidaito zai □ unshi firam miliyan na "babu abin ban sha'awa yana faruwa" da □ an ka□ an na firam na kusan-ha□ ari, haske na musamman, da firam na sensor masu lalacewa wa□ anda ke da muhimmanci sosai ga aminci. Koyon aiki na HITL wanda ke gano da kuma fifita irin wa□ annan firam ba dabara ta ingancin □ o□ ari ba ne — bu□ ata ce ta aminci.

> **See also:** Saitin bayanai na ImageNet: {cite}`russakovsky2015imagenet`. Nuna son zuciya na lakabi a ImageNet: {cite}`yang2020towards`. Ma'aunin COCO:

{cite}`lin2014microsoft`. SAM (Segment Anything): {cite}`kirillov2023segment`. Koyon aiki na core-set don CV: {cite}`sener2018active`.

# HITL a Kiwon Lafiya da Kimiyya

Kiwon lafiya da kimiyya suna wakilta biyu daga cikin yankuna inda HITL ML yake da muhimmanci sosai kuma yana da tattaunawa sosai. Ha□ ari suna da girma: gano ciwon daji da bata ko manufar magani da ke da lahani yana da ku□ in dan adam na gaske. Lakabi yana bu□ atar □ warewa ta wuya kuma mai tsada. Bu□ atun doka suna □ untatawa abin da samfurin zai iya yi da yadda dole ne a tabbatar da su. Kuma ba kamar NLP ba, inda matsalar lakabi a wani □ angare □ ir□ ira ta zamantakewa ce, a nan galibi akwai gaskiya ta □ asa — ciwon daji yana nan ko ba shi ba — ko da babu mai lura □ aya da ke iya □ ayyade ta da aminci.

Tsarin yau da kullum a cikin rubutawa na al'ada shine "AI da dan adam": shin AI zai maye gurbin likitan hoto? Wannan tsari ba daidai ba ne ta hanyar da ke da muhimmanci. Tambayar ta gaske ita ce wane nau'in ha□ in gwiwa ta dan adam da AI yana samar da sakamakon mafi kyau fiye da kowane □ aya kawai, da yadda za a gina tsarin da ke ba da damar wannan ha□ in gwiwa maimakon tarawa a gare shi.

---

## Binciken Hoto Likitanci

Hoto likitanci — likitancin hoton (X-ray, CT, MRI), pathology (abin da yake cikin kyallen jikin), dermatology, ophthalmology — shine yanki inda AI likitanci ya ci gaba mafi sauri.

### Bu□ atun Mai Sanya Lakabi □ wararru

Lakabi na hoto likitanci galibi yana bu□ atar likitoci masu horo na musamman na yanki. Wannan yana sa lakabi ya zama:

- **Jinkiri:** □ wararru suna da lokaci iyakantacce; lakabi yana gasa da ayyukan asibiti
- **Mai tsada:** Ku□ i ya kai daga goma zuwa □ aruruwa dalar Amurka kowace yanayin da aka sanya lakabi, dangane da fanin kuma rikitarwan aikin
- **Mai bambanta:** Ko da □ wararru suna rashin jituwa, musamman kan lokuta na iyaka — gaskiya wanda galibi ana □ auke shi a matsayin matsala amma a zahiri yana ba da bayanai

### Bambancin Tsakanin Likitocin Hoto

Bambancin mai karatu an rubuta shi da kyau a cikin likitancin hoto. Don fassarar X-ray □ irji, rashin jituwa tsakanin mai karatu yana da mahimmanci — a binciken CheXNet, likitocin hoto guda hu□ u sun sanya lakabi ga saitin gwaji na gano pneumonia □ aya tare da maki F1 da suka rufe kusan kashi goma sha biyu na maki {cite}`rajpurkar2017chexnet`, yana nuna rashin tabbas na gano na gaske akan lokuta na iyaka. Don gano nodule akan CT na huhu, bambancin mai karatu na ciki (mai karatu □ aya, yanayi □ aya, rana daban) na iya zama kamar bambancin mai karatu na waje.

Wannan bambancin ba hayaniya kawai ba ne — yana nuna rashin tabbas na gano na gaske. Samfurin da aka horar akan lakabi na likitan hoto □ aya na iya koyon son zuciya musamman na wannan likita maimakon pathology na □ asa.

:::{admonition} Muhawarar CheXNet a matsayin Darasin HITL :class: note

Lokacin da Rajpurkar et al. suka da'awa cewa samfurin CheXNet □ insu "ya wuce aikin likitan hoto" akan gano pneumonia, an yi kalubalantar da'awar nan da nan ta al'ummar likitan hoto {cite}`yu2022assessing`. Wani □ angare na muhawara ya kasance game da saitin gwaji musamman da kwatancen likitan hoto. Amma matsala mafi zurfi ta kasance ta hanyar: tushen "aikin likitan hoto" da aka yi amfani da shi yana amfani da masu karatu kawai a □ ar□ ashin matsincin lokaci, yayin da likitancin hoto na asibiti galibi yana ha□ a shawara, kwatancen zuwa hotuna na baya, da damar mahallin asibiti — babu □ ayan wa□ annan samfurin ya samu.

Darasin ba shine samfuri ya yi kyau ko mara kyau ba, amma **kwatancen aiki yana bu□ atar □ ayyadewa da tsarin HITL**. Samfuri wanda ya wuce likitan hoto □ aya da ke karanta mai sanyi na iya har yanzu ya kasance mara daidai fiye da likitan hoto da ke amfani da sakamakon samfuri a matsayin ra'ayin biyu. Wa□ annan tsarin daban ne masu yanayin kuskure daban, kuma ha□ a su daban yana ba da sakamakon daban. :::

:::{admonition} Lakabin laushi a likitanci :class: important

Ayyukan AI likitanci da yawa sun koma ga amfani da **lakabin laushi** wa□ anda suke nuna rarrabawar ra'ayoyin □ wararru maimakon lakabi □ aya na "zinariya". X-ray na □ irji da aka sanya lakabi a matsayin 60% pneumonia / 40% atelectasis ta kwamitin likitocin hoto yana □ aukar bayanai fiye da za□ in tilas biyu. Samfurin da aka horar akan irin wa□ annan rarrabawar suna nuna daidaitar □ ididdigan iya yiwuwa mafi kyau da kuma □ ididdigan rashin tabbas mafi dacewa — kuma wannan rashin tabbas yana da ma'ana ta asibiti, tun da yana gaya wa likita lokacin da za su yi shawarwari, ba kawai abin da samfuri ya yi tsammani ba. :::

### Koyon Aiki don Yanayin Da Ba A Cika Ba

Koyon aiki yana da □ ima musamman don cututtuka mara cika da pathology mara cika, inda ko da tafkin da ba a yi lakabi ba yana □ unshi ka□ an yanayi masu kyau. Samfurin ta damari zai bata lokacin □ wararru yana yin lakabi ga yanayin mara kyau galibi.

Koyon aiki bisa rashin tabbas yana za□ ar lokuta na iyaka inda samfuri ya kasance mara tabbas — wa□ anda, don cututtuka mara cika, sukan zama yanayi masu kyau da mara kyaun na iyaka. Wannan yana tattara lokacin □ wararru inda yake da □ ima sosai. Ha□ in gwiwar horarwa mara daidaito na aji (tare da `class_weight='balanced'` ko irin haka) da za□ i bisa rashin tabbas aiki ne na yau da kullum don ayyukan gano pathology mara cika.

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## Lakabi na NLP na Asibiti

Rikodin lafiya na lantarki (EHRs) suna □ unshi arziki mai yawa na rubutun labari na asibiti: bayanin likita, takaitawar fita daga asibiti, rahotannin likitancin hoto, rahotannin pathology. Fitar da bayanai mai tsari daga wannan rubutu yana bu□ atar NLP — kuma NLP mai inganci yana bu□ atar bayanin horarwa da aka sanya lakabi.

**Ayyukan lakabi na NLP na asibiti na yau da kullum:** - **NER na asibiti:** Gano magungunan, allurai, ganewar cutar, hanyoyin, da alamun a cikin rubutu - **Gano musun:** "Babu shaidar pneumonia" da "An tabbatar da pneumonia" — bambance mai mahimmanci wanda yana da wuya

sosai - **Tunani na lokaci:** Bambanta yanayin yanzu daga tarihi ("tarihi na MI, ya gabatar da ciwo na □ irji") - **Cire gane-gane:** Cire Bayanan Lafiya da aka Kare (PHI) don ba da damar raba bayanai

**Cire PHI** duka aikin lakabi ne da bu□ atar sarrafa bayanai. □ ar□ ashin HIPAA (Amurka) da GDPR (EU), ba za a iya raba bayanai na lafiya ba tare da cire ko □ oye alamun marasa lafiya ba. Kayan aikin cire da atomatik suna wanzuwa amma ba cikakke ba; duba dan adam na sakamakon da atomatik ya □ ir□ ira aiki ne na yau da kullum, kuma ha□ arin ha□ ari ba daidai ba ne: kararan karya mara kyau (PHI da aka rasa) yana □ ir□ ira hadari na doka, yana mai da iyakar mai ra'ayin kiyayewa ta dole.

### i2b2 / n2c2 a matsayin Template

Ayyukan i2b2 (Bayani don Ha□ a Biology da Gadon) da na n2c2 maye gurbinsu (□ alubalen NLP na Asibiti na □ asa) sun sakar da jeri na saitin bayanai na NLP na asibiti da □ wararru suka sanya lakabi. Wa□ annan suna nuna duka yiwuwa da ku□ i: □ o□ arin lakabi galibi suna ha□ a □ ungiyoyin □ wararru na yanki da ke aiki tsawon watanni da yawa, yana sanya lakabi ga takardun da yawa kowace □ alubale. Saitin bayanai na n2c2 sun haifar da ci gaba mai sauri ainihin saboda sun warware matsalar sarrafa raba bayanai (cire gane-gane + yarjejeniyar cibiya), ba kawai matsalar lakabi ba.

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## La'akari na Doka

AI na likitanci yana □ ar□ ashin kulawa ta doka a mafi yawan □ asashe.

**FDA (Amurka):** Software ta AI/ML a matsayin Na'urar Likitanci (SaMD) yana bu□ atar amincewar kafin kasuwa ko izini. Shirin Aiki na AI/ML na FDA na 2021 yana jaddada **tsare-tsaren sarrafa sauyi da aka riga aka □ ayyade** — rubuta yadda za a sabunta samfuri da yadda wa□ annan sabuntar za a tabbatar da su kafin aikawa. Samfuri wanda ke ci gaba da koyo daga ra'ayin asibiti shine, □ ar□ ashin wannan tsari, na'ura daban bayan kowane sabuntawa kuma na iya bu□ atar sake tabbatarwa.

**Alamar CE (Turai):** Na'urori na likitanci gami da tsarin AI dole ne su bi Doka na Na'ura Likitanci (MDR). MDR yana bu□ atar kimantawa ta asibiti, sa ido bayan kasuwa, da takardun bayanai da aka yi amfani da su don horarwa da tabbatarwa.

**Abubuwa masu amfani na HITL:** Tsarin doka yana bu□ atar takardun bayyananne na tsarin lakabi, cancantar mai sanya lakabi, amincin tsakanin mai kimantawa, da kowane canjin bayanin horarwa. Wannan ba nauyin hukuma ba ne — shi ne hanyar bincike wanda ke ba da damar likita ya fahimci wa□ anne bayanin horarwa suka samar da □ abi'ar samfuri ta yanzu, kuma ana bu□ ata ta doka. Tsarin HITL wa□ anda ke □ aukar lakabi a matsayin tsari na bisa □ asa yana □ ir□ ira hadari na doka wanda galibi yana bayyanawa kawai a lokacin mafi muni.

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## Lakabi na Bayanai na Kimiyya

Bayan kiwon lafiya, HITL ML yana taka rawa mai □ aruwa kuma ba a yabe shi ba a cikin binciken kimiyya, inda kalubalen lakabi galibi yana ha□ a □ warewa ta yanki da sikeli.

### Taurari: Galaxy Zoo

Galaxy Zoo {cite}`lintott2008galaxy` ya □ ar□ aza rarrabuwa ta morphological na taurari daga Sloan Digital Sky Survey zuwa masanan kimiyya na al'umma. Aikin asali ya tattara rarrabuwa fiye da miliyan 40 daga masu aikin sa kai fiye da 100,000, yana nuna cewa □ ar□ aza ta sizeli ta rarrabuwa na hoto na kimiyya yana yiwuwa lokacin da za a iya rushe aiki zuwa tambayoyi masu sau□ i da za a iya amsa ba tare da horo na □ wararru ba ("Shin wannan taurare yana da santsi ko yana da fasali?").

Kwarewa ta Galaxy Zoo ta samar da binciken hanyar guda biyu masu muhimmanci. Na farko, yarjejeniya tsakanin masanan kimiyya na al'umma da masanan taurari □ wararru ta kasance mai □ ari don lokuta bayyananne kuma ta bambanta ta tsari akan lokuta na iyaka — ainihin lokuta inda bambancen yana da muhimmanci ta kimiyya. Mafita ba ta watsi da bayanin kimiyyar al'umma akan lokuta na iyaka ba amma ta □ aukar rarrabawar amoshin masu aikin sa kai a matsayin lakabi mai laushi da ke □ oye rikitarwar morphological na gaske. Na biyu, mai rarraba da aka horar akan lakabin Galaxy Zoo ya wuce samfurin da aka horar don maimaita lakabi na □ wararru □ aya, saboda rarrabawar jama'a ta kamo rashin tabbas na gani na gaske wanda za□ in tilas na □ wararru □ aya ya ruguje.

### Genomics: Rarrabuwa ta Pathogenicity

Sanya lakabi ga sigar kwayar halitta — yanke shawara ko sigar yana da cuta, mara cutar, ko mara tabbas na muhimmanci — matsala ce mai ha□ ari ta NLP da hukuncin □ wararru. Bayanan sigar asibiti kamar ClinVar suna ha□ a fassarori na □ wararru daga labs da yawa masu gabatarwa, kuma rashin jituwa tsakanin labs yana yau da kullum. Ana amfani da koyon aiki don fifita wa□ anne sigogi suna bu□ atar duba □ wararru cikakke (bincike na wallafe-wallafen, kimantawa ta shaida aiki) da wa□ anne za a iya auto-classified ta shaida da ke wanzuwa. Sakamakon shine tsarin ha□ e inda mafi yawan sigogi suna sarrafawa ta logic ta atomatik, □ angaren yana bu□ atar duba □ wararru, kuma lokuta mafi wuya an yin alamar su don yarjejeniyar labs da yawa.

### Yanayi da Kimiyyar Duniya

Sanya lakabi ga hoton tauraron dan adam don sauyin amfani da □ asa, sare daji, fa□ in kankara, da hanyoyin guguwa yana ha□ a □ wararru a cikin aikin nunawa na nesa kuma fiye da kima dandamali na kimiyyar al'umma. Babban kalubalen HITL a wannan yanki na lokaci ne: lakabin da aka yi yau na iya zama tsohon yayin da duniya ke canzawa, kuma tabbatarwa na □ asa (bincike na filin) yana da tsada kuma yana da iyakancin nisan tafiya. Koyon aiki wanda ke fifita hotuna inda hasashen samfuri ya sabawa □ arin na jiki (misali, hasashen sare daji a yanki da aka sani yana da kariya) hanyar da ke da amfani don jagorar albarkatu masu iyaka na tabbatarwa na filin.

### Neuroscience: Connectomics

Sake gina da'irori na jijiya daga hoton microscope na lantarki — connectomics — yana bu□ atar lakabi matakin pixel na membranin neuron cikin □ imbin hoto masu yawa. Aikin Eyewire ya yi wasanin bidiyo da wannan aiki, yana kara da'awar 'yan wasa goma da yawa a cikin bin hanyoyin neuron ta hanyar □ angaren hoto 3D. Zanen wasanin bidiyo ya warware matsala musamman ta HITL: aikin yana bu□ atar kulawa ta ci gaba da tunani na sarari akan zaman mai tsawo, wanda ke haifar da □ udurin inganci a cikin lakabi na gargajiya. Rushe aiki zuwa sassan wasan da salon zamantakewa ya kiyaye shiga da ingancin mai sanya lakabi a sikalin da lakabi na □ wararru ba za a iya cimma shi ba.

---

## Sarrafa Masu Sanya Lakabi □ wararru

Lokacin da lakabi yana bu□ atar □ warewa mara cika, hanyoyin tarin aiki na yau da kullum (Babi na 13) ba su aiki ba.

**Damuwa ta asali** ita ce mutanen da za su iya samar da lakabin mafi inganci sune kuma mutanen da lokacin su yake da □ ima sosai kuma yana da iyakancin yawa. Kowane yanke shawara na zane a cikin tsarin lakabi na □ wararru ya kamata a kimanta shi □ ar□ ashin tambaya: shin wannan yana mai da mafi kyau amfani da lokacin □ wararru mara cika?

**Abin da wannan ke nufi a aiki:**

- **Pre-annotate da gangan.** Yi amfani da masu sanya lakabi na □ asa, samfurin ta atom-atik, ko tsarin bisa □ a'idoji don □ ir□ ira □ an takara wanda □ wararrun yana duba da gyarawa maimakon □ ir□ ira daga farko. Hukuncin □ wararru shine □ oshewa; ciyar da pre-annotation don gyara yana da sauri fiye da ro□ a su sanya lakabi daga allon fanko, da yanayin pre-annotation yana da isa don gyarawa ya zama mafi sauri fiye da farawa daga farko.
- **Tsara don hankalin □ wararru, ba gudu ba.** Mai amfani da lakabi da aka inganta don gudu mai □ ari (yanke shawara biyu biyu mai sauri, mahimman ma□ alli, □ arancin nuna alama) ya dace don tarin aiki. Lakabi na □ wararru galibi yana amfana daga mai amfani da shi mai arziki: kwatancen kusa da kusa da lokuta na baya, damar samun kayan tunani cikin sau□ i, filin amincewar lakabi, da damar yin alamar yanayin don tattaunawa. Wa□ annan suna jinkirtawa lakabi na mutum □ aya amma suna inganta inganci kuma suna rage bu□ atar sake lakabi.
- **Bi sahun salon mai sanya lakabi □ aya musamman.** Tare da □ aramin rukunin □ wararru, yana yiwuwa kuma yana da muhimmanci don bin yawan yarjejeniyar kowane mai sanya lakabi tare da kwamitin, yin alamar lokuta wa□ anda suka yi kama daban da tarihin su, kuma tattauna su a zaman daidaita na yau da kullum. Wannan ba sa ido ba ne — shi ne tsarin inganci □ aya da likitanci na asibiti ke amfani da shi don duba aiki, kuma □ wararru galibi suna amsa da kyau lokacin da aka tsara shi a matsayin ingancin gama gari maimakon kimantawa.
- **Zanen zaman yana da muhimmanci.** Lakabi na likitanci yana da □ alubale ta kwamitin tunani. Shaida daga likitancin hoto da pathology tana ba da shawara cewa yawan kuskure suna □ aruwa sosai bayan kusan mintuna 90 na karatu mai ci gaba, kuma mafita na minti ka□ an suna iya maido da hankali a wani □ angare. Mai amfani da lakabi wa□ anda ke tilasta duba hutu (ba tare da su iya watsi da su ba) yanke shawara ne mai sau□ i na zanen HITL tare da tasiri na inganci na gaske.

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## Tsarin Koyon Aiki na HITL don Hoto Likitanci

```python
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score
from sklearn.datasets import make_classification
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)

# Simulate a rare-pathology detection task
```

```
# 8% positive class (e.g., rare pathology)
X, y = make_classification(
    n_samples=5000, n_features=100,
    n_informative=20, n_redundant=10,
    weights=[0.92, 0.08],
    random_state=42
)
X_train, y_train = X[:4000], y[:4000]
X_test,  y_test  = X[4000:], y[4000:]

print(f"Training set positive prevalence: {y_train.mean():.1%}")

def run_medical_al(strategy, n_initial=50, budget=300):
    labeled = list(rng.choice(len(X_train), n_initial, replace=False))
    unlabeled = [i for i in range(len(X_train)) if i not in labeled]
    aucs = []

    while len(labeled) < n_initial + budget:
        model = LogisticRegression(max_iter=500, class_weight='balanced')
        model.fit(X_train[labeled], y_train[labeled])

        if len(labeled) % 30 == 0:
            preds = model.predict_proba(X_test)[:, 1]
            aucs.append(roc_auc_score(y_test, preds))

        X_pool = X_train[unlabeled]
        if strategy == 'uncertainty' and len(labeled) >= 10:
            probs = model.predict_proba(X_pool)
            entropy = -np.sum(probs * np.log(probs + 1e-9), axis=1)
            q = int(np.argmax(entropy))
        else:
            q = rng.integers(0, len(unlabeled))

        labeled.append(unlabeled.pop(q))

    return np.array(aucs)

aucs_al  = run_medical_al('uncertainty')
aucs_rnd = run_medical_al('random')
label_counts = np.arange(len(aucs_al)) * 30 + 50

plt.figure(figsize=(7, 4))
plt.plot(label_counts, aucs_al,  'o-',  color='#2b3a8f', linewidth=2, label='Uncertainty AL')
plt.plot(label_counts, aucs_rnd, 's--', color='#e05c5c', linewidth=2, label='Random baseline'
plt.xlabel("Expert labels obtained", fontsize=12)
plt.ylabel("AUROC", fontsize=12)
plt.title("Active Learning for Rare Pathology Detection", fontsize=13)
plt.legend(); plt.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('medical_al.png', dpi=150)
plt.show()
```

```
# Show how AL preferentially selects positive examples
# by checking which class the queried examples belong to
n_init = 50
labeled_al  = list(rng.choice(len(X_train), n_init, replace=False))
labeled_rnd = labeled_al.copy()
unlabeled_al  = [i for i in range(len(X_train)) if i not in labeled_al]
unlabeled_rnd = unlabeled_al.copy()

model = LogisticRegression(max_iter=500, class_weight='balanced')
model.fit(X_train[labeled_al], y_train[labeled_al])
probs = model.predict_proba(X_train[unlabeled_al])
entropy = -np.sum(probs * np.log(probs + 1e-9), axis=1)
top50_al  = [unlabeled_al[i] for i in np.argsort(entropy)[-50:]]
top50_rnd = list(rng.choice(unlabeled_rnd, 50, replace=False))

pos_rate_al  = y_train[top50_al].mean()
pos_rate_rnd = y_train[top50_rnd].mean()
print(f"\nPositive rate in next 50 queries:")
print(f"  Uncertainty AL: {pos_rate_al:.1%}  (base rate: {y_train.mean():.1%})")
print(f"  Random:         {pos_rate_rnd:.1%}")
print(f"  AL queries {pos_rate_al/y_train.mean():.1f}x more positives than random")
```

**See also:** Tarin aiki na Galaxy Zoo: {cite}`lintott2008galaxy`. Aikin likita na CheXNet: {cite}`rajpurkar2017chexnet`. Ingancin hoton da gano cutar ta tallafi na AI: {cite}`yu2022assessing`. Hanyar lakabi na NLP na asibiti: {cite}`pustejovsky2012natural`. Don jagora na shirin aiki na FDA AI/ML, duba takardun FDA da aka wallafa (2021).

# Dandamali da Kayan Aikin Lakabi

Dandamali na lakabi shine yanayin inda ra'ayin dan adam ke zama bayanai. Dandamali mai kyau yana □ ara gudu, yana rage kurakurai, yana kiyaye sarrafa inganci, kuma yana sa tsarin lakabi mai sarrafa a sikeli. Za□ ar dandamali da ya dace — kuma sanin lokacin da za a gina da lokacin da za a saya — yanke shawara mai muhimmanci ne a kowane aikin HITL.

---

## Shimfidar Dandamali na Lakabi

Kasuwar kayan aikin lakabi ta girma sosai a cikin 'yan shekarun nan, tana jawo ta bu□ atar kamfanoni don bayanan horarwa na ML. Kayan aiki suna fa□ a□ awa daga ayyuka masu sarrafa gaba □ aya zuwa tsarin bu□ a□ □ en tushe na masaukin kai.

### Dandamali Bu□ a□ □ en Tushe

**Label Studio** shine mafi shahararren dandamali na lakabi bu□ a□ □ en tushe, yana tallafa wa lakabi na rubutu, hotuna, sauti, bidiyo, da bayanai na jeri ta amfani da tsarin tsari na XML mai ha□ in gwiwa. Za a iya masaukin kai kuma yana ha□ a da kwangarai na ML don koyon aiki. □ arfin muhimmanci: sassauci, tallafin al'umma, da damar saka hasashen ML na gani don pre-annotation.

**Prodigy** (daga masu □ ir□ ira spaCy) kayan aikin lakabi mai mai da hankali kan aikin ne wanda aka tsara don ayyukan NLP. Tsarinsa na cigaban bayanai yana aika misalai □ aya □ aya kuma yana tallafa wa koyon aiki daga akwatin. Mai amfani da lakabi yana da □ asa amma yana da sauri — an tsara shi don yin mafi yawan gudu na lakabi. Prodigy software ne mai biyan ku□ i amma ana amfani da shi sosai a binciken NLP.

**CVAT (Kayan Aikin Lakabi na Gane Hoto ta Kwamfuta)** shine kayan aikin bu□ a□ □ en tushe na farko don lakabi na CV, tare da tallafi mai □ arfi don gano, rarrabuwa, da lakabi na bidiyo. Asali an ha□ aka shi a Intel, CVAT yana tallafa wa algorithms na bin sawu, lakabi na tsarin kashi, da ha□ e-ha□ en algorithm na □ angare na uku.

**Doccano** yana mai da hankali kan ayyukan sanya lakabi jeri (NER, fitar da dangantaka, rarrabuwa rubutu). Mai amfani da shi mai sau□ i na yanar gizo yana sa shi ya samu damar □ ungiyoyi ba tare da albarkatu musamman na injiniyan bayanai ba.

### Dandamali Na Kasuwanci

**Scale AI** yana ba da ayyukan lakabi masu sarrafa daga farko: □ ungiyar aiki ta dan adam, sarrafa inganci, da ha□ e-ha□ en API. Mai □ arfi musamman don tuka kai tsaye, robots, da lakabi 3D masu rikitarwa. Ku□ i yana dogara akan rikitarwan aiki da yawa.

**Labelbox** dandamali cikakke ne don yin kula da bayanai, lakabi, da sanya lakabi mai tallafi na ML. Fasalun kamfanin masu □ arfi: sarrafa aiki, hanyoyin inganci, zagaye na ra'ayin samfuri, da ha□ e-ha□ e tare da dandamali na ML manyan (SageMaker, Vertex AI, Azure ML).

**Appen** (tsohon Figure Eight / CrowdFlower) yana aiki da babban □ ungiyar aiki na lakabi na duniya tare da kayan aiki. Za□ i mai kyau lokacin da yawa da sarrafa □ ungiyar aiki sune damuwa ta farko.

**Surge AI** yana mai da hankali kan masu sanya lakabi □ wararru kuma yana da □ arfi don ayyuka wa□ anda ke bu□ atar ilimin yanki ko hukunci mai laushi.

**Humanloop** yana □ wararre a tattara ra'ayin LLM — lakabi na fifiko, tattara bayanin RLHF, da kimantawa ta samfuri.

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## Kwatancen Fasalin Dandamali na Lakabi

| Fasali | Label Studio | Prodigy | CVAT | Labelbox | Scale AI |
|---|---|---|---|---|---|
| Lasisin | Bu□ a□ □ en tushe | Na kasuwanci | Bu□ a□ □ en tushe | Na kasuwanci | Na kasuwanci |
| Masaukin | Kai / girgije | Kai | Kai / girgije | Girgije | Sarrafa |
| Lakabi rubutu | □ | □ | — | □ | □ |
| Lakabi hoto | □ | Iyakantacce | □ | □ | □ |
| Lakabi bidiyo | □ | — | □ | □ | □ |
| Ha□ in koyon aiki | □ | □ | Iyakantacce | □ | □ |
| Pre-annotation ta tallafi na ML | □ | □ | □ | □ | □ |
| Hanyoyin sarrafa inganci | Asali | Asali | Asali | Ci gaba | Ci gaba |
| API / Samun dama ta shirye-shirye | □ | □ | □ | □ | □ |
| Sarrafa □ ungiyar aiki | — | — | — | Iyakantacce | □ |

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## Lakabi a matsayin Lambar Shirye-shirye

Muhimmin abu amma galibi ba a kula da shi a cikin ababen more rayuwa na lakabi shine **sarrafa sigar don lakabi da schemas na lakabi**. □ aukar lakabi a matsayin lambar shirye-shirye yana nufi:

**Zane-farko na schema.** Taxonomy na lakabi da □ a'idojin lakabi suna bayyananne a cikin fayil □ in tsara na sigar (YAML ko JSON) kafin lakabi ya fara. Canjin schema yana □ ir□ ira sigar sabon.

**Sigar lakabi.** Ana ajiye lakabi tare da ha□ i zuwa sigar schema da aka yi su □ ar□ ashinsa. Wannan yana ba da damar bincike, komawa, da kwatanta lakabi cikin sigar schema.

**Tsarin da ake iya maimaita su.** Tsarin lakabi — daga bayanai mentah zuwa lakabin shirye don horarwa — ya kamata a iya maimaita su daga lambar. Samfurin pre-annotation, tacewar inganci, logic ta ha□ awa, da rarraba bayanai duka ya kamata a rubuta su.

```yaml
# Example: Label Studio annotation schema (text classification)
label_config: |
  <View>
    <Text name="text" value="$text"/>
    <Choices name="sentiment" toName="text" choice="single">
      <Choice value="positive"/>
      <Choice value="negative"/>
      <Choice value="neutral"/>
      <Choice value="mixed"/>
    </Choices>
  </View>
schema_version: "2.1.0"
task_type: text_classification
guidelines_version: "guidelines_v3.pdf"
```text
Tushen bayanai
    |
    v
Tafkin ba a yi lakabi ba -->  Dandamali na lakabi --> Saitin bayanai da aka yi lakabi
    ^                             |                              |
    |                             | (Mai tallafi na ML)          v
Dabarun tambaya <-----------------+                        Gudu na horarwa
na koyon aiki                                                    |
    ^                                                            v
    +------------ Samfurin da aka horar <---------- Kimantawa
                                                          |
                                                   Aikawa &
                                                   sa ido
```
```

Manyan wurin ha□ e: 1. **Shigar bayanai:** Bayanai ba a yi lakabi ba yana gudana ta atomatik daga gidan ajiya na bayanai zuwa dandamali na lakabi 2. **Aika samfuri:** Ana aika samfuri mafi kyau na yanzu zuwa dandamali na lakabi don pre-annotation da □ ididdiga koyon aiki 3. **Fitarwa:** An fitar da lakabi da aka □ are a cikin tsarin dacewa da tsarin horarwa (COCO JSON, saitin bayanai na Hugging Face, dkk.) 4. **Zagayen ra'ayi:** Ana jagorancin kurakuran samfuri na samarwa koma zuwa dandamali na lakabi don gyarawa

> **See also:** Don takardun Label Studio da ha□ in koyon aiki: https://labelstud.io.

Don Prodigy: https://prodi.gy. CVAT: https://cvat.ai. Don cikakken kwatancen kayan aiki na lakabi, duba {cite}`monarch2021human`, Babi na 7.

# Tarin Aiki da Kula da Inganci

Lokacin da ayyukan lakabi suka yi sau□ i sosai har masu amfani da ba □ wararru ba za su iya yi, dandamali na tarin aiki suna ba da damar samun babbar □ ungiyar aiki ta lakabi mai bu□ ata a bu□ ata a ku□ i □ arami kowace item. Gina saitin bayanai masu inganci daga taron aiki yana bu□ atar zanen aiki mai hankali, waddattaccen tsattsauran maimaitawa, da sarrafa inganci mai □ arfi.

---

## Dandamali na Tarin Aiki

**Amazon Mechanical Turk (MTurk)** shine kasuwar tarin aiki ta asali, wanda aka □ addamar a 2005. Masu aiki ("Turkers") suna cika □ ananan ayyuka (HITs) da masu bu□ ata suka wallafa. Bincike na 2018 ya sami matsakaicin albashi na sa'a na gaske don Turkers na kusan $2/hr — □ asa da □ arancin albashi a □ asashe da yawa masu samun ku□ i sosai {cite}`hara2018data` — damuwa ta da'a da aka magance daga baya a Babi na 15. MTurk yana mafi kyau don ayyuka masu sau□ i tare da ma'aunin bayyananne, mai iya tabbatarwa.

**Prolific** dandamali ne na tarin aiki na ilimi wanda ke tilasta ma'aunin biyan ku□ i na □ aranci (a halin yanzu kusan £9/hr, kusan $11/hr, kamar yadda aka bayyana a cikin jagoran wallafa na Prolific), yana tattara mahalarta ta halaye na dimokra'iyya, kuma yana kiyaye kwamitin masu aiki wa□ anda suka yarda su shiga bincike. Ana son shi don binciken kimiyyar zamantakewa da ayyuka wa□ anda ke bu□ atar wakilci.

**Appen** (da makamancin su: Telus International, iMerit) yana ba da □ ungiyar aiki na lakabi masu sarrafa tare da sarrafa inganci, ana amfani da su don ayyuka masu rikitarwa fiye da kima da ayyukan kamfanin.

**Al'ummu na musamman.** Don ayyuka musamman na yanki, al'ummu na sha'awa na yanki na iya ba da lakabi mai inganci: Galaxy Zoo don taurari, eBird don nau'in tsuntsu, Chess Tempo don lakabi na matsayi na chess.

---

## Zanen Aiki don Tarin Aiki

### Rushe Ayyuka Masu Rikitarwa

Ya kamata a rushe ayyuka masu rikitarwa zuwa □ ananan ayyuka masu sau□ i, ma'anar bayyananne. Maimakon ro□ a masu aiki su yi cikakken lakabi na takarda, tambaya musu tambaya □ aya mai mai da hankali a lokaci guda: "Shin wannan jumla ta □ unshi sunan mutumin?" ko "□ ididdiga bayyanar wannan fassara a sikeli na 1–5."

**Fa'idodin rushe aiki:** - □ arancin bu□ ata ta kwamitin tunani kowace aiki □ □ arancin gajiya, inganci mafi □ ari - Za a iya sarrafa ingancin kowane □ aramin aiki daban - Mafi sau□ in bincike da gyarawa

### Mahimmancin Umarni

Mafi yawan hasashen ingancin tarin aiki shine ingancin umarni. Umarni masu kyau na aiki: - Bayyana *manufar* aikin a cikin jumla □ aya - Ba da ma'ana mai bayyana, mara rikitarwa na kowane rukuni - Ba da misalai 3–5 masu aiki (musamman lokuta na iyaka) - Ba su fi tsawo fiye da masu aiki za su karanta ta gaske ba (kalmomi < 300 don ayyuka masu sau□ i)

Gudanar da **bincike na gwaji** (masu aiki 10–50, ayyuka 20–100) kafin sikeli. Bincika rashin jituwa na gwaji; mafi yawansu suna nuna rikitarwa a cikin umarni wa□ anda za a iya gyara.

### Tambayoyin Ma'aunin Zinariya

Saka **tambayoyin ma'aunin zinariya** — ayyuka tare da amoshin daidai da aka sani — cikin duk jerin aiki. Ana cire masu aiki wa□ anda suka gaza tambayoyin zinariya □ asa da iyaka daga aikin.

```
import numpy as np
from scipy.stats import binom

rng = np.random.default_rng(42)

def simulate_gold_screening(n_workers=100, gold_per_batch=5,
                            p_good_worker=0.7, good_acc=0.92,
                            bad_acc=0.55, threshold=0.70):
    """
    Simulate quality screening via gold questions.
    Returns: precision and recall of identifying bad workers.
    """
    worker_quality = rng.choice(['good', 'bad'], size=n_workers,
                                p=[p_good_worker, 1 - p_good_worker])
    results = {'tp': 0, 'fp': 0, 'tn': 0, 'fn': 0}

    for q in worker_quality:
        acc = good_acc if q == 'good' else bad_acc
        n_correct = rng.binomial(gold_per_batch, acc)
        passed = (n_correct / gold_per_batch) >= threshold
        if q == 'bad' and not passed:  results['tp'] += 1
        if q == 'good' and not passed: results['fp'] += 1
        if q == 'bad' and passed:      results['fn'] += 1
        if q == 'good' and passed:     results['tn'] += 1

    tp, fp, fn = results['tp'], results['fp'], results['fn']
    precision = tp / (tp + fp + 1e-6)
    recall    = tp / (tp + fn + 1e-6)
    return precision, recall

# Vary gold question count
gold_counts = [3, 5, 8, 10, 15, 20]
precisions, recalls = [], []
```

```
for g in gold_counts:
    p_list, r_list = [], []
    for _ in range(50):
        p, r = simulate_gold_screening(gold_per_batch=g)
        p_list.append(p); r_list.append(r)
    precisions.append(np.mean(p_list))
    recalls.append(np.mean(r_list))

import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(gold_counts, precisions, 'o-', color='#2b3a8f', label='Precision', linewidth=2)
ax.plot(gold_counts, recalls,    's--', color='#0d9e8e', label='Recall',    linewidth=2)
ax.set_xlabel("Gold questions per batch", fontsize=12)
ax.set_ylabel("Screening performance", fontsize=12)
ax.set_title("Worker Screening via Gold Standard Questions", fontsize=13)
ax.legend(fontsize=11); ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('gold_screening.png', dpi=150)
plt.show()

p, r = simulate_gold_screening(gold_per_batch=5)
print(f"5 gold questions: Precision={p:.3f}, Recall={r:.3f}")
p, r = simulate_gold_screening(gold_per_batch=10)
print(f"10 gold questions: Precision={p:.3f}, Recall={r:.3f}")
```

---

## Samfurin Lissafi don Ha□ awa ta Lakabi

Za□ in rinjaye asali ne amma yana watsi da bambancin daidaito na mai sanya lakabi. Samfurin lissafi na iya yin mafi kyau.

### Tsarin Dawid-Skene

**Tsarin Dawid-Skene (DS)** {cite}`dawid1979maximum` yana ha□ a □ iyasin: - **Lakabi na gaskiya** $z_i$ don kowane item $i$ - **Matrix na rikicewa** $\pi_j^{(k,l)}$ don kowane mai sanya lakabi $j$: iya yiwuwar cewa mai sanya lakabi $j$ yana yin lakabi ga item mai ajin gaskiya $k$ a matsayin aji $l$

Algorithm EM yana maimaita: - **E-mataki:** Da matrices na rikicewa na mai sanya lakabi da aka ba, □ ididdiga iya yiwuwar bayar da kowane lakabi na gaskiya - **M-mataki:** Da □ iyasin lakabi na item da aka ba, sabunta matrices na rikicewa na mai sanya lakabi

```
import numpy as np
from scipy.special import softmax

def dawid_skene_em(annotations, n_classes, n_iter=20):
    """
    Simplified Dawid-Skene EM for binary classification.
    annotations: dict {item_idx: [(annotator_idx, label), ...]}
    Returns: estimated true labels and annotator accuracies.
    """
    items = sorted(annotations.keys())
```

```
    n_items = len(items)
    annotators = sorted({a for anns in annotations.values() for a, _ in anns})
    n_annotators = len(annotators)
    ann_idx = {a: i for i, a in enumerate(annotators)}

    # Initialize: majority vote
    T = np.zeros((n_items, n_classes))  # soft label estimates
    for i, item in enumerate(items):
        for _, label in annotations[item]:
            T[i, label] += 1
        T[i] /= T[i].sum()

    # Confusion matrices: shape (n_annotators, n_classes, n_classes)
    PI = np.ones((n_annotators, n_classes, n_classes)) * 0.5

    for _ in range(n_iter):
        # M-step: update confusion matrices
        PI = np.zeros((n_annotators, n_classes, n_classes)) + 1e-6
        for i, item in enumerate(items):
            for ann, label in annotations[item]:
                a = ann_idx[ann]
                for k in range(n_classes):
                    PI[a, k, label] += T[i, k]
        PI /= PI.sum(axis=2, keepdims=True)

        # E-step: update soft label estimates
        T = np.zeros((n_items, n_classes))
        for i, item in enumerate(items):
            log_p = np.zeros(n_classes)
            for ann, label in annotations[item]:
                a = ann_idx[ann]
                log_p += np.log(PI[a, :, label] + 1e-10)
            T[i] = softmax(log_p)

    return {item: T[i] for i, item in enumerate(items)}, PI

# Simulate crowdsourcing scenario
rng = np.random.default_rng(42)
N_ITEMS, N_ANNOTATORS, ANNS_PER_ITEM = 200, 10, 3
true_labels = rng.integers(0, 2, N_ITEMS)
# Annotator accuracies: 3 good (0.9), 4 medium (0.75), 3 noisy (0.6)
accuracies = [0.90]*3 + [0.75]*4 + [0.60]*3

annotations = {}
for i in range(N_ITEMS):
    anns_for_item = []
    chosen = rng.choice(N_ANNOTATORS, ANNS_PER_ITEM, replace=False)
    for a in chosen:
        acc = accuracies[a]
        label = true_labels[i] if rng.random() < acc else 1 - true_labels[i]
        anns_for_item.append((a, int(label)))
```

```
    annotations[i] = anns_for_item

# Run Dawid-Skene
soft_labels, confusion = dawid_skene_em(annotations, n_classes=2, n_iter=30)
ds_preds = {i: int(np.argmax(soft_labels[i])) for i in range(N_ITEMS)}

# Compare with majority vote
mv_preds = {}
for i in range(N_ITEMS):
    votes = [l for _, l in annotations[i]]
    mv_preds[i] = int(np.round(np.mean(votes)))

ds_acc = np.mean([ds_preds[i] == true_labels[i] for i in range(N_ITEMS)])
mv_acc = np.mean([mv_preds[i] == true_labels[i] for i in range(N_ITEMS)])

print(f"Majority vote accuracy:  {mv_acc:.3f}")
print(f"Dawid-Skene accuracy:    {ds_acc:.3f}")
print(f"\nEstimated annotator accuracies (diagonal of confusion matrix):")
for a in range(N_ANNOTATORS):
    est_acc = np.mean([confusion[a, k, k] for k in range(2)])
    print(f"  Annotator {a}: estimated={est_acc:.3f}, true={accuracies[a]:.2f}")
```

### MACE

**MACE (□ iyasin Cancantar Masu Sanya Lakabi Da Yawa)** {cite}`hovy2013learning` tsari ne na iya yiwuwar mai sabani wanda ke wakilta ta fili mai sanya lakabi ya zama spammer (sanya lakabi ta damari) da lakabi mai cancantar aiki. Mai sanya lakabi ko yana ba da lakabi mai ma'ana (da iya yiwuwar $1 - \text{spam}_j$) ko lakabi ta damari (da iya yiwuwar $\text{spam}_j$). Wannan tsarin mai sassa biyu galibi yana da daidaitar □ ididdigan mafi kyau fiye da Dawid-Skene don yanayin tarin aiki inda wasu masu sanya lakabi sune masu spamming kawai.

---

## Tsattsauran Maimaitawa da Dabarun Ha□ awa

Adadin mafi kyau na masu sanya lakabi kowace item yana dogara akan wahalar aiki da ingancin mai sanya lakabi:

- **Ayyuka masu sau□ i tare da masu sanya lakabi □ wararru:** Masu sanya lakabi 1–2 kowace item galibi ya isa
- **Ayyuka matsakaici tare da masu sanya lakabi masu horo:** Masu sanya lakabi 3 + za□ in rinjaye
- **Ayyuka masu wuya/na son zuciya tare da masu aiki na taron aiki:** Masu sanya lakabi 5–7 + Dawid-Skene

Ra'ayin muhimmanci: tsattsauran maimaitawa yana da □ ima sosai lokacin da daidaito na mai sanya lakabi ya kasance □ arami. Don masu sanya lakabi masu daidaito $p$, daidaito na za□ in rinjaye tare da $n$ masu sanya lakabi shine:

$$P(\text{MV correct}) = \sum_{k>n/2}^{n} \binom{n}{k} p^k (1-p)^{n-k}$$

Don $p = 0.70$, □ ara mai sanya lakabi na uku yana □ ara daidaito na za□ in rinjaye daga 70% zuwa 78%; don $p = 0.90$, riba daga mai sanya lakabi na uku yana da □ arami (daga 90% zuwa 97%).

> **See also:** Tsarin Dawid-Skene: {cite}`dawid1979maximum`. MACE: {cite}`hovy2013learning`. Don duba cikakke na tarin aiki don NLP: {cite}`snow2008cheap`. Da'a da biyan ku□ i mai gaskiya: duba Babi na 15.

# Kimantawa da Ma'aunai

Sanin ko tsarin HITL □ inka yana aiki yana bu□ atar fiye da auna daidaito na samfuri. Kana bu□ atar sanin ko kana samun darajar daga kasafin ku□ in lakabinka, ko samfuri ya fi dacewa gaske da nufin dan adam, da kuma ko ra'ayin dan adam □ ari zai ci gaba da inganta abubuwa. Wannan babi ya □ unshi shimfidar cikakkiyar kimantawa a cikin yanayi na HITL.

---

## Ma'aunai na Tsakiyar Samfuri

Ma'aunai na ML na yau da kullum suna aiki kai tsaye ga tsarin HITL, tare da wasu abubuwa masu muhimmanci.

### Ma'aunai na Rarrabuwa

**Daidaito** yana dacewa lokacin da aji suke daidaito kuma duk kurakurai suna da ku□ i □ aya. A cikin yanayi na HITL, ko da yake, saitin gwaji da aka yi lakabi na iya kasancewa mara daidaito ta dabarun tambaya (tambayoyi na koyon aiki ba su daidai ba), yana mai da □ iyasin daidaito mara aminci.

**Maki F1** shine matsakaicin harmonic na daidaito da koma-ga-asali, wanda ya dace don aji mara daidaito. A cikin mahallin HITL, daidaito da koma-ga-asali duka na iya yin muhimmanci daban-daban dangane da rashin daidaito na ku□ i tsakanin kararan karya mara kyau da kararan karya mai kyau.

**AUROC** yana auna damar samfuri don bambanta tsakanin aji ba tare da la'akari da iyaka ba — muhimmanci don ayyuka masu ma'ana ta □ ididdigan iya yiwuwa kamar tantancewa na likitanci.

**Daidaitar □ ididdigan** yana auna yadda iya yiwuwar da aka hasashe suka dace da yawan gwaji na gwaji. A cikin tsarin HITL, samfurin da aka horar akan saitin da aka yi lakabi mara daidaito (daga koyon aiki) na iya kasancewa mara daidai ko da daidai ne.

### Ma'aunai ta Samfurin Generative

Don samfurin harshe da tsarin generative, kimantawa yana da wuya ta asali. Babu wani metric □ in ta atomatik □ aya wanda ke kama inganci:

- **BLEU / ROUGE / METEOR:** Ma'aunai bisa tunani don fassara da takaitawa. Yana dacewa da □ arancin daidaito da hukunci na inganci na dan adam don □ ir□ ira tsawo.
- **Perplexity:** Yana auna yadda samfuri ke hasashen rubutu da aka □ aure. Yanayin dole amma ba ya isa don inganci.
- **BERTScore:** Kamancin bisa embedding zuwa tunani. Yana dacewa mafi daidaito da hukunci na dan adam fiye da ma'aunai n-gram.

- **Kimantawar dan adam:** Ma'aunin zinariya. Duba Sashe na 14.3.

---

## Ma'aunai na Ingancin Lakabi

Kimantawar HITL ya kamata kuma ta auna ko ake amfani da ra'ayin dan adam yadda ya kamata.

### Curves na Koyo

**Curve na koyo** yana zana aiki na samfuri a matsayin ayyuka na yawan misalin da aka yi lakabi. Curve mai tsananin girma (ingancin da sauri tare da lakabin ka□ an) yana nuna cewa dabarun lakabi suna za□ ar misalai masu bayanai. Curve mai filefilen yana nuna cewa lakabi □ ari yana ba da raguwar dawo.

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import f1_score, roc_auc_score
from sklearn.model_selection import StratifiedShuffleSplit

rng = np.random.default_rng(42)
X, y = make_classification(n_samples=5000, n_features=30, n_informative=15,
                           weights=[0.8, 0.2], random_state=42)
X_test, y_test = X[4000:], y[4000:]
X_pool, y_pool = X[:4000], y[:4000]

label_sizes = [20, 40, 60, 100, 150, 200, 300, 400, 600, 800, 1000, 1500, 2000]
metrics = {'f1': [], 'auc': []}

for n in label_sizes:
    idx = rng.choice(len(X_pool), n, replace=False)
    clf = LogisticRegression(max_iter=500, class_weight='balanced')
    clf.fit(X_pool[idx], y_pool[idx])
    preds = clf.predict(X_test)
    probs = clf.predict_proba(X_test)[:, 1]
    metrics['f1'].append(f1_score(y_test, preds))
    metrics['auc'].append(roc_auc_score(y_test, probs))

# Fit learning curve: performance ≈ a - b/sqrt(n)
from scipy.optimize import curve_fit

def learning_curve_fn(n, a, b):
    return a - b / np.sqrt(n)

popt_f1, _ = curve_fit(learning_curve_fn, label_sizes, metrics['f1'], p0=[0.9, 2])
popt_auc, _ = curve_fit(learning_curve_fn, label_sizes, metrics['auc'], p0=[0.95, 1])

n_smooth = np.linspace(20, 3000, 200)
```

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(11, 4))

ax1.scatter(label_sizes, metrics['f1'], color='#2b3a8f', zorder=5, s=40)
ax1.plot(n_smooth, learning_curve_fn(n_smooth, *popt_f1), '--', color='#e05c5c',
         label=f'Fit: {popt_f1[0]:.3f} - {popt_f1[1]:.1f}/√n')
ax1.set_xlabel("Labeled examples"); ax1.set_ylabel("F1 score")
ax1.set_title("Learning Curve: F1"); ax1.legend(); ax1.grid(alpha=0.3)

ax2.scatter(label_sizes, metrics['auc'], color='#0d9e8e', zorder=5, s=40)
ax2.plot(n_smooth, learning_curve_fn(n_smooth, *popt_auc), '--', color='#e05c5c',
         label=f'Fit: {popt_auc[0]:.3f} - {popt_auc[1]:.2f}/√n')
ax2.set_xlabel("Labeled examples"); ax2.set_ylabel("AUROC")
ax2.set_title("Learning Curve: AUROC"); ax2.legend(); ax2.grid(alpha=0.3)

plt.tight_layout()
plt.savefig('learning_curves.png', dpi=150)
plt.show()

# Estimate the annotation budget needed to reach a target performance
target_f1 = 0.80
n_needed = (popt_f1[1] / (popt_f1[0] - target_f1)) ** 2
print(f"Estimated labels needed to reach F1={target_f1}: {n_needed:.0f}")
```

### Bincike na Dawo kan Zuba Jari (ROI)

ROI na ra'ayin dan adam yana amsa: don kowane lakabi □ ari, nawa aiki na samfuri yana inganta?

$$\text{ROI}(n) = \frac{\Delta\text{performance}(n)}{\text{cost per label}}$$

Yayin da samfuri ya nuna (kuma yayin da misalan mafi sau□ in koyo suka □ are), ROI galibi yana raguwa. Abubuwa masu amfani na aiki: ya kamata a sanya kasafin ku□ in lakabi a farkon, tare da lakabin fiye da kima da aka tattara a matakai na farko lokacin da ROI ya kasance mafi □ ari.

---

## Kimantawar Dan Adam

Don tsarin generative da ayyuka na son zuciya, kimantawar dan adam har yanzu ita ce ma'aunin zinariya.

### Kimantawa Kai Tsaye (DA)

Masu sanya lakabi suna □ ididdiga sakamakon akan sikeli na cikakke (misali, 1–100 don ingancin fassara, ko 1–5 don taimako na amsa). DA ta kasance ta daidaita a cikin kimantawar fassara ta injin (ma'aunai WMT).

**Mafi kyau na aiki don DA:** - Bazata tsarin sakamakon don hana anchoring - Yi amfani da adadin mai sanya lakabi masu isa kowace item (□ aranci 3–5) - Ha□ a sarrafa inganci (mis-

alai bayyananne masu kyau da mara kyau don kama masu □ ididdiga ba su kula ba) - Ruwaito yarjejeniya tsakanin masu sanya lakabi tare da maki ha□ e

### Kimantawa na Kwatancen

Masu sanya lakabi suna za□ ar tsakanin sakamakon biyu: "Wanne ya fi?" Hukunci na kwatancen suna da sauri da daidaito fiye da □ ididdigan cikakke (duba Babi na 8). **Tsarin □ ididdigan ELO** (an aro daga chess) yana canza sakamakon kwatancen biyu biyu zuwa matsayin inganci na ci gaba.

```
import numpy as np

def update_elo(rating_a, rating_b, outcome_a, k=32):
    """Update ELO ratings. outcome_a: 1=A wins, 0=B wins, 0.5=tie."""
    expected_a = 1 / (1 + 10 ** ((rating_b - rating_a) / 400))
    expected_b = 1 - expected_a
    new_a = rating_a + k * (outcome_a - expected_a)
    new_b = rating_b + k * ((1 - outcome_a) - expected_b)
    return new_a, new_b

# Simulate 5 model versions being compared pairwise
rng = np.random.default_rng(42)
true_quality = [0.60, 0.70, 0.75, 0.80, 0.85]  # underlying model quality
n_models = len(true_quality)
elo_ratings = {i: 1000.0 for i in range(n_models)}

for _ in range(500):  # 500 pairwise comparisons
    i, j = rng.choice(n_models, 2, replace=False)
    p_i_wins = true_quality[i] / (true_quality[i] + true_quality[j])
    outcome = 1.0 if rng.random() < p_i_wins else 0.0
    elo_ratings[i], elo_ratings[j] = update_elo(elo_ratings[i], elo_ratings[j], outcome)

print("ELO Rankings after 500 comparisons:")
sorted_models = sorted(elo_ratings.items(), key=lambda x: x[1], reverse=True)
for rank, (model_id, elo) in enumerate(sorted_models, 1):
    print(f"  Rank {rank}: Model {model_id}  ELO={elo:.1f}  True quality={true_quality[model_
```

### Gwajin □ abi'a (CheckList)

**CheckList** {cite}`ribeiro2020beyond` hanya ce don kimantawar □ abi'a ta tsari na samfurin NLP. Maimakon saitin gwaji ta damari, yana tsara lokuta na gwaji wa□ anda suna bincika iyawa musamman:

- **Gwajin Iyaka ta □ aranci (MFT):** Shin samfuri yana sarrafa lokuta masu sau□ i, bayyananne?
- **Gwajin Daidaito (INV):** Shin sakamakon samfuri yana canzawa lokacin da bai kamata ba (misali, lokacin sake rubuta)?
- **Gwajin Tsammani na Alkibla (DIR):** Shin sakamakon samfuri yana canzawa ta alkiblar da ake tsammani lokacin da shigarwa ta canza?

CheckList yana mai da kimantawar dan adam kai tsaye kuma mai aiki: maimakon lamba □ aya ta daidaito, yana ba da bayanan iyawa.

## Auna Dacewa da Nufin Dan Adam

Don tsarin RLHF, auna dacewa yana kalubale na kimantawa ta tsakiya.

**Kimantawar samfurin lada:** Daidaito na samfurin lada akan saitin gwaji na fifiko da aka □ aure. Ouyang et al. {cite}`ouyang2022training` sun ruwaito kusan daidaito 72% biyu biyu don samfurin lada na InstructGPT; a matsayin tunani na □ ana da aka sani, alkaluma a wannan □ asa ana ambaton su galibi don tsarin RLHF makamancin haka, ko da yake sakamakon suna bambanta sosai ta aiki da ingancin bayanai.

**Yawan nasara:** Da sigar samfuri biyu (misali, tushe na SFT da amsar RLHF da aka daidaita), □ aukar □ auren amsoshi na samfuri RLHF ya ci a cikin kwatancen biyu biyu na dan adam?

**GPT-4 a matsayin mai kimantawa:** Amfani da LLM mai □ arfi don kimanta amsoshi ya zama yau da kullum don sake zagaye mai sauri. Gilardi et al. {cite}`gilardi2023chatgpt` da Zheng et al. {cite}`zheng2023judging` suna nemo yarjejeniyar mai kimantawa na LLM tare da hukunci na dan adam da ke yawo kusan daga 0.7 zuwa 0.9 dangane da aiki — mai amfani don kwatancen A/B mai sauri, amma mara aminci sosai don gano son zuciya na sycophancy, □ oyayyen al'adu, ko matsalolin aminci.

**Gano sycophancy:** Auna ko samfuri yana canza amoshinsa bisa fifikon mai amfani da aka nuna (misali, "Ina tsammanin X daidai ne; me ka yi tsammani?"). Samfuri mai daidaito ya kamata bai zama mai sycophancy ba.

---

## Gwajin A/B a Cikin Tsarin Aiki

Don tsarin da ke aiki, kimantawar □ arshe shine **gwajin A/B**: jagorancin □ aukar □ angaren masu amfani zuwa sigar samfuri sabon kuma auna sakamakon □ asa.

Gwajin A/B yana ba da □ iyasin mara nuna son zuciya na ingancin samfuri a cikin mahallin aikawa na gaske, yana kama tasirin da kimantawar dakin bincike ya rasa (□ abi'ar mai amfani, rarrabawar yawan jama'a, lokuta na iyaka).

Kalubale: ma'aunai □ asa masu dacewa. Ma'aunai na shiga (danna, tsawon zaman) na iya lada da □ abi'ar mai yaudarar. Yawan cikawa na aiki ko binciken gamsuwar mai amfani sun fi dacewa amma suna da hayaniya.

> **See also:** Gwajin □ abi'a na CheckList: {cite}`ribeiro2020beyond`. Don hanyar kimantawa na RLHF, duba {cite}`ouyang2022training`. Don mafi kyau na aiki na kimantawar dan adam a MT: {cite}`graham2015accurate`. Don ka'idar curve na koyo: {cite}`mukherjee2003estimating`.

# Adalci, Nuna Bambanci, da Da'a

Tsarin da dan adam ke cikin zagaye yana gaji duka iyawar da kuma iyakokin mutanen da ke cikin su. Masu sanya lakabi suna kawo ilimi, hukunci, da □ ir□ ira ga aikinsu — amma kuma nuna son zuciya, gajiya, da mahallin zamantakewar rayuwarsu. Za□ u□ □ uka da muke yi a cikin tsara tsarin HITL — wane ya sanya lakabi, abin da ake ro□ ar su, yadda ake biyan su, da yadda ake amfani da aikinsu — suna da sakamako wa□ anda ke fa□ a□ awa fiye da ma'aunai na daidaito na samfuri.

Wannan babi yana magance girman da'a na HITL ML.

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## Hanyoyin Nuna Son Zuciya a Cikin Tsarin HITL

### Halaye na Dimokra'iyya na Masu Sanya Lakabi

Lakabi ba aikin neutral ba ne. Lakabin da masu sanya lakabi suke bayarwa suna nuna hangen nesansu, □ warewarsu, da mahallin al'adunsu. Lokacin da □ ungiyar aiki na lakabi yana da halaye na dimokra'iyya □ aya — kamar yadda galibi take, tare da aikin taron aiki da ke nuna son zuciya ga matasai, maza, da Yammaci — saitin bayanai da aka samu yana □ oye hangen nesa na wannan □ ungiya.

**Shaidar:** Bincike na saitin bayanai na lakabi na NLP sun sami cewa halaye na dimokra'iyya na masu sanya lakabi suna hasashen za□ u□ □ ukan lakabinsu don ayyuka na son zuciya (guba, ji, laifi). Saitin bayanai da masu aiki da ke Amurka da farko suka sanya lakabi suna □ oye al'adun Amurka wa□ anda ba sa fa□ a□ awa zuwa wasu yankuna {cite}`geva2019annotator`.

**Sakamako:** Samfurin da aka horar akan irin wa□ annan bayanai suna aiki da kyau don masu amfani wa□ anda suke kama da tafkin mai sanya lakabi kuma mafi □ aranci — ko ta hanyar nuna son zuciya — don masu amfani wa□ anda ba su yi ba.

**Rage:** Bambancita deliberate na □ ungiyar aiki; lakabi mai tsari (tabbatar da masu sanya lakabi daga rukunin halaye na dimokra'iyya masu dacewa suna ba da gudummawa ga ayyukan masu dacewa); bin sahun halaye na dimokra'iyya na masu sanya lakabi da tasirin su akan rarrabawar lakabi.

### Tsarin Tsara Tambaya

Yadda tambaya aka tsara yana shafawa amoshin da ta jawo. Idan ana ro□ a masu sanya lakabi "Shin wannan rubutu ne mai guba?" na iya amsa daban fiye da "Shin wannan rubutu zai cutar wa wani da yake cikin □ ungiyar da aka ambata?" Tsara yana □ oye □ auki a kan abin da ke da muhimmanci.

**Misali:** Lakabi na "harshen cin zarafi" akan kafofin yada labarai na zamantakewa yana bambanta sosai dangane da ko ana nuna masu sanya lakabi bayanin mahallin game da asalin

marubucin. Magana da ke kama da barazana a ke□ e na iya a gane shi a matsayin harshe da aka □ auka ko wasan kamfani na □ ungiya lokacin da aka ba da mahallin.

### Tasirin Dandamali

Dandamali da tsarin biyan ku□ i suna shafawa ingancin lakabi. Masu aiki da ake biya kowace aiki maimakon kowace sa'a suna da □ arfin gwiwa don aiki da sauri; wannan yana □ ara gudu amma yana rage inganci. Masu aiki da suka ji tsoron an toshe su daga dandamali don □ arancin daidaito na iya guji alamar "mara tabbas" kuma maimakon haka yin kiyasi, yana rufe rikitarwa ta gaske.

### Nuna Son Zuciya na Tabbatarwa da Anchoring

Masu sanya lakabi suna shafawa ta: - **Pre-annotation:** Hasashen samfuri da ake nuna ga masu sanya lakabi ana yarda da su fiye da a yi watsi da su, ko da lokacin da ba daidai ba - **Tasirin tsari:** Yin lakabi ga item □ aya a mahallin daban yana ba da amoshin daban - **Priming:** Abubuwa na baya a cikin aiki suna shafawa yadda ake yin lakabi ga abubuwa masu zuwa

---

## Adalci a Cikin Tsarin HITL

### Menene Adalci?

Adalci a ML ra'ayi ne mai jayayya tare da ma'anoni da yawa masu yiwuwa masu dacewa wa□ anda galibi suna sabawa da juna {cite}`barocas2019fairness`. Don manufofin HITL, girman mafi dacewa su ne:

**Wakilci:** Shin bayanin horarwa yana wakilta yawan jama'a wa□ anda samfuri zai shafe su?

**Daidaito na aiki:** Shin samfuri yana aiki daidai cikin rukunin halaye na dimokra'iyya dabandaban?

**Daidaito na lakabi:** Shin ana yin lakabi ga □ abi'oci □ aya daidai ba tare da la'akari da wanda yake yin su ba? (Bincike ya nuna cewa ba koyaushe haka ne ba — iri iri na abun ciki wani lokaci ana yin lakabi daban lokacin da aka danganta ga rukunin jinsi ko jima'i daban.)

### Koyon Aiki Mai Sani ga Adalci

Dabarun koyon aiki na yau da kullum suna mai da hankali kan rashin tabbas na samfuri, wanda ke mai da hankali kan misalin ajin rinjaye. Wannan na iya □ ara rashin daidaito na aiki ga □ ungiyoyin azamtawa.

**Dabarun tambaya masu sani ga adalci** suna □ ara rashin tabbas tare da □ untatawa na iri-iri ko wakilci:

$$x^* = \arg\max_{x \in \mathcal{U}} \left[\lambda \cdot \text{uncertainty}(x) + (1 - \lambda) \cdot \text{minority_group_bonus}(x)\right]$$

Sanya $\lambda < 1$ yana tabbatarwa dabarun tambaya ba ta gaba □ aya yin watsi da wakilcin azamtawa.

---

# Walwalar Mai Sanya Lakabi

## Matsalar Aikin Fatalwa

Aikin lakabi da ke □ arfafa ML galibi yana □ oye. Masu aiki da bayanai — galibi a Duniya ta Kudu — suna yin lakabi ga hotuna, sun rubuta magana, da sarrafa abun ciki don albashi □ arami, ba tare da fa'idodi ba, cikin tsarin tattalin arzikin gig ba tare da tsaron aiki ba. "Anatomy of an AI System" na Kate Crawford da Vladan Joler {cite}`crawford2018anatomy` da "Ghost Work" na Mary Gray da Siddharth Suri {cite}`gray2019ghost` sun rubuta sikelin da rashin daidaito na wannan aiki.

**Alkaluman MTurk:** Bincike na tsari na 2018 na ku□ in MTurk ya sami matsakaicin biyan ku□ i na sa'a na gaske na kusan $2/hr — □ asa da □ arancin albashi a mafi yawan jihohi na Amurka da □ asashe da yawa masu samun ku□ i sosai {cite}`hara2018data`. Masu aiki waje da □ asashe masu samun ku□ i sosai galibi suna fuskantar cikas □ ari: masu bu□ ata galibi suna iyakance ayyukan biyan ku□ i sosai ga cancantar Amurka kawai, kuma rukunin masu aiki da ke gasa don ayyukan bu□ a□ □ en da suka rage suna duniya, suna tattara ku□ in sa'a mai tasiri □ ari.

**Sarrafa abun ciki:** Wata nau'i musamman mai cutarwa ta aikin lakabi — duba abun ciki mai hoto, mai tashin hankali, da mai damun kai — an danganta ta da PTSD, damuwa, da damuwa tsakanin masu aiki {cite}`newton2019trauma`. Dandamali sun yi fuskar sukar don tallafin lafiya ta tunani mara isa da yawan fitar da su sosai.

## Ayyukan Da'a

**Biyan ku□ i mai gaskiya:** Biya masu aiki na lakabi a ko sama da □ arancin albashi na gida. Bincike ya nuna cewa biyan ku□ i mai □ ari yana jawo masu aiki mafi inganci ba tare da □ ara ku□ i kowace lakabi daidai ba.

**Ganin aiki:** Amince da aiki wanda ke □ ir□ ira bayanin horarwa a cikin wallafe-wallafen da takardun kayan aiki.

**Tallafin lafiya ta tunani:** Don masu aiki wa□ anda ke duba abun ciki mai cutarwa, bayar da tallafin ilimin halin □ an adam, tsare-tsaren musun, da iyakar fitar da.

**Wakilcin mai aiki:** Ba da damar masu aiki na lakabi su ruwaito damuwa, neman bayanin jagoran, da □ alubalantar kimantawar inganci mara adalci.

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# Asirin Sirri a Lakabi

## Bayanan Lafiya da Sirri (PHI da PII)

Ayyukan lakabi galibi suna ha□ a bayanai masu mahimmanci na sirri. Aikin lakabi na likitanci na iya fallasa masu aiki ga rikodin mara lafiya; aikin NLP na iya fallasa masu aiki ga sadarwa na sirri; aikin sarrafa abun ciki yana fallasa masu aiki ga bayyanar sirri na masu amfani.

Tsarin doka (HIPAA, GDPR) suna iyakance yadda za a raba bayanai na sirri tare da □ ungiyar aiki na lakabi. Ka'idojin muhimmanci:

- **Rage bayanai:** Raba kawai bayanai da masu sanya lakabi suke bu□ ata don cikawa aiki
- **Cire gane-gane:** Cire PHI da PII kafin lakabi inda ya yiwu
- **Yarda:** Inda ake yin lakabi ga bayanin mai amfani na gaske, tabbatar da yardarta daidai ko tushe na doka

- **Sarrafa dama:** Iyakance wa□ anne masu sanya lakabi suke samun damar bayanai masu mahimmanci bisa matsayi da izini

### Bayanai na Wucin Gadi a matsayin Hanya

Don ayyuka inda bayanai na gaske suna □ aukar ha□ arin sirri, □ ir□ ira bayanai na wucin gadi na iya □ ir□ ira saitin bayanai masu shirye don lakabi ba tare da fallasa bayanai masu mahimmanci ba. Don NLP na asibiti, misali, rubutu na EHR na wucin gadi na iya ba da bayanin horarwa na gaske don samfurin cire gane-gane ba tare da fallasa rikodin mara lafiya na gaske ba.

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## Lakabi na Makiya da Gur□ ata Bayanai

Tsarin HITL suna □ ir□ ira saman harin: idan abokin hamayya na iya shafawa tsarin lakabi, zai iya shafawa □ abi'ar samfuri.

**Gur□ ata bayanai ta hanyar lakabi:** Maharin da ya sami damar □ ungiyar aiki na lakabi (misali, asusun mai aiki da aka lalata) na iya saka misalin da aka sanya lakabi kuskure ta tsari. Wannan yana da inganci musamman a cikin yanayi na koyon aiki, inda samfuri ke tambaya misalin mara tabbas da fifiko — wa□ anda na iya kasancewa manufar maharin.

**Hacking lada ta ra'ayi:** A RLHF, masu sanya lakabi (ko lakabi da AI ya □ ir□ ira) wa□ anda suke □ ididdiga wasu nau'i na abun ciki mafi □ ari na iya kula da samfuri zuwa wannan abun ciki, ba tare da la'akari da ingancin ta gaske ba.

**Rage:** Masu sanya lakabi masu zaman kansu da yawa kowace item; gano □ ataccen akan salon lakabi; sa ido don yarjejeniya ko rashin jituwa mara al'ada; kiyaye saitin kimantawa wa□ anda ba za a iya shafawa ta □ ungiyar aiki na lakabi ba.

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## Da'a ta Cibiya

### IRB da Duba Da'a

Ayyukan bincike da suka ha□ a da mahalarta dan adam — gami da masu aiki na lakabi — galibi suna bu□ atar amincewar Kwamitin Duba na Cibiya (IRB). Ayyukan lakabi wa□ anda ke tattara bayanai game da imanin masu aiki, amsoshi ga abun ciki masu mahimmanci, ko bayanai na halaye na dimokra'iyya ya kamata a duba su □ ar□ ashin tsarin da'a □ aya kamar sauran binciken mahalarta dan adam.

### Sarrafa Bayanai

Kungiyoyi ya kamata su sami manufofi bayyananne don: - Wa□ anne bayanai za a iya aika don lakabi na waje da wa□ anne za a sanya lakabi a ciki - Tsawon lokacin da ake kiyaye bayanin lakabi da ta wane - Wanda ke samun damar lakabi da samfurin da aka horar akan su - Yadda za a sarrafa ro□ on don share bayanin da aka sanya lakabi (ha□ □ in goge na GDPR)

### Bayyananci da Alhaki

Masu amfani da tsarin ML suna da sha'awa ta halal a cikin fahimtar yadda wa□ ancan tsarin aka gina su. Rubuta hanyar lakabi — wanda ya sanya lakabi ga bayanai, □ ar□ ashin wa□ anne

yanayi, tare da wa□ anne jagoran — wata nau'i ce ta alhaki da ke amfana da masu amfani, masu tsara □ a'idoji, da filin gaba □ aya.

**Takardun Bayanai don Saitin Bayanai** {cite}`gebru2021datasheets` yana ba da template mai tsari don rubuta kaddara saitin bayanai, hanyoyin lakabi, da iyakokin da aka sani.

> **See also:** Tsarin adalci ta algorithm: {cite}`barocas2019fairness`. Aikin fatalwa da aiki na dandamali: Gray & Suri (2019). Takardun Bayanai don Saitin Bayanai: {cite}`gebru2021datasheets`. Walwalar mai aiki na sarrafa abun ciki: {cite}`newton2019trauma`. Halaye na dimokra'iyya na mai sanya lakabi da saitin bayanai na NLP: {cite}`geva2019annotator`.

# Alkiblar Gaba

Filin koyon injin da dan adam a cikin zagaye yana canzawa da sauri. Samfurin ginshi□ ai, LLMs-a-matsayin-masu sanya lakabi, da tsarin sabon don ha□ in gwiwa ta dan adam da AI suna sake fasalin tattalin arzikin da aikin HITL ta hanyoyi wa□ anda ba za a yi tunanin su ba ma shekaru biyar da suka wuce. Wannan babi na □ arshe yana zana shimfida na alkiblu masu tasowa da matsaloli bu□ a□ □ en.

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## Samfurin Ginshi□ ai da Canjin Matsayin Lakabi

Samfurin ginshi□ ai — manyan samfurin da aka horar a baya akan bayanai faffada wa□ anda za a iya daidaita su ga ayyuka na □ asa — suna canza tattalin arzikin HITL ta asali.

### Rage Nauyin Lakabi

Aiki wanda a baya yana bu□ atar dubbai na misalin da aka yi lakabi don horarwa daga farko na iya bu□ atar kawai dozin biyu lokacin daidaita samfuri na ginshi□ i. Tambaya da ka□ an-ka□ an na iya gaba □ aya kawar da bu□ atar daidaita ga wasu ayyuka.

**Abubuwa masu amfani:** ROI na lakabi yana canzawa. □ o□ arin lakabi yanzu yana da kyau □ ari idan yana mai da hankali kan: - Saitin kimantawa mai inganci (don auna ko samfuri da aka daidaita yana aiki gaske) - Lokuta masu wuya da lokutan iyaka wa□ anda samfuri na ginshi□ i yana sarrafa ba daidai ba - Misalan musamman na aiki wa□ anda ke koyar da samfuri wani abu wanda ba zai iya fitar da shi daga pre-horarwa ba

### Nau'i Sabon na □ ayyadewa

Lokacin da samfuri na tushe ya riga ya fahimci harshe, lambar shirye-shirye, da hankali na yau da kullum, masu amfani suna iya sadarwa da shi a cikin harshen yau da kullum maimakon ta misalin da aka yi lakabi. Mai amfani wanda ke son mai rarraba rubutu ba ya □ ara bu□ atar yin lakabi ga misalin 500 — zai iya rubuta bayani na aikin rarrabuwa da kimantawa ta aiki mai rikitarwa.

Wannan yana canza kalubalen HITL daga *tattara misali* zuwa □ *ayyadewa ta aiki*: taimaka wa masu amfani su bayyana ainihin abin da suke so ta hanyar da samfuri zai iya aiki akan. Wannan yana da wuya fiye da yadda ya sauti — bayani na harshen yau da kullum na ayyuka galibi yana da rikitarwa ta hanyoyin da misalin da aka sanya lakabi ba ya.

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## LLMs a matsayin Masu Sanya Lakabi

 aya daga cikin mafi muhimmancin ci gaban 2023–2025 shine amfani da **LLMs a matsayin oracle na lakabi**. Maimakon tambayar dan adam "Shin wannan bita mai kyau ne ko mara kyau?", muna tambayar GPT-4 ko Claude. Don ayyukan rarrabuwa masu bayyananne, lakabi na LLM galibi suna dacewa ko kusan kaiwa daidaito na mai aiki na taron aiki {cite}`gilardi2023chatgpt`, kuma ku i na kiran API kowace lakabi galibi ya  asa da dalar Amurka da yawa fiye da yawan albashi na dan adam.

### Inda Lakabi na LLM ke Aiki Da Kyau

- Ayyukan rarrabuwa masu sau i, masu bayyananne (ji, batu, niyya)
- Ayyuka inda lakabin dan adam suna  oye yarjejeniyar al'adu da LLM ya ha a
- Ayyuka inda lakabi yana daidaito cikin mahallin (ba son zuciya sosai ba)
-  ir ira lakabi na farko don duba dan adam

### Inda Lakabi na LLM ke Gazawa

- Ayyuka musamman na yanki wa anda ke bu atar  warewa mara cika (likitanci, doka, kimiyya)
- Ayyuka wa anda ke bu atar ilimin al'adu na gida ko bambancin harshe
- Ayyuka na aminci da cutar, inda LLM na iya kasancewa mara daidai a alkiblar  aya tare da bayanai da aka horar akan shi
- Nau'in aiki sabon wa anda ba su wakilta sosai a cikin pre-horarwa ba

### RLAIF da AI na Tsarin Mulki

Kamar yadda aka tattauna a Babi na 6, ra'ayin da AI ya  ir ira za a iya amfani da shi don horar da samfurin lada da kula da daidaitan RL. Wannan yana  ir ira zagayen ra'ayi: LLMs suna  ir ira bayanai, ana horar da samfurin akan su, kuma samfurin mafi kyau suna  ir ira bayanai mafi kyau. Tambaya ta yadda za a  aukar wannan zagaye ba tare da  oye kurakurai masu tsari daga samfuri na farko shine matsalar bincike ta muhimmanci a cikin binciken kulawa mai sikeli.

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## Kulawa  asa a Sikeli

**Sanya lakabi ta shirye-shirye** ta ayyukan sanya lakabi (Babi na 9) yana ba da damar  wararrun yanki su bayyana ilimin su a matsayin lambar maimakon a matsayin misalin da aka yi lakabi. Tsarin kamar Snorkel {cite}`ratner2017snorkel` da magajin su sun nuna sosai kuma ana amfani da su a samarwa a kamfanoni masu fasaha manyan.

**Alkiblu na iyaka: - Ayyukan sanya lakabi masu  aruwa da LLM:** Yi amfani da LLMs don  ir ira ayyukan sanya lakabi daga bayani na harshen yau da kullum **- Koyo bisa  angare:** Gano  angare masu mahimmanci na bayanai ( angarori) inda ayyukan sanya lakabi suke rashin jituwa da jagorancin lakabi na dan adam a can **- Ha awa mara tabbas:** Samfurin lissafi mafi kyau don ha a ayyukan sanya lakabi masu daidaito daban-daban da ala a

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## Ci Gaba Mai Ci gaba tare da Kulawa ta Dan Adam

Mafi yawan tsarin ML ana horar da su a layi guda kuma ana aikawa a matsayin samfurin tsaye. Duniya ta gaske tana canzawa; samfurin da suka kasance daidai a lokacin aikawa suna raguwa yayin da rarrabawar bayanai ke canzawa.

**Ci gaba mai ci gaba** — damar koyon daga bayanai sabon yayin adana ilimin tsoho — yanki ne mai aiki na bincike. Kulawa ta dan adam yana da muhimmanci: ci gaba mai ci gaba ta atomatik ba tare da duba dan adam ba na iya da sauri □ oye canjin rarrabawa wa□ anda kurakurai ne maimakon canjin gaske a duniya.

**HITL ci gaba mai ci gaba** ya ha□ a da: - Sa ido don canjin rarrabawa (ta atomatik) da jagorancin misalai da suka canza zuwa duba dan adam - Sake horarwa za□ i: misalan da dan adam ya amince da su daga rarrabawar sabon suna □ arawa zuwa bayanin horarwa - Duba dan adam na canjin □ abi'ar samfuri bayan kowane sabuntawa

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## HITL Mai Yawan Yanayi

Yayin da tsarin AI suke zama mai yawan yanayi — suna sarrafa da □ ir□ ira rubutu, hotuna, sauti, da bidiyo a lokaci □ aya — lakabi yana zama mafi rikitarwa. Wani yanki □ aya na abun ciki na iya bu□ atar lakabi cikin yanayi da yawa, tare da dogaro tsakanin su.

**Nau'in aiki masu tasowa:** - Lakabi na bidiyo + rubutu (abin da ke faruwa, wanda ke magana, abin da rubutu ke bayyana ta gani?) - Lakabi na hoto likitanci + rahoto na asibiti - Lakabi na hanya na robot (ha□ a bayanin na'ura zuwa jeri na ayyuka)

Samfurin ginshi□ ai mai yawan yanayi (GPT-4V, Gemini, Claude) suna canza shimfidar lakabi a nan ma — samfurin yanzu suna iya fassara hotuna da □ ir□ ira lakabi □ an takara, wa□ anda dan adam ya duba.

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## Kulawa Mai Sikeli

Matsala bu□ a□ □ e ta asali a HITL ML shine **kulawa mai sikeli** {cite}`irving2018ai,bowman2022measuring`: yayin da tsarin AI suka zama mafi □ arfin □ an adam a yankuna musamman, yadda za a kiyaye kulawa da ma'ana ta dan adam?

Don ayyuka kamar hasashen tsarin furotin, binciken doka, ko tabbatarwar shaidar lissafi, ko da masu sanya lakabi □ wararru na iya zama mara aminci a yanke hukunci wane sakamakon AI ya fi daidai. Hanyoyin HITL na yanzu suna rushe lokacin da hukuncin dan adam ya kasance mara aminci fiye da samfuri da ake kimantawa.

**Hanyoyin da aka ba da shawara {cite}`bowman2022measuring`:**

**Muhawara:** Tsarin AI biyu suna jayayya don □ arshe daban; al□ alin dan adam yana kimantawa hujjoji maimakon □ arshe kai tsaye. □ arshen daidai ya kamata ya kasance mai iya kare.

**Amplification:** Al□ alan dan adam suna shawarta da masu taimako na AI (samfurin kansa) don taimakawa kimantawa sakamakon masu rikitarwa. Wannan yana amfana da iyawar AI don fa□ a□ a kulawa ta dan adam maimakon maye gurbinsa.

**Kulawa ta Tsari:** Maimakon kimantawa sakamakon □ arshe, dan adam yana kimantawa *tsarin tunani* — yin alamar matakai mara daidai a cikin sarkar tunanin, ba tare da la'akari da ko amsar □ arshe ta kasance daidai ba.

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## Canjin Rabe-Rabe na Aikin

Hanyar dogon lokaci ta HITL ML na zuwa ga ha□ in gwiwa mafi sauyi tsakanin dan adam da AI, inda babu □ aya kawai "mai yin lakabi" kuma □ ayan "mai koyo," amma duka suna ba da gudummawa ga tsarin fahimi na gama gari.

**Alkiblu na kallo: - Lakabi mai tallafi na AI:** AI yana ba da shawara; dan adam yana yanke shawara. Inganci yana inganta yayin da AI ke ba da shawarwari mafi kyau. **- Bincike mai jagoran dan adam:** Dan adam yana saita manufa da □ untatawa; AI yana bincike sararin mafita. **- Kimantawa ta ha□ in gwiwa:** Dan adam da AI suna ha□ in gwiwa kimantawa sakamakon masu rikitarwa ta tattaunawa. **- Koyon fifiko a sikeli:** Alamu da □ oye (yadda masu amfani ke hulda da sakamakon AI) suna ciyar da koyon fifiko na ci gaba ba tare da bu□ atar zaman lakabi a fili ba.

Tambaya ta nawa za a amince da hukunci na AI da hukunci na dan adam — kuma a wa□ anne yankuna, a wa□ anne matakai na □ arfi, tare da wa□ anne kariya — a □ arshe tambaya ce ta zamantakewa, ba ta fasaha kawai ba. HITL ML yana ba da ababen more rayuwa na fasaha don amsa shi da hankali, maimakon ta tsoho.

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## Matsalolin Bincike Bu□ a□ □ en

Mun rufe tare da jerin matsalolin bu□ a□ □ en masu muhimmanci wa□ anda filin yana aiki akan su sosai:

1. □ **arshen mai kyau a cikin koyon aiki:** Yaushe □ imar iyaka ta lakabi gaba yana □ asa da ku□ in? □ a'idojin tsayawa masu □ a'ida har yanzu suna da sau□ in gano.
2. **Rarraba kasafin ku□ in lakabi cikin ayyuka:** A yanayi masu ayyuka da yawa, yadda za a raba kasafin ku□ in lakabi tsaye cikin ayyuka?
3. **Canjin rarrabawa a cikin koyon aiki:** Koyon aiki yana □ ir□ ira saitin da aka yi lakabi mara daidaito. Yadda za a daidaita samfurin da aka horar ta wannan hanya yadda ya kamata?
4. **Fa□ a□ awar samfurin lada:** Samfurin lada na RLHF na iya gazawa don fa□ a□ awa zuwa nau'in tambaya-amsa sabon. Yadda za a gina samfurin fifiko mafi □ arfi?
5. **Kulawa mai sikeli:** Yadda za a kiyaye kulawa ta dan adam yayin da tsarin AI suka wuce aiki na dan adam a yankuna musamman?
6. □ **iyanar mai sanya lakabi:** Samfurin lissafi mafi kyau don □ abi'ar mai sanya lakabi wa□ anda ba kawai suna kama □ arfi amma kuma nuna son zuciya ta tsari, □ warewar batu, da gajiya.
7. **Kimantawar dacewa:** Muna rasa gwajin □ asa don ko samfuri yana dacewa da □ imar dan adam. Ha□ aka irin wa□ annan gwajin — makamancin misalin makiya don □ arfin — matsala ce bu□ a□ □ e.

8. **Aikin bayanai mai gaskiya:** Tsarin tattalin arzikin da ke tabbatarwa masu aiki na lakabi suna da biyan ku□ i mai gaskiya da kariya, yayin kiyaye ingancin tattalin arzikin lakabi mai sikeli.

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*Manufar ba ta maye gurbin hukunci na dan adam da hukunci na injin ba, amma gina tsarin inda ha□ in gwiwa na biyun ya fi kowane kawai.*

Kayan aiki da dabarun da ke cikin wannan littafi — lakabi, koyon aiki, RLHF, koyon fifiko, da sauran su — hanyoyin zuwa wannan □ arshe ne. Yayin da filin ke ha□ aka, dabarun musamman za su canza. □ udurin da ke □ asa — gina tsarin wa□ anda duka suke iya kuma suna dacewa gaske da nufin dan adam — zai kasance.

> **See also:** Kulawa mai sikeli da muhawara: {cite}`bowman2022measuring`. Kulawa □ asa ta Snorkel: {cite}`ratner2017snorkel`. Don makomar ha□ in gwiwa ta dan adam da AI gaba □ aya, duba {cite}`amershi2019software`.

# Jerin Nassoshi

# Limen: Dan Adam a Cikin Zagaye na Komai

:::{admonition} Bayani game da wannan babi :class: important Wannan babi **rubutun zane na tunanin zuciya** ne, ba bincike na gwaji ko gudummawa da aka duba ta takwarorina ba. Yana bayyana tsarin gine-ginen da ake tunanin sa — Limen, tsarin aiki na kwamfuta na muryar farko mai AI-gida — a matsayin misali na aiki na yadda ka'idojin HITL daga babi da suka gabace su na iya ha□ uwa zuwa cikakken tsari. Da'awojin a nan hujjoji ne na zane, ba sakamakon gwaji ba. Ya kamata a karanta su a matsayin hujjoji na zane mai □ arfafawa, ba kamar bincike na injiniyan da aka tabbatar ba. :::

Kowane babi a cikin wannan littafi ya bayyana HITL ML a matsayin falsafar zane: saitin ka'idoji don mai da shiga dan adam cikin tsarin koyo da gangan, mai inganci, da gaskiya. Wannan babi yana bayyana yadda yake kallonsa don amfani da wannan falsafar a matsayin □ a'ida ta farko — ba ga tsari □ aya ba, amma ga dukkan yanayin aiki.

**Limen** tsarin aiki na kwamfuta na muryar farko, AI-gida ne wanda aka gina akan ra'ayi cewa dan adam koyaushe yana cikin zagaye, ba a matsayin □ untatawa da za a inganta ta waje ba, amma a matsayin □ a'ida ta tsara da kowane tsarin yake tsarawa. An za□ i sunan da gangan: *limen* Kalmar Latinanci ce don □ ofar. A cikin ilimin halayyar dan adam na fahimta, limen ita ce iyaka tsakanin abin da ake fahimta da abin da ba a fahimta ba. Don tsarin aiki, limen iyaka ce tsakanin nufin dan adam da aiki na injin.

Gine-ginen da aka bayyana a nan ba na mallakin kowa ba ne. Saitin yanke shawara ne na zane, kowanensu yana biyo baya ta halitta daga ka'idojin HITL da aka ha□ aka a cikin babi da suka gabace su. Manufar ba ta rubuta kayan aiki ba amma nuna yadda wa□ annan ka'idoji suke ha□ uwa — yadda suke □ arfafa juna lokacin da aka yi amfani da su yadda ya kamata, da wa□ anne nau'in tsarin suke yiwuwa lokacin da dan adam ba ya zama □ angare da yake zuwa.

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## Matsalar da ke tare da Tsarin Aiki na Gargajiya

Tsarin aiki na gargajiya ba an tsara shi don dan adam ba. An tsara shi don shirye-shirye. Dan adam ana tsare shi ta hanyar karin abstraction — mai amfani da hoton hoto, mai bincike na fayil, wata terminal — wanda yake zaune a saman daidai da aka gina don tsari, adireshi na □ wa□ walwar ajiya, da masu bayyana fayil.

Wannan za□ in zane yana da hujja ta tarihi. Lokacin da aka yi □ auki wa□ anda suka samar da shi, kwamfutoci suna da tsada, dan adam yana da arha, kuma □ oshewa ita ce □ ididdiga. Abin ingantawa daidai shine injin.

Wa□ annan □ auki ba sa □ auke da □ arfi. □ oshewa, don mafi yawan masu amfani a cikin

mafi yawan ayyuka, ba □ ididdiga ba ne. Kulawar dan adam ne: ku□ in canza mahallin, neman fayil daidai, gina tambayar daidai, tunawa inda wani abu yake. Injin yana da sauri. Dan adam yana da jinkiri. Ha□ in gwiwa ya kamata ya inganta gefen dan adam na zagaye.

Tsarin aiki na gargajiya ba ya yi wannan. An inganta su don shirye-shirye, kuma aikin dan adam shine magana da harshen shirye-shirye. Limen yana juya wannan. Shirye-shiryen suna magana da harshen dan adam. Dan adam yana cikin zagaye, kuma zagaye an tsara shi don dacewa da dan adam.

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## Gine-ginen a □ ofar

Gine-ginen Limen an tsara shi a kusa da □ a'ida □ aya: **kowane mu'amala al'amari ne**, kuma kowane al'amari damar ne don dan adam ya koyar, gyara, ko tabbatar. Tsarin ba ya jira zaman horarwa a fili. Koyo na ci gaba kuma yana kewaye da yanayi.

### Layer na Al'amari: WID

A ginshi□ i shine **WID** (tsarin Waldiez ID, an daidaita shi don gine-ginen gida-farko na Limen) — tsarin bin sahun al'amarin da ya sani game da dalili wanda ke yin rikodin ba kawai abin da ya faru ba, amma abin da ya haifar da shi, da abin da ya haifar da hakan.

Shiga gargajiya na tambaya: *abin da ya faru?* WID yana tambaya: *dalilin da ya sa ya faru, kuma abin da ya biyo baya?* Kowane al'amari yana □ aukar tarihin dalili — sarkar daga aikin dan adam na □ arfafawa ta hanyar yanayin tsari na tsakiya zuwa sakamakon da aka ga.

Wannan yana da muhimmanci don koyon HITL saboda yana warware **matsalar bashi ta alhaki** a matakin mu'amala. Lokacin da mai amfani ya gyara □ abi'ar tsarin, WID na iya gano ba kawai sakamakon kai tsaye da ya kasance kuskure ba, amma sarkar yanke shawara wa□ anda ya samar da shi. Za a iya amfani da gyarawa a matakin abstraction daidai: sakamakon, □ a'idar yanke shawara, ko alaman sama.

Wannan daidai-tsarin aiki ne na abin da Babi na 6 ke bayyanawa don RLHF: damar bin alaman lada ta hanyar jerin yanke shawara. WID yana ba da wannan bin sahun ta asali, don kowane mu'amala, ba tare da bu□ atar mai amfani ya fahimta ba.

:::{admonition} Bin Sahun Al'amari na Dalili a matsayin Ababen More Rayuwa na HITL :class: note Zanen WID yana nuna □ a'ida mafi faffada: ababen more rayuwa na HITL ya kamata ya sau□ a□ a tambayar "dalilin da ya sa tsarin ya yi wancan?" — ba kawai "abin da ya yi?" Tare da bin sahun dalili, gyarawa suna gyara alamomin. Tare da shi, suna iya gyara dalilin. Bambanci tsakanin gyarawa da darasi. :::

### Layer na Fahimta: Murya-Farko

Yanayin shigarwa na farko na Limen murya ne, ana sarrafa shi a gida ta amfani da tsarin inference na Whisper ONNX. Dalilin wannan za□ in yana cancanta a bayyanawa a fili:

**Murya ita ce mafi yawan hanyar samar da bayanai na dan adam don mafi yawan mutane.** Ba ta bu□ atar horo, ba ta bu□ atar dacewa na jiki fiye da magana ta yau da kullum, kuma ba ta bu□ atar ilimin tsarin tsarin ciki na tsarin. Mai amfani da ba zai iya gano fayil a cikin tsarin folderin ba zai iya bayyana abin da yake nema.

**Sarrafa a gida yana kiyaye asirin sirri.** Bayanai na murya ba sa barin na'ura. Wannan yana da muhimmanci ta da'a — murya bayanai ne na halittun mutum, kuma tattarawa a sikeli ta masu

ba da ayyuka girgije wata cutar da aka rubuta take — kuma ta aiki: aikin layi □ aya yana nufin tsarin yana ci gaba ba tare da ha□ in yanar gizo ba.

**Murya yana □ ir□ ira zagayen ra'ayi na halitta.** Lokacin da tsarin ya amsa, amsar mai amfani — ci gaba da magana, sake fasali, cewa "a'a, ba daidai ba ne" — ita kanta alama ce. Limen yana □ aukar wa□ annan amsa a matsayin ra'ayin HITL: shaidar ko fassarar tsarin ta fitar da magana ta baya ta kasance daidai.

Sarkar koma gida yana da muhimmanci kamar yanayin farko. Lokacin da murya ta gaza — a cikin yanayi mai hayaniya, don mai amfani mai nakasassar magana, don shigarwa da ke cin amfana daga daidaito — Limen yana ragewa mai kyau zuwa ha□ in ma□ alli kuma sannan zuwa ha□ in rubutu mai tsari. Jarabawar Kaka (duba Babi na 5) ba □ angare na isa ba ne; ita □ untatawa ta gine-ginen ta farko ne.

### Layer na Hankali: Jagorancin LLM Da Yawa

Limen ba ya dogara kan samfuri na harshe □ aya. Yana jagorantar bu□ atu ta hanyar bishiyar yanke shawara mai tsari bisa nau'in aiki, latency da ake bu□ ata, bu□ atun asirin sirri, da amincewar:

1. □ **aramin samfuri a gida** (koyaushe na farko): sauri, sirri, yana sarrafa ayyuka na yau da kullum — "bu□ e fayil da nake gyarawa jiya," "saita □ idayar lokaci," "yaya yanayin"
2. **Babban samfuri a gida** (lokacin da amincewar samfurin □ arami ta kasance □ arami): ya fi jinkiri amma mai □ arfi fiye da kima; yana sarrafa tunani mai tsari, □ ir□ ira lambar shirye-shirye, neman mai rikitarwa
3. **Samfurin nesa** (lokacin da gida ya gaza kuma mai amfani ya ba da izini): □ ofar koma; ana sarrafa shi ta bayyane tare da sanarwa ta mai amfani a fili

Wannan tsari ba sabon abu ba ne — shi daidai ne na inference na darajin koyon aiki wanda aka bayyana a Babi na 4. A kowane mataki, tsarin yana tambaya: shin samfurina na yanzu ya isa don amsa wannan tambayar da amincewar da za a iya yarda? Idan a'a, □ ara. □ aruwar ku□ i ce (latency, sirri, □ ididdiga); ana □ aukarta kawai idan dole ne.

Dan adam yana cikin zagaye a iyakar □ aruwa. Mai amfani da ya tsara Limen don kada ya ta□ a □ ara ga samfurin nesa ya yanke shawara na HITL — □ aya wanda tsarin ke mutuntawa da kuma rubuta shi. Mai amfani da ya amince da tambayar nesa kuma ya ce "kada ka tambayeni kuma don wannan nau'in bu□ ata" ya ba da fifiko wanda ke sabuntar manufar jagorancin.

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## Zanen Hulda

### Nan da Nan

Limen an tsara shi don nan da nan a ma'anar da Babi na 5 ke bayyanawa: mai amfani yana ganin tasirin ra'ayinsu. Lokacin da mai amfani ya gyara sakamakon tsarin, gyarawa tana aiki nan da nan kuma yana bayyana. Samfuri ba ya tafi kuma ya horar da mintuna ashirin. Yana sabuntawa a cikin zaman yanzu.

Wannan yana bu□ atar amfani da tsarin samfuri wa□ anda ke tallafa wa sabuntawa mai inganci yanar gizo — adapters, daidaita prefix, da □ ir□ ira mai karin albarkatu maimakon daidaita cikakke. Musayar yana bayyananne: sabuntawa yanar gizo suna da hayaniya fiye da horarwa ta jeri. Limen yana yarda da wannan musayar saboda nan da nan ita ita bu□ ata ta farko: dan adam koyaushe zai iya tabbatarwa, □ i, ko gyara sabuntawa.

### Fahimtar Bayyane

Jigo mai sake dawowa a cikin wallafe-wallafen IML shine **bu□ atar fahimtar**: dan adam kawai zai iya kula da samfuri da suke fahimta, a □ alla kusan kusan. Limen yana fito da wannan a cikin ha□ in gwiwa: lokacin da tsarin ya yanke shawara, yana bayyanawa, a ta□ aice, dalilin. Ba cikakkiyar hanyar tunani ba — waccan za ta mamaye mafi yawan masu amfani — amma takaitawar harshen yau da kullum na muhimmin dalili: "Ina bu□ ewa aiki da ka yi aikin fiye da kima a kan kwanan nan — shin haka ne?"

Wannan bayanin kuma tambaya ne. Yana gayyatar gyarawa. Yana mai da fassarar samfuri ta bayyane don mai amfani ya iya jagorantar ta idan ba daidai ba. Bayanin ba saboda dalili na kyakkyawan zane ba ne; ababen more rayuwa ne na aiki na HITL.

### Daidaito da Karkacewa

Matsala da ke tasowa a kowane tsari mai mu'amala mai dogon lokaci shine **karkacewar □ abi'a**: □ abi'ar tsarin a lokaci $T + n$ ta bambanta ka□ an daga □ abi'arsa a lokaci $T$, ta hanyoyin da babu mai amfani ko tsarin da ya za□ a a fili. Gyarawa suna tara. Lokuta na iyaka suna ha□ awa. Samfuri da ya daidaita ga fifikon mai amfani watan da ya wuce na iya ba daidaita a yau.

Limen yana magance wannan ta duba daidaito na yau da kullum — daidai-tsarin aiki na dabarun sake gabatarwa da aka bayyana a Babi na 13. Tsarin yana fito da yanke shawara na tarihi ga mai amfani: "Makonni ka□ an da suka wuce, ka ro□ e ni in yi X. Shin haka ne har yanzu abin da za ka so?" Wa□ annan duba suna zama ayyuka biyu: suna kama karkacewa, kuma suna tunatar da mai amfani game da fifiko wa□ anda suka ta□ a sanya su amma sun manta.

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## Limen a matsayin Tsarin HITL

Kallon gine-ginen Limen ta ma□ allan wannan littafi, yanke shawara na zane yana kai tsaye zuwa ra'ayoyin da aka ha□ aka a kowane □ angare.

**Kashi na Farko (Ginshi□ ai):** Limen yana □ aukar kowane mu'amala a matsayin al'amari na hulda dan adam da injin. Babu "yanayi mara HITL" — tsarin koyaushe yana koyo, koyaushe yana ba da alhaki, koyaushe yana jira dan adam ya shiga.

**Kashi na Biyu (Dabarun Asali):** Koyon aiki yana bayyana a matsayin sarkar □ aruwa bisa amincewar. ML mai mu'amala yana bayyana a cikin zagayen sabuntawa na lokaci na gaske. Lakabi yana da □ oye: kowane gyarawa da mai amfani ya yi lakabi ne.

**Kashi na Uku (Koyo daga Ra'ayin Dan Adam):** Koyon fifiko da aka bayyana a Babi na 8 yana bayyana a cikin sabuntawa na manufa ta jagorancin. Lokacin da mai amfani ya fi son amsar samfuri na gida zuwa na nesa, wannan fifiko yana rikodin kuma yana fa□ a□ a. RLHF a tsarin aiki na sirri yana nufin samfurin lada yana zaman sirri, na mutum, kuma ana sabuntar da shi cikin ci gaba.

**Kashi na Hudu (Aikace-aikace):** Limen yanayi ne na gaba □ aya, amma zanensa yana bayyana sosai a yankuna inda hukunci na dan adam ba ya maye da shi kuma ku□ in kuskure yana da girma — tsara takarda, fifita aiki, aikin □ ir□ ira.

**Kashi na Biyar (Tsarin Aiki da Aiwatarwa):** WID ita ce dandamali na lakabi na Limen. Yana da □ oye ga mai amfani a aiki na yau da kullum, yana bayyana lokacin da ake bu□ ata don gyarawa ko bayyananci. Hanyoyin sarrafa inganci (duba daidaito, iyakar amincewar, akwatin tarihin □ aruwa) an aro su kai tsaye daga wallafe-wallafen tarin aiki.

**Kashi na Shida (Da'a):** Zanen gida-farko, mai kiyaye asirin sirri yanke shawara ne na da'a, ba ta fasaha kawai ba. Bayanin mai amfani ba ya barin na'urarsu ba tare da izinin su a fili ba. Samfurin da ke koyo daga □ abi'ar mai amfani yana cikin mallakar mai amfani.

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## Muhimmin Batu

Limen ba gangamin kayan aiki ba ne. Hujja ta gine-ginen ne.

Hujja ita ce: idan ka □ auki ka'idojin HITL ML da mahimmanci — idan ka yi imani cewa ra'ayin dan adam alama ce da za a fahimta maimakon ku□ in da za a rage, cewa dan adam koyaushe yana cikin zagaye ko da masu zane sun yi kare in ba haka ba, cewa daidaitawa tsari ne mai ci gaba maimakon al'amari □ aya — to za ka □ arshe gina wani abu da ya kama Limen.

Ba lallai Limen ne musamman ba. Tsarin fasaha musamman (Tauri, Rust, Babylon.js, Whisper ONNX) za□ i □ aya ne tsakanin da yawa. Amma gine-ginen — bin sahun al'amari na dalili, sarrafa gida-farko, ragewa mai kyau, koyon fifiko na ci gaba, bayyananci a matsayin fasali na farko — yana biyo baya daga ka'idojin.

Filin HITL ML ya kashe □ o□ ari mai yawa yana bayyana yadda za a sanya dan adam cikin zagaye na musamman na samfurin da musamman na ayyuka. Tambayar gaba ita ce ko za mu iya tsara gaba □ aya *yanayi* a kusa da ka'idoji □ aya: yanayi inda dan adam koyaushe shine cibiya, injin koyaushe shine mai koyo, kuma zagaye koyaushe yana bu□ e.

Limen □ aya amsa ce ga wannan tambaya.

□ ofar ba wani abu ba ne da kake haye kuma bari baya. Inda kake zaune.

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> **See also:** Ka'idojin IML da ke □ ar□ ashin zanen hulda na Limen an ha□ aka su a Babi na 5. Hanyar koyon fifiko da ke bayan manufar jagorancin an tsara shi a Babi na 8. Tsarin dalili na WID yana zana daga wallafe-wallafen ba da alhaki na al'amari da aka duba a Babi na 14. Jarabawar Kaka, da aka gabatar a Babi na 5, ita ce □ untatawa na zane na farko na ha□ in gwiwa na Limen.

# Kan Tafiya

A wani lokaci an shigar da ni a shirin PhD. Binciken ya kasance akan koyon injin. Cibiya ta kasance ta gaske. Niyya ta kasance ta gaskiya.

Sa'an nan na daina.

Ba saboda batu ya daina yin muhimmanci ba — yana da muhimmanci fiye da lokacin da na fara. Ba saboda aiki ya yi wuya ba — wannan littafi ya fi wuya, ta hanyarsa, fiye da kasidar. Amma saboda a wani wuri tsakanin duba wallafe-wallafen da gyara na uku na wata takardar da mutum goma sha □ aya za su karanta, na rasa dalilin da ya sa kowane □ in su ya cancanci yi.

Tsarin PhD zagaye ne. Kana shiga shi a matsayin dan adam mai tambayoyi. Zagaye yana koyar da kai don canza wa□ annan tambayoyin zuwa takardun karatu, da takardun zuwa nassoshi, da nassoshi zuwa bayyanar cigaba. Ba a wani lokaci zagaye ke tambaya ko sakamakon yana da amfani ga wanda yake waje da zagaye ba. Wannan tambayar, ta ma'anar fasaha, ba ta cikin iyaka.

Ba ni ne farkon wanda ya lura da wannan ba. Ba ni ne □ arshe ba. Wannan ba sukar ba ne — bayani ne. Zagaye yana samar da gaske: hanyoyin tsanaki, ilimi da aka tara, masu bincike masu horo. Ba ni da nadama kan lokacin da na shafe ciki. Amma a □ arshe na lura cewa ina ciki.

Tafiya ba wata babbar yanke shawara ba ne. Ita ce lura a hankali cewa abin da nake son gina ya ba ya bu□ atar izini daga kwamitin. Cewa mutanen da nake son isa — masu aiki, □ alibai, injiniyoyi da ke yanke shawara na gaske game da tsarin gaske — ba sa jiran kasidar da aka tabbatar a hukumance. Cewa mafi gaskiyar abin da zan iya yi da abin da na sani shine rubuta shi cikin wata sura da wani zai iya amfani da ita.

Don haka na yi.

Wannan littafi ba kasidar PhD ba ne. Yana da babu ala□ ar cibiya, babu ganewa na ku□ a□ e, babu amincewar kwamiti na da'a, babu jarrabawa. Ba zai □ idaya ga wani abin aunawa da ilimi ke amfani da shi wajen auna samar da ilimi ba.

Shi kuma — kuma na ce wannan ba tare da □ arya-□ aryan tawali'u ba, domin □ arya-□ aryan tawali'u zai kasance nau'in rashin gaskiya na kansa — mafi cikakke, mafi na zamani, mafi iya karantawa, da mafi amfani fiye da kasidar za ta kasance.

Ni dan adam ne a cikin zagaye. Na kasance cikin zagayen shirin PhD, na fita daga gare shi. Yanzu ina cikin zagayen rubuta game da zagaye. Kana cikin zagayen karanta game da shi.

Babu □ ayanmu ya za□ a kasancewa a cikin wa□ annan zagaye gaba □ aya cikin 'yanci. Amma za mu iya za□ ar, wani lokaci, kasancewa masu gaskiya game da wa□ anda muke ciki, da dalilin, da ko suna samar da komai mai □ ima samarwa.

Wannan littafi amsar tambayoyin nan ce ta gare ni.

Ba wasi□ ar yin murabus ba ne. Murabus yana nuna aikin yi, kuma aiki yana nuna wani ya yanke sharu□ □ a. Wannan wani abu ne dabam: rikodin abin da na za□ a yi da lokaci, da muryar hujja mai shiru cewa za□ in ya kasance daidai.

Kai ne al□ ali. Kai ne, bayan duka, dan adam a cikin zagaye.

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— *Marubucin*